

## AUTOMATING IN HIGH-EXPERTISE, LOW-LABEL ENVIRONMENTS: EVIDENCE-BASED MEDICINE BY EXPERT-AUGMENTED FEW-SHOT LEARNING<sup>1</sup>

**Rong Liu**

College of Business, Florida International University  
Miami, Florida, U.S.A. {rong.liu2@fiu.edu}

**Jingjing Li**

Information Technology and Innovation, McIntire School of Commerce  
University of Virginia, Charlottesville, VA, U.S.A. {jl9rf@virginia.edu}

**Marko Zivkovic**

Genesis Research Group  
Hoboken, NJ, U.S.A. {marko@genesissrg.com}

**Ahmed Abbasi**

Department of IT, Analytics, and Operations, Mendoza College of Business  
University of Notre Dame, Notre Dame, IN, U.S.A. {aabbasi@nd.edu}

*Many real-world process automation environments are rife with high-expertise and limited labeled data. We propose a computational design science artifact to automate systematic review (SR) in such an environment. SR is a manual process that collects and synthesizes data from medical literature to inform medical decisions and improve clinical practice. Existing machine learning solutions for SR automation suffer from a lack of labeled data and a misrepresentation of the high-expertise manual process. Motivated by humans' impressive capability to learn from limited examples, we propose a principled and generalizable few-shot learning framework—FastSR—to automate the multistep, expertise-intensive SR process using minimal training data. Informed by SR experts' annotation logic, FastSR extends the traditional few-shot learning framework by including (1) various representations to account for diverse SR knowledge, (2) attention mechanisms to reflect semantic correspondence of medical text fragments, and (3) shared representations to jointly learn interrelated tasks (i.e., sentence classification and sequence tagging). We instantiated and evaluated FastSR on three test beds: full-text articles from Wilson disease (WD) and COVID-19, as well as a public dataset (EBM-NLP) containing clinical trial abstracts on a wide range of diseases. Our experiments demonstrate that FastSR significantly outperforms several benchmarking solutions and expedites the SR project by up to 65%. We critically examine the SR outcomes and practical advantages of FastSR compared to other ML and manual SR solutions and propose a new FastSR-augmented protocol. Overall, our multifaceted evaluation quantitatively and qualitatively underscores the efficacy and applicability of FastSR in expediting SR. Our results have important implications for designing computational artifacts for automating/augmenting processes in high-expertise, low-label environments.*

**Keywords:** Evidence-based clinical practices, systematic review, data extraction, few-shot learning, natural language processing, attention, deep learning, joint learning

<sup>1</sup> Balaji Padmanabhan was the accepting senior editor for this paper. Jing Wang served as the associate editor. The transparency materials for this paper can be found at [https://osf.io/tsdza/?view\\_only=1c0507e9e88d4bce8f7b63c530880e78](https://osf.io/tsdza/?view_only=1c0507e9e88d4bce8f7b63c530880e78)



©2025. The Authors. This work is licensed under the terms of the Creative Commons Attribution CC BY 4.0 License (<https://creativecommons.org/licenses/by/4.0/>)

## Introduction

Computational design artifacts have garnered considerable attention in recent years as a mechanism for automating/augmenting manually intensive tasks and processes using machine learning (Abbasi et al., 2016; Padmanabhan et al., 2022; Rai, 2017). However, prior IS work has often focused on contexts involving abundant labeled training data and well-defined machine learning (ML) tasks (Kitchens et al., 2018; J. Li et al., 2020). In this study, we explore the critical and timely problem of designing computational artifacts to automate the systematic review (SR) process crucial to evidence-based medicine (EBM)—a context in which labeled training data is at a premium and routine ML models cannot fully represent the high-expertise nature of the manual tasks.

SR is an explicit, reproducible, and bias-reducing process to extract and summarize medical evidence from prior studies (Higgins et al., 2019). It can enable EBM, an important paradigm in modern medical practice that advocates the conscientious, explicit, and judicious use of current best evidence for medical decisions (Djulfbegovic & Guyatt, 2017; Guyatt et al., 2002). SR has proven instrumental in deriving insights from historical medical successes and failures (Clarke et al., 2007; Higgins et al., 2019) and guiding clinical decisions, such as validating treatment options and determining the best interventions (Jones et al., 2013). For instance, during the COVID-19 pandemic, a government-coordinated crowdsourced SR played a crucial role in accelerating vaccine research and development (Wang et al., 2020a). Overall, it is estimated that SR-enabled EBM could prevent up to a third of deaths and reduce healthcare spending by at least 30% (Djulfbegovic & Guyatt, 2017; Pronovost, 2013).

SR is time-consuming, typically performed manually based on a comprehensive protocol, and requires independent work by multiple experts (Higgins et al., 2019; Page et al., 2021). The arduous nature of SR stems from the need to extract *highly granular, disease-specific, and constantly changing* PICO elements (*patient, intervention, comparison, and outcome*)<sup>2</sup> from articles (Higgins et al., 2019). For example, the SR for a single disease may involve extracting hundreds of granular PICO elements (See Appendix A), most of which are not applicable to other diseases (Smith et al., 2015). A case in point is that 126 distinct Outcome (O) elements from COVID-19 clinical trials are markedly different from those associated with other coronaviruses (Qiu et al., 2020). Moreover, the constant evolution of medical interventions and the emergence of new symptoms mean that PICO elements, even within a disease, frequently change. For instance, 37% of the O elements

reported by Cochrane reviews in 2007 had become outdated in 2011 due to advances in clinical practices or disease progression (Smith et al., 2015). This variability complicates the creation of a standardized database or taxonomy for PICO elements, which could otherwise streamline the SR process.<sup>3</sup> Consequently, the average completion time for a manual SR project is 67.3 weeks (Borah et al., 2017), leading to many studies being outdated by the time they are published (Marshall & Wallace, 2019).

ML and natural language processing (NLP) solutions have been proposed to automate the extraction of sentences annotated with PICO elements from medical publications, often known as *data extraction* (Marshall & Wallace, 2019; Schmidt et al., 2025). Typically, such solutions are divided into two supervised machine learning tasks: sentence classification to detect sentences with PICO elements in articles, and sequence tagging to identify PICO-related text fragments (or *fragments* for simplicity) within those sentences (Nye et al., 2018; Schmidt et al., 2020). Due to the sheer number of PICO elements involved and the need for sufficient labeled samples per element (Richter & Khoshgoftaar, 2020), data labeling for ML SR can be as time-consuming as manual SR (Spasic & Nenadic, 2020). To mitigate labeling efforts, current ML solutions typically limit the number of extracted PICO elements, focus on specific study types such as randomized clinical trials (RCT), or extract from abstracts instead of full texts (Schmidt et al., 2025). Such workarounds can significantly restrict the medical evidence that can be accessed by ML solutions (DeYoung et al., 2020). Moreover, the performance of these ML solutions is also less satisfactory, primarily because they do not adequately incorporate the comprehensive semantic, contextual, and medical knowledge that human experts use in SR (Kiritchenko et al., 2010; Marshall & Wallace, 2019; Schmidt et al., 2020).

In order to address these issues, we propose *FastSR*, a principled and generalizable machine learning framework. Guided by compositionality theory (Lake et al., 2015; Pelletier, 1994), FastSR incorporates design components to address task complexities stemming from the two aforementioned impediments that have inhibited IT support artifacts for SR data extraction: (1) the need to mitigate extensive manual labeling efforts, and (2) the need to account for rich human expertise in sensemaking. More specifically, FastSR employs a few-shot learning (FSL) framework that leverages compositionality theoretic insights into how humans can efficiently learn and understand language from limited examples by systematically and synergistically constructing complex meanings from simpler elements (namely, parts, subparts, and their interrelations). To represent the expertise-intensive nature of

<sup>2</sup> In PICO, P = patient, problem, or population (e.g., “pediatric patients”), I = intervention (e.g., “zinc 150 mg/day for 1 year”), C = comparison (e.g., “gender-matched controls”), and O = outcomes (e.g., “one-year patient survival”).

<sup>3</sup> Existing evidence-based databases, such as the Cochrane Database and Unified Medical Language Systems (UMLS), primarily provide high-level information rather than granular PICO elements that are required by an SR project.

SR, we extend FSL by adding (1) section, sentence, and fragment representations to account for diverse SR knowledge; (2) attention mechanisms to reflect semantic correspondence of PICO text fragments; and (3) shared representations to jointly learn sentence classification and sequence tagging.

We instantiated and evaluated FastSR on three test beds: full-text articles on Wilson disease (WD) and COVID-19, and abstracts from a public dataset (EBM-NLP). We acquired the WD dataset from a company that conducted a 1,500-hour SR project to inform a Phase III clinical trial of a new medicine for Wilson disease. The COVID dataset was obtained from an open research challenge (Wang et al., 2020a), and we assembled a team of researchers with a combined 30 years of medical experience to annotate the dataset (see Appendix B). The EBM-NLP dataset, crowd-labeled by experts, included annotated abstracts from randomized clinical trials (RCTs) focused on cardiovascular diseases, cancer, and autism (Nye et al., 2018).

We compared the performance of FastSR with alternative ML solutions and observed a significant performance gain from both sentence classification and sequence tagging across all three datasets. An ablation analysis further demonstrated the utility of our proposed design components. To assess FastSR's applicability in the real-world SR process, we collaborated with an SR company and integrated FastSR and alternative ML models into their existing manual SR workflow. By comparing the SR reports facilitated by FastSR with those from other ML solutions, we found that FastSR produced results comparable to human-generated reports, whereas other models often had critical errors and omissions due to lower performance. Additionally, we found that FastSR's superior performance reduces the need for manual validation of ML errors, leading to significant time savings and economic benefits. We also interviewed SR experts to validate FastSR's applicability and gained insights into integrating FastSR into the SR protocol for a seamless ML-augmented process.

Our study, which follows the computational design genre (Padmanabhan et al., 2022; Rai, 2017; Samtani et al., 2023; Abbasi et al., 2024), makes several contributions to the literature and practice. At a high level, we contribute to the emerging IS body of literature on design research for artificial intelligence (AI) contexts (Abbasi et al., 2024). Our pathway includes developing predictive ML artifacts that follow a *theory-guided design* abstraction, which involves rich domain adaptation to the SR context, novel and principled integration of ML/deep learning components into an end-to-end architecture, and a sociotechnical design and evaluation process encompassing expert consultations and applicability checks. More specifically, using a compositionality theoretic perspective, we identify four design requirements to satisfy task complexities pertaining to the high-expertise, low-label nature of automated SR. We used these requirements to guide

the design of FastSR and demonstrated its effectiveness in three real-world SR contexts. The design proved superior in terms of data quality and accuracy, time-saving economic benefits, and qualitative indicators of usefulness when compared to fine-tuned deep learning and LLM-based FSL methods. Finally, we propose a new SR process augmented by FastSR, which can save up to 65% in time and \$73,500 per project compared to manual SR. With over 4,000 SR reports published annually (Grimshaw et al., 2013), tools like FastSR could yield substantial economic gains. Overall, given the paucity of IS computational design work on high-expertise, low-label problems, our work has important implications for future design research on human-machine process automation/augmentation in such environments.

## SR Automation: Task Complexity, Design Requirements, and Related Literature

Computational design research in IS focuses on creating ML artifacts situated in an *application environment*, necessitating a profound understanding of task processes and complexities (Li et al., 2020; Samtani et al., 2023). We conducted a comprehensive literature review (e.g., Guyatt et al., 2002; Higgins et al., 2019; Marshall & Wallace, 2019) on the critical steps in SR and their suitability for automation, as in Table 1. We found that the most labor-intensive step in SR is data extraction, yet it is also the least supported by extant IT artifacts. Hence, this paper focuses on automating the data extraction step using ML.

After investigating the SR automation literature and consulting with several SR experts, we discovered that the complexity of data extraction stems from the (1) massive quantity of PICO elements for labeling, (2) need for local and global context of sentences, (3) semantic correspondence of PICO text fragments, and (4) multifaceted knowledge jointly required in an iterative annotation process. Notably, task complexity (1) relates to the importance of mitigating extensive manual labeling efforts, whereas complexities (2)-(4) relate to the need to account for rich human expertise in sensemaking (Li et al., 2020). Theory-guided design is a form of abstraction that can connect gaps/impediments from the application environment to design requirements (Abbasi et al., 2024). We use compositionality theory (Lake et al., 2015; Pelletier, 1994) to guide our artifact design. This theory posits that the meaning of a complex expression is determined by its individual components and the rules for combining them. "Rich concepts can be built 'compositionally' from simpler primitives," such as parts, subparts, and their relations, allowing learning from a few examples (Lake et al., 2015, p. 1333). Sensemaking is the process of searching for a representation and encoding data into that representation in

order to answer task-specific questions (Klein et al., 2006; Russell et al., 1993). SR data extraction involves sensemaking across multiple linguistic components (text fragments, sentences, and paragraphs) and understanding their interrelations—all in a fast-evolving environment (Higgins et

al., 2019; Schmidt et al., 2025; Smith et al., 2015). Based on this discussion, we present four identified task complexities, design requirements, existing consideration sets, and adopted design components in FastSR in Table 2 and use the PICO examples in Table 3 to elaborate on these components.

**Table 1. Steps in SR Process and Automation Potential**

|   | Steps                                                                          | Labor intensity | Supporting it artifact                                                                                                                                                                                                             |
|---|--------------------------------------------------------------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Plan:</b> Defining the research question and deciding on inclusion criteria | Varies by case  | Low. Driven by search strategy and protocol defined by humans.                                                                                                                                                                     |
| 2 | <b>Identification:</b> Searching (e.g., keyword-based search)                  | Low             | Search engines and medical databases (e.g., MEDLINE, PubMed)                                                                                                                                                                       |
| 3 | <b>Screening:</b> filter by titles and abstracts)                              | Moderate        | ML systems for text classification have matured and accomplished a satisfactory performance (e.g., Marshall & Wallace, 2019).                                                                                                      |
| 4 | <b>Data extraction</b> (e.g., PICO data extraction from full text)             | High            | Existing solutions are confined to a limited number of PICO elements and article types, and their performance was less satisfactory (DeYoung et al., 2020; Kiritchenko et al., 2010; Marshall et al., 2016; Schmidt et al., 2025). |
| 5 | <b>Evidence synthesis</b> (e.g., meta-analysis, dashboard)                     | Moderate        | Extensive human assessment and decisions are needed.                                                                                                                                                                               |

**Table 2. Task Complexity, Design Requirements, and Related Literature**

| A. Task complexity for data extraction                             | B. Design requirements                                                            | C. Related literature                                                                                                                                                                   | E. Benchmarking / ablation models                                                                                                                                      | D. FastSR design components                                                                   |
|--------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| 1. Massive amount of PICO elements for labeling                    | A model with minimal labeling effort                                              | <ul style="list-style-type: none"> <li>Large language models (LLMs) <sup>[1,5]</sup></li> <li>Semi-supervised learning (SSL) <sup>[2]</sup></li> <li>FSL <sup>[3]</sup></li> </ul>      | Fine-tuned BioBERT (LM) <sup>[4]</sup> ; GPT models <sup>[5]</sup> , virtual adversarial training (SSL) <sup>[6]</sup> ; FSL baselines <sup>[7]</sup> (see Table 4)    | Adapted ProtoNet with diverse prototypes                                                      |
| 2. Need for local and global context of sentences                  | Capture contextualized word embedding of sentences                                | <ul style="list-style-type: none"> <li>Sentence classification <sup>[8]</sup></li> <li>Contextualized word embedding <sup>[1]</sup></li> </ul>                                          | BERT <sup>[1]</sup> ; SciBERT <sup>[9]</sup> , RNN <sup>[10]</sup> ; RNN + CNN <sup>[8]</sup>                                                                          | Contextualized sentence repr. (BioBERT + CNN)                                                 |
|                                                                    | Consider global context of sentences                                              | <ul style="list-style-type: none"> <li>Document classification <sup>[8]</sup></li> <li>Summarization <sup>[11]</sup></li> <li>Cross-sentence representations <sup>[12]</sup></li> </ul> | Graph-based document summarization <sup>[11]</sup>                                                                                                                     | Section classification using hierarchical embedding and transfer learning                     |
| 3. Semantic correspondence of PICO text fragments                  | Tagging fragments based on semantic correspondence rather than lexical matching.  | <ul style="list-style-type: none"> <li>Sequence tagging <sup>[13]</sup></li> <li>Attention mechanisms <sup>[14]</sup></li> </ul>                                                        | Traditional taggers (BERT <sup>[1]</sup> , BiLSTM <sup>[13]</sup> , CRF <sup>[15]</sup> ); FSL taggers (ProtoNER <sup>[16]</sup> , nearest neighbors <sup>[17]</sup> ) | Adapted FSL tagger; Attention-based fragment repr. (global attention, word-to-word attention) |
| 4. Multifaceted knowledge jointly required in iterative annotation | Jointly identify sentences and fragments via the integration of diverse knowledge | Multitask learning: tree-like, cascading, and modular architectures <sup>[18]</sup>                                                                                                     | Hierarchical architecture <sup>[18]</sup>                                                                                                                              | Integrating semantics, context, and fragment repr.; joint learning in modular architecture    |

**Note:** <sup>[1]</sup> (Devlin et al., 2019); <sup>[2]</sup> (van Engelen & Hoos, 2020); <sup>[3]</sup> (Wang et al., 2020b); <sup>[4]</sup> (Lee et al., 2020); <sup>[5]</sup> (Brown et al., 2020; Luo et al., 2022; Peng et al., 2023); <sup>[6]</sup> (Miyato et al., 2017); <sup>[7]</sup> (Snell et al., 2017); <sup>[8]</sup> (Xiao & Carenini, 2019); <sup>[9]</sup> (Beltagy et al., 2019); <sup>[10]</sup> (Hochreiter & Schmidhuber, 1997); <sup>[11]</sup> (Zheng & Lapata, 2019); <sup>[12]</sup> (Cohan et al., 2018); <sup>[13]</sup> (He et al., 2020); <sup>[14]</sup> (Luong et al., 2015); <sup>[15]</sup> (Lafferty et al., 2001); <sup>[16]</sup> (Huang et al., 2020); <sup>[17]</sup> (Wiseman & Stratos, 2019); <sup>[18]</sup> (Chen et al., 2024b)

**Table 3. PICO Elements and Example Sentences for SR Data Extraction**

| PICO       | Elements                                          | Article section                 | Example sentences                                                                                                                                                                                                             |
|------------|---------------------------------------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P          | Perspective study design                          | Introduction                    | 1. The objective of this [retrospective study] <sup>Perspective</sup> was to [characterize the prevalence] <sup>Study_Design</sup> of HD in residents of US SNFs...                                                           |
|            | Population (eligibility)                          | Methods                         | 2. [Adults with atherosclerotic cardiovascular disease, heterozygous familial hypercholesterolemia, or both] <sup>Population</sup> were eligible to participate...                                                            |
| I          | Intervention dosage, intensity delivery, duration | Methods                         | 3. Patients were randomly assigned 1:1 to [vosaroxin] <sup>Intervention</sup> [(90 mg/m2 intravenously on days 1 and 4 in a first cycle; ...)] <sup>Dosage_Intensity_Delivery_Duration</sup>                                  |
| C          | Comparator                                        | Clinical course under treatment | 4. Wilson's disease when compared with [a control population matched for age and sex] <sup>Comparator</sup> ....                                                                                                              |
| P, I, O    | Population efficacy safety intervention           | Results                         | 5. The objective was to assess the [efficacy] <sup>Efficacy</sup> and [safety] <sup>Safety</sup> of [Zinc therapy] <sup>Intervention</sup> in [pediatric patients with WD] <sup>Population</sup> who were under 14 years old. |
| Irrelevant | NA                                                | Discussion                      | 6. These 21 patients were part of a cohort of 141 WD patients, 30 of whom had pre-symptomatic WD and 90 had neurological WD ... In another study...                                                                           |

### Task Complexity 1: Massive Amount of PICO Elements for Labeling

**Task complexity:** A standard SR project could extract hundreds of PICO elements, translating to hundreds of classes in a supervised ML context. Thus, the volume of labels can be daunting. Moreover, labeling is a highly specialized and demanding task, with SR experts needing to painstakingly analyze every sentence in an article, as PICO elements may be scattered throughout. Furthermore, PICO elements are disease-specific and constantly evolving, preventing the reuse of labels across projects or over time. As such, each new SR project generally starts with labeling from the ground up. Hence, automation solutions must minimize labeling efforts, leading to our first design requirement:

**DR1:** *The proposed ML automation solution for data extraction should require minimal labeling effort.*

**Related literature:** Existing supervised ML models with minimal labeling effort include large language models (LLMs), semi-supervised learning, and few-shot learning. LLMs include masked language models (MLMs) like BERT (Devlin et al., 2019), which use Transformer encoders (Vaswani et al., 2017) to predict masked tokens with bidirectional self-attention, and autoregressive language models (ALMs) like GPT (Radford et al., 2018), which use Transformer decoders for text generation with unidirectional self-attention. The former excel in extractive tasks (Chen et al., 2024a), while the latter are ideal for generative tasks (Brown et al., 2020). Both models can be pretrained or fine-tuned with domain-specific data to adapt to particular tasks.

For instance, BioBERT (Lee et al., 2020) and SciBERT (Beltagy et al., 2019) are pretrained on biomedical texts and can be fine-tuned for BioNLP tasks. BioGPT (Luo et al., 2022) and GatorTronGPT (Peng et al., 2023) are pretrained on medical or clinical texts and can be fine-tuned for medical content generation. Fine-tuning large models like GPT-4 (Brown et al., 2020) typically requires extensive, high-quality labeled data. However, SR tasks often have limited labels per class, making fine-tuning less effective (Schmidt et al., 2025). ALMs can also perform tasks without further training through few-shot prompting, but they are highly sensitive to prompt wording, example selection, and domain specificity, making them more challenging for precise tasks (Liu et al., 2023). Additionally, ALMs are more susceptible to data contamination than MLMs—where test data overlaps with pretraining data, inflating performance (Dong et al., 2024). This is because ALMs, with their sequential generation process, are more likely to reuse memorized content, especially in open-domain datasets. To illustrate these limitations, we benchmarked FastSR against fine-tuned BioBERT, fine-tuned and few-shot prompted GPT across both open-domain and closed-domain datasets.

Semi-supervised learning (SSL) leverages unlabeled data to enhance the performance of models trained on a smaller labeled dataset, operating on the principle that similar data points are likely to share the same labels (Oliver et al., 2018; van Engelen & Hoos, 2020; Yang et al., 2022). Among various SSL approaches, such as graph-based, wrapper-based, and deep generative methods, consistency regularization methods, which enforce the model to produce consistent predictions under small perturbations, are more practical for the SR context due to their

applicability to limited labeled data and less stringent requirement for data structure. Virtual adversarial training (VAT) (Miyato et al., 2017) emerges as a strong method in this category. However, VAT may struggle when the distributions of labeled and unlabeled are misaligned, a common scenario in the SR context given the diverse distributions across PICO classes (Oliver et al., 2018). To illustrate this limitation, we included VAT as a benchmarking model.

Different from LM and SSL, FSL addresses label scarcity by making more efficient use of labeled samples without requiring a large labeled dataset to be effective. FSL is motivated by the aforementioned compositionality theory in cognitive science, which documents that humans use compositional skills, i.e., combining and contrasting a limited number of samples, to learn in principled and generalizable ways (Lake et al., 2015). FSL accomplishes label efficiency by generating multiple classification tasks comprised of support sets and queries (test instances). These support sets contain small, representative labeled samples from each class, often randomly chosen to ensure representation. In each specific task, FSL classifies queries based on a few labeled samples in each support set (Wang et al., 2020b).

FSL models can be divided into optimization-based and metric-based learning (Yu et al., 2018), each with its own models and strategies, as described in Table 4. The basic idea of optimization-based FSL is to “build an internal representation that is broadly suitable for many tasks” (Finn et al., 2017, p. 1126), allowing new tasks to be trained with just a few gradient updates. However, in the SR context, the classification tasks are highly diverse, each targeting a unique set of PICO elements. For instance, “outcome” and “participant” share less than 1% of tag words. The diversity diminishes the advantage of using a shared initialization. Metric-based models, which construct distinct class embeddings and predict instances by their distances to the embeddings, are better suited to these diverse tasks. Building on *ProtoNet* (Snell et al., 2017) and *matching net* (Vinyals et al., 2016), various solutions have been proposed to enhance support set construction (Rios & Kavuluru, 2018b; Xu et al., 2019), embedding networks (Bao et al., 2019; Rios & Kavuluru, 2018b), and prediction functions (Bao et al., 2019; Xu et al., 2019).

Moreover, we chose ProtoNet over matching net due to its computational efficiency, superior performance, and better context alignment. ProtoNet classifies a test instance based on the shortest Euclidean distance to class prototypes, which are the average representation of the support instances (Snell et al., 2017). In contrast, matching net classifies based on a weighted sum of support set labels, with weights determined by cosine similarity between the test instance and each support instance (Vinyals et al., 2016). ProtoNet is more computationally efficient, avoiding the pairwise distance calculations required

by matching net (Wang et al., 2020b). ProtoNet’s prototypes provide maximum likelihood estimates of the posterior class distribution (Banerjee et al., 2005), enabling simultaneous prototype calibration and distribution estimation for improved performance (Dopierre et al., 2021; Snell et al., 2017). Prototypes can better represent our multifaceted knowledge, unlike matching net, which spreads knowledge across individual instances. We included FSL benchmarking in Table 4 for justification.

## **Task Complexity 2: Need for the Local and Global Context of Sentences**

**Task complexity:** When identifying PICO-related sentences, SR experts need to examine the local and global context of a sentence (Kiritchenko et al., 2010; Marshall & Wallace, 2019; Schmidt et al., 2020). Local context involves word-level or sentence-level meanings, such as understanding that “prevalence” in Sentence 1 from Table 3 indicates a cross-sectional study design based on nearby terms like “retrospective study.” Global context concerns the sentence’s location within the article and can be obtained by examining the section heading (e.g., Method), encompassing paragraphs, or adjacent sentences (e.g., “In another study...”). Global context is instrumental in discerning sentence relevance. For example, both Sentences 2 and 5 in Table 3 describe patient groups for the Population element; however, Sentence 5 is omitted because the study is a benchmarking study in the results section. Hence, we propose:

**DR2:** *The proposed ML automation solution for data extraction should consider both the local and global context of sentences, in that a sentence’s relevance to PICO elements is determined by its contextualized meanings and encompassing section.*

**Related literature:** Popular classifiers that can capture the local context of a sentence include support vector machines (SVM), naïve Bayes, recurrent neural network (RNN) (Hochreiter & Schmidhuber, 1997), convolutional neural network (CNN), and Transformer-based classifiers (Schmidt et al., 2025). RNNs, which capture word-level context by processing sequences of words, often struggle with long-range dependencies (Cohan et al., 2018). In contrast, BERT’s bidirectional self-attention mechanism allows it to capture contextual relationships across the entire sentence, providing a richer representation of each word (Devlin et al., 2019). Specifically, BioBERT, which fine-tunes BERT on biomedical texts, can effectively grasp biomedical terminologies and contextual nuances (Beltagy et al., 2019). For example, in Sentence 1 from Table 3, BioBERT can link the word “prevalence” to cross-sectional study design. Thus, we prefer BioBERT over RNN for contextualized word embeddings.

**Table 4. Survey of FSL Models Used in NLP**

| Model                                                                               | Description                                                                                                                                                                                                                                                        | Used as benchmark? |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>Sentence classification: Metric-based learning (learn to embed)</b>              |                                                                                                                                                                                                                                                                    |                    |
| Matching bet (Vinyals et al., 2016)                                                 | Learns embedding functions for support sets and queries. A query is classified based on its average cosine distance to the support set instances.                                                                                                                  | Y                  |
| Prototypical network (Snell et al., 2017)                                           | Similar to matching net, except that the query is classified by its Euclidean distances to the prototypes (the mean of a support set representation).                                                                                                              | Y                  |
| FSL with distributional signatures (Bao et al., 2019)                               | Generates word attention per class by TF-IDF weights of support sets and then creates attention-weighted sample representations. A ridge regressor is trained to classify queries.                                                                                 | Y                  |
| Robust task clustering FSL (Yu et al., 2018)                                        | Evaluates task similarity and cluster tasks. Then, a matching net model is trained for each cluster.                                                                                                                                                               | N <sup>[1]</sup>   |
| Learning to accept class (L2AC) (Xu et al., 2019)                                   | Selects the most similar support set instances in each class and use them to train a classifier to predict queries. With a threshold, the model can reject instances from unseen classes.                                                                          | Y                  |
| EMR multihead matching networks (Rios & Kavuluru, 2018a)                            | Extends matching net for multilabel classification. It first searches for relevant support sets for a query using different representations, then constructs auxiliary features from the support sets, and finally trains a classifier to predict multiple labels. | N <sup>[2]</sup>   |
| Few-shot attentive GCNN (Rios & Kavuluru, 2018b)                                    | This model derives label representations using a graph convolution network (GCN). Then, it uses label representations to attend to relevant n-grams in the document.                                                                                               | N <sup>[3]</sup>   |
| <b>Sentence classification: Optimization-based learning (learning to fine-tune)</b> |                                                                                                                                                                                                                                                                    |                    |
| Model-agnostic meta-learning (MAML) (Finn et al., 2017)                             | Trains a generic model on multiple tasks so that it can be fine-tuned to new tasks with only a small number of training samples through a few adaptation steps.                                                                                                    | Y                  |
| FSL across diverse NLP tasks (LEOPARD) (Bansal et al., 2020a) <sup>[4]</sup>        | This model extends MAML with (1) a shared input encoder based on BERT, (2) task-specific and task-agnostic parameters separated, and (3) the number of classes in the final prediction layer can vary by a new task.                                               | N <sup>[4]</sup>   |
| Self-supervised FSL tasks (SMLMT) (Bansal et al., 2020b) <sup>[4]</sup>             | Similar to MAML, this model tries to solve the data inefficiency issue. It proposes a self-supervised approach, e.g., masked words, to generate learning tasks from the unlabeled text for training the MAML model.                                                | N <sup>[4]</sup>   |
| <b>Sequence tagging: Metric-based learning</b>                                      |                                                                                                                                                                                                                                                                    |                    |
| Prototypical networks for NER (ProtoNER) (Huang et al., 2020)                       | This model applies prototypical networks to NER, where the label of each word in a sentence is determined by its distances to entity prototypes.                                                                                                                   | Y                  |
| Nearest neighbors (Wiseman & Stratos, 2019)                                         | This model calculates the distance between a word and each labeled token in support sets to find the nearest neighbor and then copies the neighbor's label as the output.                                                                                          | Y                  |

**Note:** <sup>[1]</sup> It applies to homogeneous tasks in diverse domains, while our tasks are diverse. <sup>[2]</sup> It performed badly in our highly overlapping multilabel dataset. <sup>[3]</sup> It requires a hierarchical label structure, such as diagnosis codes. <sup>[4]</sup> Extensions proposed in these models are irrelevant to our task.

The word-level embeddings can be mapped into sentence-level representations using either CNN or RNN layers. We opted for CNNs over RNNs because CNNs are particularly effective at capturing key phrases, such as “characterize the prevalence,” by applying filters that detect local patterns in the text (Kim, 2014). Various CNN architectures, including word-level (Kim, 2014) and character-level CNNs (Zhang et al., 2015), have been

developed. Given our small dataset, a word-based CNN with minimal parameters is more appropriate than a character-level CNN, which requires larger datasets to learn language patterns effectively (Zhang et al., 2015). Thus, we used BioBERT with CNN layers to capture the local context of sentences while benchmarking with other alternatives, as in Table 2.

Global context-related NLP methods include document-level, cross-sentence, or hierarchical embedding. Document-level embeddings are typically generated by summarization models (Xiao & Carenini, 2019). Unsupervised graph-based approaches, which create summaries by selecting sentences with high semantic centrality without relying on large-scale labeled data, are better suited for the SR context (Zheng & Lapata, 2019). However, they often overemphasize dominant classes and may overlook the full diversity of information, a critical issue in SR contexts with class imbalance. Cross-sentence embedding (Xiao & Carenini, 2019) uses a BiLSTM to encode sections but may suffer from memory loss in encoding longer sections (Cohan et al., 2018). Hierarchical embedding (Cohan et al., 2018) mitigates this by using two BiLSTM layers: The first layer embeds individual sentences, and the second encodes the sequence of the sentence embeddings, offering a more robust context representation over longer texts.

Considering these factors, we selected hierarchical embedding as the most suitable method. However, our limited labeled data precludes directly training a hierarchical model. We employed a transfer learning approach (Wang et al., 2020b) for global context representation, comparing it with an unsupervised graph-based summarization model to ensure robustness.

### **Task Complexity 3: Semantic Correspondence of Text Fragments**

**Task complexity:** When tagging fragments in a sentence, from a compositionality theoretic perspective, SR experts often engage in *semantic sensemaking* to assess whether the *core concepts* of a text fragment relate to PICO elements (Nye et al., 2018). In the SR context, human sensemaking pertains to finding text fragments (data) that are semantically related to a PICO element (representation), via abstraction and disambiguation, to answer questions about whether a sentence contains a PICO element. In essence, the semantic correspondence of a text fragment to a PICO element is more important than exact lexical matching. For instance, in Sentence 5 from Table 3, the core concept that relates to a P (population) element is “pediatric patients,” and other fragments that entail this concept, such as “patients ... under 14 years old” or “pediatric patients ... under 14 years old,” should be deemed acceptable text fragments. Hence:

**DR3:** *The proposed ML automation solution for data extraction should incorporate ML models capable of mimicking human semantic sensemaking of PICO elements, in that when identifying PICO text fragments, semantic correspondence is prioritized over lexical matching.*

**Related literature:** Many ML solutions approach PICO fragment tagging as a sequence tagging problem, aiming to assign each word an IOB label (i.e., inside, outside, and beginning of a PICO element) in a sequence (Schmidt et al., 2025). This method focuses on precisely delineating text segments that match predefined labels, relying on exact lexical matches to identify PICO elements in sentences. Widely used models in this domain include conditional random fields (CRF) (Lafferty et al., 2001), RNN, and various configurations of Transformer-based models (He et al., 2020; Marshall & Wallace, 2019; Schmidt et al., 2025). Within the FSL framework, the ProtoNER model, derived from the prototypical network, labels each word in a sentence based on its proximity to entity prototypes (Huang et al., 2020). Additionally, the nearest neighbor (NN) model, an adaptation of matching net, determines the label of a word by its distance to each labeled token in support sets (Wiseman & Stratos, 2019).

However, in the context of SR, human sensemaking extends beyond tagging exact matches, involving abstraction and disambiguation to identify semantically relevant text fragments that convey PICO elements, even if they differ lexically. For instance, in Sentence 5 from Table 3, “pediatric patients with WD” might be tagged as a Population (P) element, while “patients with WD who were under 14 years old,” which conveys the same concept, would perform poorly under a rigid IOB tagging system. Consequently, existing models that struggle to abstract core concepts from labeled PICO fragments often result in poor tagging performance in SR contexts (Kiritchenko et al., 2010; Schmidt et al., 2025).

To address these issues, we implemented a global attention mechanism (Luong et al., 2015) to move beyond the limitations of rigid IOB tagging. This mechanism enables the model to focus on semantically relevant words, aligning them with core PICO concepts rather than relying solely on exact matches. We also introduced BERTScore (Zhang et al., 2020) as a supplementary evaluation metric to better capture semantic alignment between fragments. For instance, in the earlier example, “pediatric” could align with “patient” and “under 14 year old” with a BERTScore above 80%, which better reflects expert judgments. To demonstrate the effectiveness of our solution, we benchmarked FastSR against ProtoNER and nearest neighbor.

### **Task Complexity 4: Multifaceted Knowledge Jointly Required in an Iterative Process**

**Task complexity:** During data extraction, SR experts utilize semantic, contextual, and medical knowledge in an iterative annotation process. They typically identify a section (e.g., Methods) as a global context, search for sentences whose local

contexts may suggest PICO elements (e.g., population), and interpret text fragments within these sentences using semantic correspondence. A sentence is identified as element-bearing only if it contains PICO text fragments and the correctness of the fragments can only be justified by their local and global context. By performing sentence and fragment identification simultaneously, SR experts gradually adjust the annotated text fragments to achieve the desired granularity and completeness for data extraction (Higgins et al., 2019). Therefore, it is essential to permit this synchronous and interrelated knowledge to be shared at each stage of the extraction process. Hence, we propose:

**DR4:** *The proposed ML automation solution for data extraction should jointly identify sentences and text fragments by sharing semantic, contextual, and medical knowledge within an FSL framework.*

**Related literature:** Existing SR ML solutions, on the other hand, divide data extraction into two tasks: identifying PICO element-bearing sentences (T1) and then extracting elements from identified sentences (T2) (Kiritchenko et al., 2010; Schmidt et al., 2020). Errors in T1 can severely compromise T2 performance. If sentences are extracted incorrectly, all subsequent text fragments will be incorrect. Likewise, if any sentences are missed, the corresponding fragments will also be absent, leading to incomplete results. This separation prevents the cross-utilization of insights and leads to error propagation across tasks (Chen et al., 2024b).

Multitask learning (MTL), also known as joint learning when tasks are equally prioritized, is a promising approach to managing the intercorrelation between SR tasks (Chen et al., 2024b). Its success hinges on the careful design of task relatedness, model architecture, and optimization strategies. In SR, sentence classification (T1) and sequence tagging (T2) are related but vary in granularity—T1 addresses broader sentence-level features, while T2 involves detailed word-level analysis. Misalignment between these tasks can lead to one task dominating or negatively influencing the other, resulting in suboptimal performance. Modular architectures (Chen et al., 2024b), where shared modules capture common features and task-specific modules address individual needs, can help reduce competition between tasks. Hierarchical architectures (Song et al., 2020), which process tasks sequentially, align with common workflows but are more prone to error propagation. Optimization strategies, including regularization techniques and dynamic task-specific loss weighting, are crucial in balancing shared learning and task-specific nuances for enhanced model performance (Li & Caragea, 2019).

We adopted a modular architecture in the FSL framework, where a shared prototype for core fragment concepts was utilized across both tasks, alongside additional prototypes

tailored to the specific needs of each task. The shared module was trained on data from both tasks to promote alignment and reduce task competition. Additionally, we implemented a regularization mechanism to enforce cross-task consistency—ensuring that if a word is identified as a PICO element in T2, then the corresponding sentence will be recognized as containing a PICO element in T1. For validation, we included a hierarchical design as a benchmark.

## FastSR: A Customized Few-Shot Deep Learning Framework

### Problem Definition and Overall Architecture

Since related PICO elements (e.g., intervention and dosage) are often collocated within one sentence, SR experts typically group multiple PICO elements into an element class and identify class-specific PICO elements within a sentence together (Kiritchenko et al., 2010). Formally, let  $C = (c_1, c_2, \dots, c_J)$  denote a set of  $J$  predefined PICO element classes. For an article  $A = (q_1, q_2, \dots, q_L)$  comprised of  $L$  sentences, each sentence  $q \in A$  is represented as  $q = (WV, FR)$ , where  $WV = (w_1, w_2, \dots, w_T)$  is the  $D$ -dimensional embeddings of the total  $T$  words in  $q$ , and  $FR = (F^1, F^2, \dots, F^J)$  contains the fragment for each class. Specifically, for class  $c_j$ , the corresponding fragment  $F^j$  is the elementwise product (denoted as  $\odot$ ) of  $WV$  and the one-hot encoded label vector ( $L_{one-hot}$ ), i.e.,  $F^j = WV \odot L_{one-hot} = (f_1^j, f_2^j, \dots, f_T^j)$ , where  $f_i^j = w_i$  if the  $i^{\text{th}}$  word is a part of the PICO fragment and  $\mathbb{0}$  (zero vector) otherwise. For example, for the sentence “the objective of this [retrospective study] perspective was to ...” (see Table 3), the one-hot encoded label vector is  $(0, 0, 0, 0, 1, 1, 0, 0, \dots)$  and the fragment is represented as  $(\mathbb{0}, \mathbb{0}, \mathbb{0}, \mathbb{0}, w_5, w_6, \dots, \mathbb{0}, \mathbb{0})$ . For  $1 \leq j \leq J, 1 \leq l \leq L, 1 \leq t \leq T$ , SR data extraction is divided into three steps:

1. Train a model to generate the probability  $ps_l^j = p(c_j|q_l)$  that sentence  $q_l$  is relevant to  $c_j$ .
2. For each sentence, obtain the probability  $pw_t^j = p(c_j|w_t)$  that word  $w_t$  belongs to class  $c_j$ .
3. For each class  $c_j$ , retrieve top- $N$  sentences with the largest  $ps_l^j$  in article  $A$ , and for each retrieved sentence, extract words with  $pw_t^j$  greater than a threshold that makes text fragments relevant to PICO elements.

As shown in Figure 1, we propose a novel FSL framework, FastSR, with several novel extensions to meet the design requirements in the previous section. Table 5 lists all notations. The SR data extraction is a multilabel classification

problem. For example, Sentence 5 in Table 3 of our paper belongs to multiple classes, such as Intervention and Population. However, metric-based FSL typically assigns a unique class label to a query based on the shortest distance to a class-specific support set, and it cannot natively support multilabel classification (Cheng et al., 2019). Hence, as suggested by Zhang et al. (2018), we used a one-versus-rest approach to convert this multilabel classification into multiple two-way,  $k$ -shot FSL models. For each task that predicts whether a query belongs to a class, the positive support set is comprised of  $K$  sentences drawn from class  $c_j$ , and the negative set contains  $K$  sentences drawn from the remaining classes  $-c_j$ .

As shown in Figure 1, the overall architecture (Figure 1e) follows a ProtoNet structure (Snell et al., 2017) for learning with minimal labeling effort (DR1). In FastSR, a query is a sentence from a medical article. By comparing a query sentence  $q$  with prototypes in support sets, FastSR tries to determine whether a query sentence  $q$  relates to PICO elements (T1: Sentence classification) and which of its words are part of PICO text fragments (T2: Sequence tagging). While traditional ProtoNet often includes only one prototype per class (e.g., word semantics), FastSR includes multiple prototypes based on DR2 and DR3. They include sentence contextualized semantics representation and global context representation (Figures 1a and 1b; DR2), as well as PICO text fragment representations (Figure 1c; DR3).

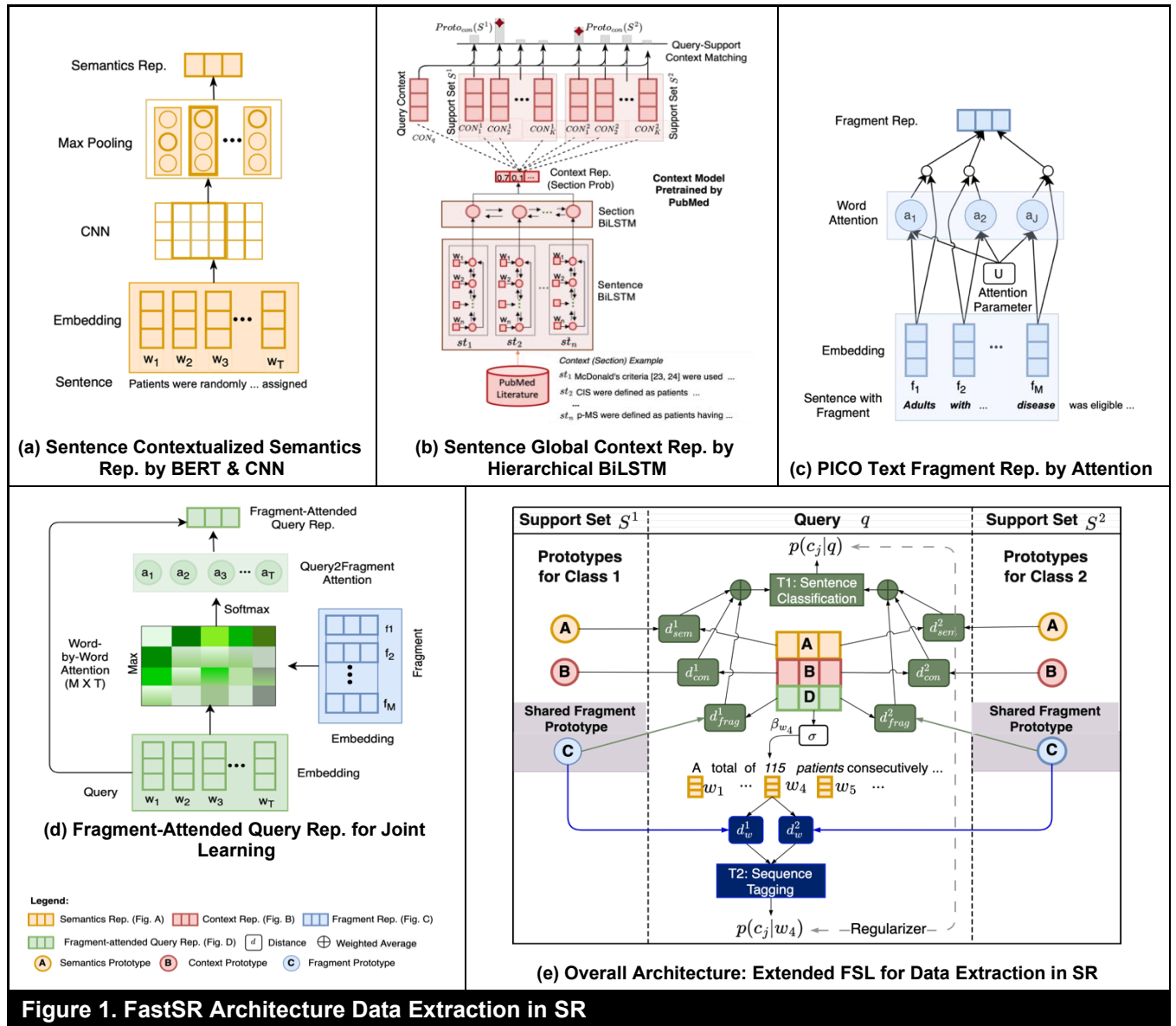


Figure 1. FastSR Architecture Data Extraction in SR

**Table 5. General Notations**

| Variable                                    | Description                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $C$                                         | $C$ is a set of $J$ predefined PICO element classes, i.e., $C = \{c_1, c_2, \dots, c_J\}$ .                                                                                                                                                                                                                                           |
| $WV, w_t, D$                                | $WV = (w_1, w_2, \dots, w_T)$ holds the embeddings of the total $T$ words in a sentence. $w_t$ is the $D$ -dimensional embedding of the $t^{\text{th}}$ word, $w_t \in \mathbb{R}^D$ , the $D$ -dimensional vector space.                                                                                                             |
| $F^j, f_m$                                  | The fragment in a labeled sentence for class $c_j$ is denoted as $F^j = (f_1^j, f_2^j, \dots, f_T^j)$ , where $f_i^j = w_i$ if the $i^{\text{th}}$ word is a part of the fragment and $\mathbb{0}$ (zero vector) otherwise. $f_m$ denotes the $m^{\text{th}}$ non-zero embedding in $F^j$ . The superscript is reserved for class id. |
| $q, q_l$                                    | A sentence $q = (WV, FR)$ , where $FR = (F^1, F^2, \dots, F^J)$ contains a target text fragment $F^j$ for each class $c_j$ . $q_l$ is the $l^{\text{th}}$ sentence in an article.                                                                                                                                                     |
| $S, S^j$                                    | $S$ denotes a support set for a class, and $S^j$ is a support set for class $c_j$ .                                                                                                                                                                                                                                                   |
| $ps_l^j, pw_t^j$                            | $ps_l^j = p(c_j q_l)$ is the probability that sentence $q_l$ is relevant to $c_j$ . $pw_t^j = p(c_j w_t)$ is the probability that word $w_t$ in $q_l$ belongs to class $c_j$ .                                                                                                                                                        |
| $a, att, ATT$                               | $a$ denotes an alignment score between two components. It is normalized across a set of components to obtain an attention score $att$ . The set of attention scores is denoted as $ATT$ .                                                                                                                                             |
| $\beta_t$                                   | $\beta_t$ is the probability of word $w_t$ being emitted for tagging. It is calculated based on the alignment of $w_t$ with fragment words.                                                                                                                                                                                           |
| $SEM_i$                                     | Contextualized semantics representation of sentence $i$ (either a support set or query sentence) obtained by the BERT + CNN module (Figure 1a). This representation captures the local contexts of sentences.                                                                                                                         |
| $CON_i$                                     | Global context representation of sentence $i$ (either support set or query sentence) obtained by the pretrained context presentation module (Figure 1b).                                                                                                                                                                              |
| $FRG_i$                                     | Representation of fragment $F_i$ in support set sentence $i$ obtained by the attention-based fragment presentation module (Figure 1c).                                                                                                                                                                                                |
| $FAQ_q$                                     | Representation of query sentence $q$ obtained by the fragment-attended query representation module (Figure 1d).                                                                                                                                                                                                                       |
| $Proto_{sem}(S)$                            | Contextualized semantics prototype of support set $S$ (Prototype A in Figure 1e).                                                                                                                                                                                                                                                     |
| $Proto_{con}(S)$                            | The global context prototype of support set $S$ (Prototype B in Figure 1e).                                                                                                                                                                                                                                                           |
| $Proto_{frag}(S)$                           | Fragment prototype of support set $S$ (Prototype C in Figure 1e).                                                                                                                                                                                                                                                                     |
| $d(q, S), d(w_t, S)$                        | $d(q, S)$ is the total distance between $q$ and all the prototypes of $S$ , and $d(w_t, S)$ is the distance between $w_t$ and the fragment prototype of $S$ .                                                                                                                                                                         |
| $\alpha_{sem}, \alpha_{con}, \alpha_{frag}$ | Weights assigned to semantics, context, and fragment, $\alpha_{sem} + \alpha_{con} + \alpha_{frag} = 1$ .                                                                                                                                                                                                                             |
| $LS(q, j), LT(q, j), LR(q, j), Loss(q, j)$  | For sentence $q$ and class $c_j$ , $LS(q, j)$ and $LT(q, j)$ are the cross-entropy losses of sentence classification and sequence tagging, respectively. $LR(q, j)$ is the loss regularizing the prediction consistency between the two tasks. $Loss(q, j)$ is the sum of these three losses.                                         |
| $U_F, U_Q, V_Q, V_G, D'$                    | $U_F, U_Q, V_Q$ , and $V_G$ are tunable parameters associated with the dense layers in the model. $D'$ is the output dimension of the layers corresponding to $U_Q, V_Q$ , and $V_G$ .                                                                                                                                                |

Finally, both tasks are jointly learned in FastSR through shared fragment prototypes (Figure 1e: DR4). For sentence classification, the prototypes are paired with a fragment-attended query (FAQ) representation (Figure 1d) to assess the extent to which the query sentence contains core fragment concepts. For sequence tagging, the prototypes are compared with each word embedding to identify words relevant to PICO elements.

### Construction of Sentence and Fragment Representations

According to our design requirements, diverse knowledge needs to be integrated into data extraction. Correctly representing this knowledge in an FSL architecture is especially critical as FSL's prediction capability hinges on a precise characterization between queries and support sets. In

this section, we introduce three representations corresponding to DR2 and DR3 and discuss how they are integrated into ProtoNet as prototypes.

### Sentence Contextualized Semantic Representations and Prototypes

This component strives to represent the local contexts of sentences via deep learning (DL) (Figure 1a; DR2). The first layer is *contextualized word embedding*, which transforms each word in a sentence into a  $D$ -dimensional vector of numbers to best represent the semantic similarity between words. The next layer concerns DL architecture, and the selection is guided by the learning paradigm, data encoding, and task characteristics (Samtani et al., 2023). As discussed in the Task Complexity 2 section, we chose the CNN design by Kim (2014) to represent the semantics of sentence  $q$  as  $SEM_q$ . We then computed a semantics prototype  $Proto_{sem}$  for a support set  $S$  by averaging the semantics representations of sentences in  $S$  (Equation 1):

$$Proto_{sem}(S) = \frac{1}{|S|} \sum_{i \in S} SEM_i \quad (1)$$

### Sentence Global Context Representations and Prototypes

This component aims to represent the global context of sentences via transfer learning (Figure 1b; DR2) and determine the extent to which a query sentence is located in a typical section for a PICO element class, defined as a global context prototype. As discussed in the Task Complexity 2 section, global context concerns the sections where a target sentence is located. However, obtaining the section information is nontrivial. A rule- or dictionary-based method is not viable because medical articles have diverse structures, varying by journals and medical domains. Certain types of articles, such as synopsis, do not always include section titles. And even when section titles are provided, they can be misaligned with sentences or lost during the PDF parsing process.

We developed a *section classifier* (Figure 1b) to classify paragraphs and sentences into several predefined section classes. Since our training dataset was relatively small, we took a *transfer learning* approach by training hierarchical embedding (Cohan et al., 2018) of medical articles using PubMed, a large corpus of over 130,000 research articles. In the first layer, sentences of a section from the PubMed corpus were processed in parallel through a shared BiLSTM unit to create multiple sentence representations. These presentations were then fed into another BiLSTM unit to generate a section representation. Finally, this representation entered a dense

layer for section title prediction. The section classifier had over 90% accuracy (see Appendix C). We transferred the pretrained section classifier to our datasets to learn a global context for each sentence  $i$ , denoted as  $CON_i$ . Specifically, we applied the classifier to the section of the sentence and obtained a probability vector of 13 section classes as  $CON_i$ ,  $CON_i \in \mathbb{R}^{13}$ .

We found the best-matching global context for a query sentence  $q$  in the support set  $S$  to obtain the context prototype, denoted as  $Proto_{con}(S)$  as shown in Figure 1b. This prototype was computed using a “pointer network” (Vinyals et al., 2015) rather than the mean representation of  $S$  (as in Equation 1). Since sentences for one PICO element (e.g., population) may appear in multiple sections (e.g., Introduction and Results) of an article, averaging the context representations of the sentences in  $S$  could be less meaningful. The pointer network, on the other hand, captures whether a support set  $S$  has at least one sentence that appears in a similar section of  $q$ . If so, the confidence in classifying  $q$  into the class label of  $S$  is high. Specifically, we used the dot product to measure the alignment between the context representations of  $q$  and sentences in  $S$  (Equation 2). The context prototype for  $S$  was then computed by the context representation  $CON_{i^*}$  for a sentence  $i^*$  that had the highest alignment with  $q$  (Equation 3).

$$a(CON_i, CON_q) = (CON_i)^T CON_q, \text{ where } i \in S \quad (2)$$

$$Proto_{con}(S) = CON_{i^*}, \text{ where } i^* = \underset{i \in S}{\operatorname{argmax}} a(CON_i, CON_q) \quad (3)$$

### PICO Text Fragment Representations and Prototypes

This component enables semantic correspondence of text fragments via attention-based fragment representations (Figure 1c; DR3). We used global attention (Luong et al., 2015) to highlight the core concepts of PICO fragments and incorporate them into ProtoNet.

Given a support set  $S$  and a sentence  $i$  in  $S$  with  $M$  tagged words for class  $c_j$ , let  $f_m$  be the  $D$ -dimensional embedding of one of the words, i.e.,  $f_m \in F^j$ ,  $1 \leq m \leq M$ . We calculated the dot product of  $f_m$  and  $U_F$ , a tunable parameter representing core concepts in  $F^j$  (Equation 4). Then, we normalized the results across all words by the Softmax function to get an attention score for each word (Equation 5). Words aligned with the core concepts of  $U_F$  were allocated with high attention scores. Finally, the fragment representation was calculated as the attention-weighted sum of the word embeddings (Equation 6). The fragment prototype for a support set  $S$  is the mean representation of the fragments in  $S$  (Equation 7).

$$a_m = (U_F)^T f_m, \text{ where } U_F \text{ is a parameter to be tuned, } U_F, f_m \in \mathbb{R}^D, 1 \leq m \leq M \quad (4)$$

$$\text{Let } ATT = \{att_1, att_2, \dots, att_M\}, att_m = \frac{\exp(a_m)}{\sum_{k=1}^M \exp(a_k)} \quad (5)$$

$$FRG_i = \sum_{m=1}^M att_m f_m \quad (6)$$

$$Proto_{frag}(S) = \frac{1}{|S|} \sum_{i \in S} FRG_i \quad (7)$$

## Integration of Representations and Joint Learning

According to DR4, sentence classification and sequence tagging need to be jointly learned to integrate diverse knowledge and accomplish optimal extraction performance. As discussed in the Task Complexity 4 section, we used a modular architecture and a cross-task regularizer for this joint learning. In this section, we first introduce a special query representation that allowed the fragment prototype  $Proto_{frag}(S)$  to be shared by both tasks and then joint learning.

### Fragment-Attended Query Representations for Joint Learning

This component shares fragment prototypes (used by sequence tagging) with sentence classification, thus facilitating joint learning (Figure 1d; DR4). Conceptually, the confidence in classifying a query sentence  $q$  into the class  $c_j$  will be enhanced by the fact that  $q$  contains some words that are aligned with the labeled fragments in a support set  $S^j$  for class  $c_j$ . In this sense, fragment tagging will inform sentence classification. To assess the extent to which  $q$  is aligned with the core concepts of labeled PICO elements in a support set, we adapted a word-to-word attention mechanism (Seo et al., 2017) to assess the alignment between each word of  $q$  and the words of a labeled fragment, generated a fragment-attended representation for  $q$ , and then measured this representation's distance to PICO text fragment prototypes.

Given a query  $q$  with word embeddings  $WV = (w_1, w_2, \dots, w_T)$ , and a support set sentence  $i$  in  $S$  with  $M$  tagged words for class  $c_j$ , let  $f_m$  be the  $D$ -dimensional embedding of one of the words, i.e.,  $f_m \in F^j$ ,  $1 \leq m \leq M$ . For each pair of  $w_t$  and  $f_m$ , we calculated their matching score  $a_{m,t}$  by concatenating  $f_m, w_t$ , and their elementwise product,  $f_m \odot w_t$ , and transforming the concatenated vector through a dense layer with trainable parameter  $U_Q$  into a  $D'$ -dimensional vector. This elementwise product emphasizes the similarity between  $w_t$  and  $f_m$ . For each word  $w_t$  in  $q$ ,  $M$  score vectors can be obtained, and the highest values ( $a_t$ ) over the fragment words indicate the

alignment between  $w_t$  and  $F^j$  (Equation 8).  $a_t$  absorbs granular information of PICO elements, rather than just general concepts, observed in the words of labeled fragments. For instance, in addition to general dosage concepts, labeled fragments may contain linguistic cues for dosage schedules. Such cues can help our model tag granular PICO elements. Thus, we transformed  $a_t$  into  $\beta_t$ , a probability of word  $w_t$  being emitted during sequence tagging, through a dense layer with a Sigmoid function and a tunable  $D'$ -dimensional parameter  $V_G$  (Equation 9). Meanwhile,  $s_t$  was transformed into a normalized attention score through a linear layer with tunable parameter  $V_Q$  and Softmax (Equation 10). Finally,  $q$  was represented as the attention-weighted sum of its word vectors, denoted as  $FAQ_q$  (Equation 11). If a support set had multiple fragments (i.e.,  $k$ -shot),  $FAQ_q$  calculated the mean of the fragment-attended representations.

$$a_t = \max_{1 \leq m \leq M} U_Q [f_m, w_t, f_m \odot w_t], \text{ where } U_Q \in \mathbb{R}^{D' \times 3 \times D} \text{ and } f_m, w_t \in \mathbb{R}^D \quad (8)$$

$$\beta_t = \text{sigmoid}((V_G)^T a_t), \text{ where } V_G \in \mathbb{R}^{D'} \quad (9)$$

$$att_t = \frac{\exp(a'_t)}{\sum_{k=1}^T \exp(a'_k)}, \text{ where } a'_t = (V_Q)^T a_t, \text{ and } V_Q \in \mathbb{R}^{D'} \quad (10)$$

$$FAQ_q = \sum_{t=1}^T att_t w_t \quad (11)$$

### Joint Learning via Representation Integration

In a prototypical network, prediction is accomplished by comparing the distances between the representation of a query  $q$  and various prototypes of the support set  $S$ . The predicted class label is determined by the class label from the nearest prototype. In FastSR, multifaceted knowledge of element extraction is captured by the proposed representations and prototypes described in Equations (1)-(11). The two tasks—sentence classification and sequence tagging—are learned jointly via the shared fragment prototypes.

First, sentence classification in FastSR represents a query sentence  $q$  by sentence contextualized semantics  $SEM_q$ , global context  $Con_q$  and fragment-attended query representations  $FAQ_q$ , and measure the distance of  $q$  to  $Proto_{sem}(S)$ ,  $Proto_{con}(S)$  and  $Proto_{frag}(S)$  prototypes. As shown in Figure 1e, we computed the Euclidean distance between representations and prototypes and took the weighted sum of the distances as the overall distance between  $q$  and  $S$ , denoted as  $d(q, S)$  (Equation 12).  $\alpha_{sem}$ ,  $\alpha_{con}$ , and  $\alpha_{frag}$  are trainable parameters determining the contributions of local and global context of sentences as well as the core concepts of fragments to the sentence classification task. In the final step, we transformed  $d(q, S)$  into a probability such that the shorter the

distance, the higher the probability. This was achieved by applying the Softmax function to the negation of  $d(q, S)$  over all  $J$  support sets (Equation 13). The final prediction for  $q$  is the class with the highest probability. During testing, a batch of classification tasks with different support set groups for a query will be created, and the final prediction will be the average output of the classification task batch.

$$d(q, S) = \alpha_{sem} * \| SEM_q - Proto_{sem}(S) \| + \alpha_{con} * \| CON_q - Proto_{con}(S) \| + \alpha_{frag} * \| FAQ_q - Proto_{frag}(S) \|, \quad \text{where } \alpha_{sem} + \alpha_{con} + \alpha_{frag} = 1 \quad (12)$$

$$ps_q^j = p(c_j|q) = \frac{\exp(-d(q, S^j))}{\sum_{k=1}^J \exp(-d(q, S^k))} \quad (13)$$

Second, as shown in Figure 1e, fragment tagging in FastSR identifies PICO text fragments by comparing the distance between a word embedding  $w_t$  in a query sentence  $q$  to fragment prototypes  $Proto_{frag}(S)$  (Equation 14). Then, as shown in Equation (15), with  $J$  support sets of PICO element classes, we obtained  $J$  distances and normalized the distances to obtain the probability of this word being relevant to class  $c_j$  and then multiplied this probability by  $\beta_t$ , the emission likelihood based on the alignment between  $w_t$  and the fragment (see Equation 9).

$$d(w_t, S) = \| w_t - Proto_{frag}(S) \| \quad (14)$$

$$pw_t^j = p(c_j|w_t) = \beta_t \frac{\exp(-d(w_t, S^j))}{\sum_{k=1}^J \exp(-d(w_t, S^k))} \quad (15)$$

Finally, we computed the loss of this joint learning model. Let  $ys_q^j$  be the ground truth indicating whether  $q$  belongs to class  $j$  and  $yw_t^j$  denotes whether the  $t^{\text{th}}$  word in  $q$  is indeed a part of a PICO element of class  $j$ . We calculated the cross-entropy losses of sentence classification  $LS(q, j)$  and fragment tagging  $LT(q, j)$  (Equations 16-17). Moreover, we added a regularization term to ensure consistent predictions from the two tasks. Intuitively, once a word in query  $q$  is predicted to be a part of a PICO element,  $q$  should be predicted to be PICO element-bearing as well. Thus, we can treat the maximum tagging probability as an imputed classification probability and compute another cross-entropy loss as the regularization term  $LR(q, j)$  (Equation 18). The total loss is the sum of  $LS(q, j)$ ,  $LT(q, j)$ , and  $LR(q, j)$  (Equation 19).

$$LS(q, j) = ys_q^j \log ps_q^j + (1 - ys_q^j) \log(1 - ps_q^j) \quad (16)$$

$$LT(q, j) = \frac{1}{T} \sum_{t=1}^T yw_t^j \log pw_t^j + (1 - yw_t^j) \log(1 - pw_t^j) \quad (17)$$

$$LR(q, j) = ys_q^j \log \max_{t \in q} pw_t^j + (1 - ys_q^j) \log(1 - \max_{t \in q} pw_t^j) \quad (18)$$

$$Loss(q, j) = LS(q, j) + LT(q, j) + LR(q, j) \quad (19)$$

## Evaluation Strategy and Datasets

### Evaluation Tasks

In computational design science research, it is critical to evaluate how well the proposed design framework satisfies design requirements (Padmanabhan et al., 2022; Rai, 2017; Samtani et al., 2023). For a problem context and design solution (e.g., SR automation) that lies at the intersection of data, theory, and machine learning (Maass et al., 2018), a multifaceted evaluation solution is often needed (Padmanabhan et al., 2022; Rai, 2017). This solution often involves benchmarking, ablation analysis, and downstream value (Padmanabhan et al., 2022).

According to Figure 2, our evaluation method consists of three components: (1) *sentence classification*, (2) *sequence tagging*, and (3) *SR automation*. For the first two, we computed the top- $N$  precision ( $P@N$ ), recall ( $R@N$ ), and F1 ( $F1@N$ ) per article, following the SR automation literature (Kiritchenko et al., 2010). Additionally, we use both lexical- and semantic-based metrics for sequence tagging. Lexical metrics measure the overlap between predicted and ground-truth word fragments (Schmidt et al., 2020), while semantic metrics use the cosine similarity between text fragments' BERT word embeddings for precision, recall, and F1 score calculation (Zhang et al., 2020). We also calculated the area under the precision-recall curve (PRC), a threshold invariant metric (Davis & Goadrich, 2006) (see Appendix D) for our metric definitions. Finally, we assessed the impact of FastSR on SR automation in terms of report generation, qualitative feedback, time reduction, and SR process.

We applied a default threshold of 0.5 across models to convert predictive probabilities into binary classifications, indicating whether an instance belongs to a particular class. Raising the threshold can increase precision but lower recall. SR experts are best positioned to determine the desired recall level, balancing the retrieval of critical rare evidence with minimizing false positives (O'Mara-Eves et al., 2015). Therefore, evaluating a model solely on precision and recall at a fixed threshold can be misleading. A more balanced assessment should consider metrics like F1 or PRC.

For sentence classification and sequence tagging, we benchmarked FastSR with a number of models summarized in Tables 2 and 4 and also conducted extensive ablation analyses to estimate the impact of each design component. We used three datasets, Wilson disease (WD), COVID, and EBM-NLP, to demonstrate both the effectiveness and generalizability of FastSR. For SR automation, we used the WD datasets due to our access to the SR experts, who manually extracted PICO elements from this dataset. The details for the datasets are presented below.

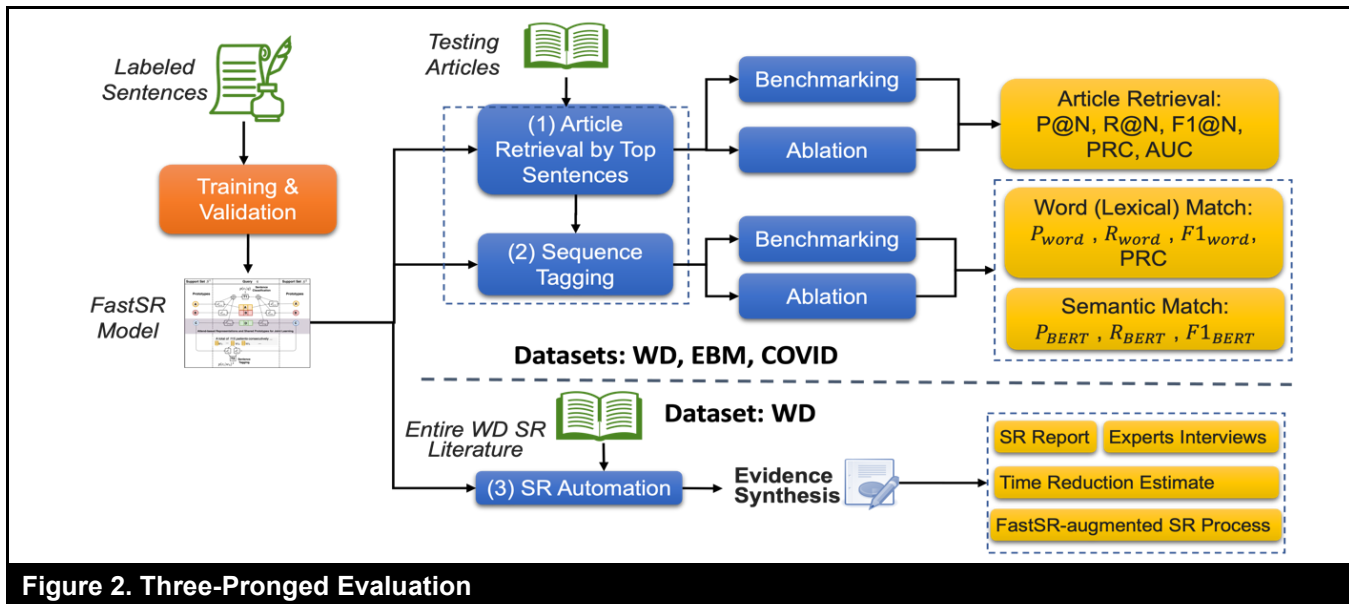


Figure 2. Three-Pronged Evaluation

## Datasets and Model Implementation

**WD dataset:** The WD dataset came from an SR project for Wilson disease conducted by a research team of seven at a global life sciences company (GLS). This project, which took about 1,500 hours over six months, was initiated in 2018 by a pharmaceutical company planning a Phase III trial for a new drug. The team's task was to summarize relevant PICO elements for the trial's design. They screened and assessed 550 articles, ultimately extracting 137 PICO elements (see Appendix A). Out of these, 27 PICO elements grouped into six classes (Table 6) were shared with us for evaluation.<sup>4</sup> Specifically, our training dataset consisted of 222 sentences,<sup>5</sup> each labeled by two SR experts from GLS. We kept no more than 100 labeled sentences in each class to challenge FastSR under labeling constraints, as other researchers have done for FSL research (Snell et al., 2017). We made a special class, NA, for irrelevant sentences. The testing dataset consisted of 110 randomly selected articles that were distinct from the aforementioned labeled data.

**COVID dataset:** We applied FastSR to a COVID-19 dataset to test its reusability and generalizability. First, we reused the trained WD models to extract sample size, country, and

population from this COVID dataset.<sup>6</sup> Second, we trained a new model to extract a new class of outcome elements—*risk factors*, in order to test the generalizability of FastSR to a different literature domain. In epidemiology, a risk factor is a variable associated with an increased risk of disease or infection measured by metrics such as odds ratio (OR) and hazard ratio (HR). The dataset was labeled by three SR experts, containing a training set of 83 positive sentences from 15 articles and a test set of 62 randomly selected articles that were different from the training set.

**EBM-NLP dataset.** We used a public dataset to further assess the generalizability of our method. The EBM-NLP dataset contained 5,000 abstracts of RCT studies fully annotated by crowdsourcing and medical experts.<sup>7</sup> Annotations included 17 PICO elements categorized into three classes: *participants*, *interventions*, and *outcomes*. 200 expert-annotated abstracts were reserved for testing and benchmarking. To preserve the flavor of FSL, we applied a stratified strategy<sup>8</sup> to randomly sample 30 training subsets, each comprising 200 sentences. Then, we trained models on each subset and tested them on the designated test set. The global context module was disabled in FastSR since this dataset only consisted of abstracts.

<sup>4</sup> Notably, FastSR can be easily extended to hundreds of PICO elements with a few labeled examples for each element, and we can expand to more elements upon request.

<sup>5</sup> The SR experts searched through a set of 51 randomly chosen articles for sentences of the six classes. Because a sentence can contain multiple PICO fragments, one sentence can appear in multiple PICO classes.

<sup>6</sup> The dataset contained 266 articles merged from <https://www.kaggle.com/allen-institute-for-ai/CORD-19-research-challenge> (143 articles) and <https://www.kaggle.com/mlconsult/summary-page-covid-19-risk-factors>. This reusability test excluded Intervention due

to the inapplicability of WD treatments (e.g., liver transplant) to COVID and also omitted Study Design and Study Period because this dataset primarily comprises observational studies conducted within a short period and thus has limited variation in these two classes.

<sup>7</sup> The dataset can be downloaded from <https://github.com/bepnye/EBM-NLP>.

<sup>8</sup> We applied stratified sampling to choose 100 instances first and then randomly selected the remaining half. This sampling procedure ensures a minimal sample size for each class and also sufficient coverage for the majority class.

For each dataset, we randomly split the training data into 75% for training and 25% for validation. The summary statistics for the training data are presented in Appendix Table F2. For our 2-way,  $k$ -shot FSL models, we conducted a sensitivity analysis for  $k$  and set  $k = 5$  for the best model performance. For most of the benchmarking models, we adopted the original implementations provided by the authors. The implementation details can be found in Appendix E. During testing, all models scanned each testing article to extract the

top three sentences for each PICO data element class. At least two SR experts checked whether each extracted sentence was associated with the correct PICO element class. The average kappa score between experts was 0.85 for the WD dataset and 0.87 for the COVID dataset, indicating strong agreement. Since the EBM-NLP test dataset was fully annotated with a kappa score of 0.67 (Nye et al., 2018), we were able to evaluate performance on all sentences, as reported next.

**Table 6. Description of Datasets**

| Disease domain                                               | Format    | PICO data                                                                  |                                                                                                                | Train sent. | Test art./ sent. | Examples                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------------|-----------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                              |           | Classes <sup>[1]</sup>                                                     | Elements                                                                                                       |             |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 1. WD dataset                                                |           |                                                                            |                                                                                                                |             |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Wilson disease                                               | Full text | Study design                                                               | Study type, problem definition, data source, perspective                                                       | 100         | 110 / 12,320     | In this pre-specified [long-term follow-up] <sub>perspective</sub> of the [Cardiac Insufficiency Bisoprolol Study] <sub>problem def.</sub> in [Elderly (CIBIS-ELD) trial] <sub>study type</sub> , we ...<br><br>[Adults with atherosclerotic cardiovascular disease, heterozygous familial hypercholesterolemia, or both] <sub>eligibility</sub> were eligible ...<br><br>Patients were randomised 1:1 to receive [vosaroxin] <sub>drug</sub> [(90 mg/m2 in cycle 1 and 70 mg/m2 in subsequent cycles by shot within 10 minutes intravenous [IV] infusion on days 1 and 4)] <sub>dose / intensity / duration / delivery</sub> OR ... |
|                                                              |           | Study period                                                               | Study period, follow-up period                                                                                 | 50          |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                              |           | Sample size                                                                | Sample size, age, gender, race, comparator sample size, intervention sample size                               | 62          |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                              |           | Country                                                                    | Country, region                                                                                                | 49          |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                              |           | Population                                                                 | Eligibility, comorbidities                                                                                     | 93          |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                              |           | Intervention                                                               | Drug name, dose, intensity, duration, delivery, monitoring, co-interventions, comparator drug & dose, schedule | 64          |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 2. COVID dataset                                             |           |                                                                            |                                                                                                                |             |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| COVID                                                        | Full text | Risk factor                                                                | Risk factor name, hazard ratio, confidence level                                                               | 83          | 62 / 8464        | Analysis showed that [age >75 years] <sub>risk factor</sub> ([hazard ratio, 7.86] <sub>HR</sub> ; [95% CI, 2.44-25.35] <sub>CI</sub> ), ... were independent risk ...                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                              |           | Population, sample size, country (reused training samples from WD dataset) |                                                                                                                |             |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 3. EBM-NLP dataset                                           |           |                                                                            |                                                                                                                |             |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| A wide range of cardio-vascular, cancer, and autism diseases | Abstract  | Participants                                                               | Age, sex, sample-size, condition                                                                               | 57          | 200 / 2044       | [Fifty] <sub>sample-size</sub> patients with [alcoholic liver disease] <sub>condition</sub> ... were randomly ...<br><br>The authors conduct... trial of [20] <sub>sample-size</sub> [healthy] <sub>condition</sub> ...<br><br>Effect of treatment with [paromomycin] <sub>pharmacological</sub> on endotoxemia in patients ...<br><br>[Endotoxin concentration] <sub>physical</sub> , [liver function tests] <sub>physical</sub> , and...                                                                                                                                                                                           |
|                                                              |           | Interventions                                                              | Surgical, physical, pharmacological, educational, psychological, other, control                                | 106         |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                              |           | Outcomes                                                                   | Physical, pain, mortality, adverse effects, mental, other                                                      | 92          |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

**Note:** <sup>[1]</sup> The NA class includes 438 irrelevant sentences in WD, 391 in COVID, and 32 in EBM-NLP

| Table 7. Sentence Classification Benchmarking on WD Dataset |       |       |              |          |            |          |
|-------------------------------------------------------------|-------|-------|--------------|----------|------------|----------|
| Models / metrics (%)                                        | P@3   | R@3   | F1@3 & diff. |          | PRC & diff |          |
| 1. Traditional classifier <sup>[1]</sup>                    |       |       |              |          |            |          |
| Naïve Bayes (NB)                                            | 20.29 | 37.66 | 25.51        | 47.49*** | 23.41      | 49.20*** |
| SVM                                                         | 31.66 | 32.98 | 32.08        | 40.92*** | 28.58      | 44.03*** |
| LSTM + CNN                                                  | 62.14 | 49.44 | 54.09        | 18.91*** | 53.88      | 18.74*** |
| LSTM                                                        | 58.34 | 52.26 | 54.88        | 18.11*** | 55.52      | 17.09*** |
| CNN                                                         | 74.56 | 54.92 | 61.98        | 11.02*** | 65.22      | 7.39***  |
| 2. Virtual adversarial training (VAT SSL) <sup>[2]</sup>    | 75.86 | 52.86 | 60.63        | 12.37*** | 63.37      | 9.24***  |
| 3. LLMs                                                     |       |       |              |          |            |          |
| Fine-Tuned BioBERT                                          | 63.63 | 66.35 | 64.17        | 8.83***  | 63.77      | 8.84***  |
| GPT-4 Few-Shot                                              | 31.22 | 55.26 | 39.23        | 33.77*** | 28.94      | 43.67*** |
| 4. FSL with BioBERT embeddings                              |       |       |              |          |            |          |
| Model-Agnostic Meta-Learning (MAML)                         | 24.09 | 31.21 | 26.36        | 46.63*** | 23.89      | 48.72*** |
| FSL with distributional signatures (FSL_DS)                 | 28.24 | 49.62 | 35.01        | 37.99*** | 35.79      | 36.82*** |
| Open-world learning to accept class (L2AC)                  | 42.59 | 46.06 | 42.80        | 30.20*** | 42.20      | 30.41*** |
| Matching net                                                | 62.82 | 62.99 | 61.98        | 11.02*** | 63.32      | 9.29***  |
| Prototypical network (ProtoNet)                             | 70.26 | 61.93 | 65.40        | 7.60***  | 66.39      | 6.22***  |
| 5. Model with different embeddings                          |       |       |              |          |            |          |
| ProtoNet + BERT                                             | 55.61 | 62.70 | 58.87        | 14.13*** | 58.01      | 14.60*** |
| ProtoNet + SciBERT                                          | 63.75 | 58.74 | 60.88        | 12.12*** | 62.21      | 10.40*** |
| 6. FastSR (ours)                                            | 72.34 | 74.09 | 73.00        | -        | 72.61      | -        |

**Note:** Significance levels are indicated as follows: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . <sup>[1]</sup> TF-IDF for NB and SVM, and BioBERT for the other three models. <sup>[2]</sup> Adversarial training for VAT SSL is conducted at the BioBERT embedding layer of the CNN model.

## Evaluation Results

### Benchmarking

#### Sentence Classification

We first present the benchmarking analyses for the WD dataset in terms of the top-three metrics and PRC <sup>9</sup> on the 110 testing articles, as shown in Table 7. We also tested the significance of differences in F1@3 and PRC <sup>10</sup> between FastSR and the benchmarking models.

As predicted in the Task Complexity 3 section, CNN emerged as the top traditional classifier, with an average F1@3 score of 0.62. LSTM, when combined with BioBERT embeddings, resulted in a lower performance. Hence, we focused on using CNN for semantic sentence representation. We further conducted VAT on the CNN model with all unlabeled sentences from the full text of the training articles. However, the model performance was barely improved. One possible reason is that the VAT assumes that the distribution of unlabeled data aligns with the labeled data (Oliver et al., 2018), but this may not hold for our diverse full-text unlabeled data.

For the LLM group, fine-tuning BioBERT with the WD dataset resulted in a marginal performance increase with an F1@3 of 64%. Likely, the limited sample size in our SR context restricted the effectiveness of such fine-tuning approaches, as highlighted by Brown et al. (2020). We extensively tested various ALM techniques, including open-source and closed-source models, along with fine-tuning and few-shot prompting (see Appendix G). GPT-4 with few-shot prompting was the best-performing model across all datasets. However, GPT-4 significantly underperformed compared to FastSR, with a 33.77% (F1) and 43.67% (PRC) gap, and fine-tuned BioBERT, with a 23.94% (F1) and 34.83% (PRC) gap. This could be due to ALMs' unidirectional self-attention, which lacks the ability to capture bidirectional dependencies among words within sentences, leading to an incomplete understanding of context. Another reason might be data contamination, as we found that when switching to the open-domain EBM-NLP data, the gap between GPT-4 and FastSR was reduced by 21.65% (F1) and 23.62% (PRC). Given GPT-4's lower sentence performance and high API cost, we did not apply GPT-4 for sequence tagging.

<sup>9</sup> PRC on top-three predictions can be ill-defined, because usually only a portion of precision/recall pairs can be drawn at high thresholds. Instead, we pooled the top three sentences from all models to form a corpus. This corpus was labeled during validation. Then, we calculated the F1(denoted as F1@all) and PRC on this corpus for each model.

<sup>10</sup> For the single WD and COVID datasets, we performed paired  $t$ -tests on the article-level metrics and followed a bootstrapping procedure (Boyd et al., 2013, n13) to compare the corpus-level PRCs for sentence classification. For the EBM-NLP dataset, paired  $t$ -tests were performed across the 30 splits.

**Table 8. Sentence Classification Performance on COVID Dataset**

| Metrics (%)                                                                                    | P@3   | R@3   | F1@3 & diff <sup>[2]</sup> |          | PRC & diff <sup>[2]</sup> |          |
|------------------------------------------------------------------------------------------------|-------|-------|----------------------------|----------|---------------------------|----------|
| Average of country/population/sample size classes (reused models trained on WD) <sup>[1]</sup> |       |       |                            |          |                           |          |
| CNN                                                                                            | 74.01 | 29.88 | 42.57                      | 35.87*** | 63.17                     | 12.59*** |
| Fine-tuned BioBERT                                                                             | 66.23 | 58.64 | 62.21                      | 16.23*** | 65.01                     | 10.75*** |
| GPT-4 Few-Shot                                                                                 | 37.24 | 50.89 | 43.01                      | 35.43*** | 33.33                     | 42.43*** |
| ProtoNet                                                                                       | 69.30 | 78.81 | 73.75                      | 4.69***  | 74.44                     | 1.32***  |
| FastSR                                                                                         | 75.59 | 81.53 | 78.44                      | -        | 75.76                     | -        |
| Risk factor (newly trained)                                                                    |       |       |                            |          |                           |          |
| CNN                                                                                            | 63.99 | 70.99 | 67.31                      | 9.65***  | 74.18                     | 1.11**   |
| Fine-tuned BioBERT                                                                             | 57.44 | 62.65 | 59.93                      | 17.03*** | 58.00                     | 17.29*** |
| GPT-4 Few-Shot                                                                                 | 45.90 | 55.86 | 50.40                      | 26.56*** | 53.43                     | 21.86*** |
| ProtoNet                                                                                       | 63.39 | 70.99 | 66.98                      | 9.98***  | 71.26                     | 4.03***  |
| FastSR                                                                                         | 71.73 | 83.02 | 76.96                      | -        | 75.29                     | -        |

**Note:** <sup>[1]</sup> Only the top three WD benchmarking models (see Table 7) plus the latest GPT-4 model were selected to test reusability. <sup>[2]</sup> Significance levels are indicated as follows: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

In the FSL category, MAML performed poorly due to the limited shared features among our six classes. FSL\_DS struggled with our small dataset, as it relies on a comprehensive training set for stable class-specific TF-IDF weight distributions. L2AC, which selects similar support instances for each query, showed bias by overusing common subsets and underusing rare ones. Matching net and ProtoNet outperformed others, with ProtoNet achieving the highest F1@3 score of over 65%. This performance was further enhanced when using domain-specific BioBERT and SciBER, as they better capture medical terminology. Overall, FastSR achieved an average F1@3 of 70%, exceeding SVM by 40%, GPT-4 by 33%, CNN by 11%, fine-tuned BioBERT by 8%, and ProtoNet by 7%. Similar gains can be observed in PRC.

Next, with the models trained by the WD dataset, we examined the reusability of FastSR and the best-performing benchmarking models by directly applying them to the COVID dataset. As shown in Table 8, among these models, FSL-based models, such as ProtoNet and FastSR, yielded better reusability than CNN and fine-tuned BioBERT, which may be overfitted to WD domain-specific features. As predicted, GPT-4 consistently struggled with performance, likely due to its autoregressive nature and data contamination. FastSR yielded the best performance, which is comparable to the WD results. For risk factors, the newly trained FastSR identified relevant articles with a 77% F1@3, 9%-26% higher than the benchmarking models.

The EBM-NLP dataset exhibited similar patterns, as shown in Table 9. Since the EBM-NLP dataset contained only abstracts, global context features were not utilized for FastSR. Notably, PICO fragments in abstracts were more coarse-grained, whereas the WD full-text dataset included fine-grained details like dosage

and delivery methods. Consequently, all models performed better on EBM-NLP than on the WD dataset. CNN and ProtoNet consistently delivered competitive results. VAT, which also incorporated unlabeled sentences from abstracts, outperformed the base model CNN by about 2%. As noted by Oliver et al. (2018), VAT can enhance model robustness when labeled and unlabeled data share similar distributions. FastSR was still the best-performing model, exceeding CNN ProtoNet by 2~4% in all the metrics. Interestingly, the gap between GPT-4 few-shot and FastSR significantly shrank by 20% in F1 and PRC, likely due to data contamination in the public EBM-NLP dataset. Given the narrowing performance gap, we also included GPT-4 in the downstream sequence tagging task benchmarking.

## Sequence Tagging Benchmarking

Building on the FastSR's promising results in sentence retrieval, we proceeded to evaluate the performance of PICO fragment tagging. This evaluation involved sequence taggers trained in conjunction with the top-performing sentence classifiers from each method group. Specifically, the CNN classifier was teamed with a linear tagger on top of BioBERT embeddings, BiLSTM, or CRF, and ProtoNet was coupled with ProtoNER and nearest neighbors. The linear tagger was used as the base model for VAT. BioBERT was fine-tuned for both tasks. GPT-4 few-shot prompting, the best model for sentence classification, was also used for sequence tagging. However, we omitted evaluation on the WD and COVID datasets due to low sentence classification performance and high API costs. For each combination, we retrieved the top three sentences and measured tagging performance by F1, BERTScore, and PRC. If a sentence was classified incorrectly, no fragments were extracted.

**Table 9. Sentence Classification Benchmarking on EBM-NLP Dataset**

| Models <sup>[1]</sup> / metrics (%) | P@3   | R@3   | F1@3 & diff. |          | PRC & diff |          |
|-------------------------------------|-------|-------|--------------|----------|------------|----------|
| 1. Traditional classifier           |       |       |              |          |            |          |
| Naïve Bayes (NB)                    | 62.26 | 56.09 | 58.18        | 21.53*** | 60.00      | 21.54*** |
| SVM                                 | 62.78 | 55.69 | 57.78        | 21.93*** | 59.53      | 22.01*** |
| LSTM + CNN                          | 75.55 | 71.99 | 73.30        | 6.40***  | 74.58      | 6.96***  |
| LSTM                                | 77.43 | 74.37 | 75.45        | 4.26***  | 76.64      | 4.90***  |
| CNN                                 | 78.78 | 74.07 | 75.47        | 4.23***  | 77.90      | 3.64***  |
| 2. VAT (SSL) <sup>[2]</sup>         | 79.02 | 76.25 | 77.09        | 2.62***  | 78.87      | 2.67***  |
| 3. LLMs                             |       |       |              |          |            |          |
| Fine-tuned BioBERT                  | 85.01 | 58.00 | 64.61        | 15.10*** | 77.65      | 3.89***  |
| GPT-4 Few-Shot                      | 63.70 | 72.19 | 67.59        | 12.12*** | 61.49      | 20.05*** |
| 4. FSL with BioBERT embeddings      |       |       |              |          |            |          |
| Matching net                        | 75.59 | 78.91 | 77.03        | 2.68***  | 76.80      | 4.74***  |
| ProtoNet                            | 77.76 | 76.95 | 77.26        | 2.45***  | 78.07      | 3.46***  |
| 5. Model with different embeddings  |       |       |              |          |            |          |
| ProtoNet + BERT                     | 73.19 | 70.51 | 71.66        | 8.05***  | 72.64      | 8.90***  |
| ProtoNet + SciBERT <sup>[3]</sup>   | 77.39 | 76.87 | 77.01        | 2.69***  | 78.10      | 3.44***  |
| 6. FastSR (ours)                    | 82.14 | 77.70 | 79.71        | -        | 81.54      | -        |

**Note:** Significance levels are indicated as follows: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . <sup>[1]</sup> Based on the results on the WD dataset (Table 7), we dropped three extremely low-performing FSL benchmarks (MAML, FSL\_DS, L2AC). <sup>[2]</sup> Adversarial training is conducted at the BioBERT embedding layer of the CNN model. <sup>[3]</sup> SciBERT performs similarly to BioBERT, perhaps because both BERT models were trained on the PubMed database, from which EBM-NLP data was extracted.

**Table 10. Sequence Tagging for PICO Fragment Extraction on WD Dataset**

| Models / metrics (%)                          | Top three <sup>[4]</sup> |                   |                            |          |                   |                   |                            |          | PRC & diff. |          |
|-----------------------------------------------|--------------------------|-------------------|----------------------------|----------|-------------------|-------------------|----------------------------|----------|-------------|----------|
|                                               | P <sub>word</sub>        | R <sub>word</sub> | F1 <sub>word</sub> & diff. |          | P <sub>BERT</sub> | R <sub>BERT</sub> | F1 <sub>BERT</sub> & diff. |          |             |          |
| 1. Traditional sequence tagger <sup>[1]</sup> |                          |                   |                            |          |                   |                   |                            |          |             |          |
| Linear layer                                  | 76.30                    | 26.51             | 35.08                      | 25.71*** | 77.81             | 37.78             | 47.75                      | 19.44*** | 24.32       | 37.75*** |
| CRF                                           | 78.48                    | 27.31             | 35.33                      | 25.46*** | 79.72             | 36.47             | 46.35                      | 20.84*** | 24.40       | 37.68*** |
| LSTM                                          | 79.48                    | 30.91             | 41.88                      | 18.91*** | 82.33             | 37.33             | 49.52                      | 17.68*** | 29.66       | 32.42*** |
| LSTM + CRF                                    | 70.80                    | 37.52             | 45.50                      | 15.29*** | 73.31             | 43.84             | 51.64                      | 15.56*** | 30.82       | 31.26*** |
| 2. VAT (SSL) <sup>[3]</sup>                   | 74.05                    | 34.48             | 44.27                      | 16.52*** | 77.23             | 44.29             | 54.19                      | 13.00*** | 49.47       | 12.61*** |
| 3. Fine-tuned BioBERT                         | 55.57                    | 44.76             | 47.83                      | 12.96*** | 62.63             | 51.74             | 55.51                      | 11.68*** | 50.04       | 12.03*** |
| 4. FSL taggers <sup>[2]</sup>                 |                          |                   |                            |          |                   |                   |                            |          |             |          |
| Nearest neighbors (NN)                        | 53.58                    | 60.06             | 56.06                      | 4.73***  | 67.59             | 62.91             | 64.70                      | 2.50***  | 50.61       | 11.46*** |
| ProtoNER                                      | 56.83                    | 58.08             | 57.17                      | 3.62***  | 66.91             | 62.36             | 64.27                      | 2.92***  | 50.12       | 11.96*** |
| 5. FastSR (ours)                              | 71.04                    | 55.77             | 60.79                      | -        | 74.48             | 63.28             | 67.19                      | -        | 62.08       | -        |

**Note:** Significance levels are indicated as follows: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . <sup>[1]</sup> Joint training with the sentence classifier CNN. <sup>[2]</sup> Joint training with the sentence classifier ProtoNet. <sup>[3]</sup> Virtual adversarial training on model "linear layer." <sup>[4]</sup> GPT models were skipped due to low sentence classification performance and high API costs.

We first present the top-three tagging metrics,  $F1_{word}@3$ , and PRC, from the WD dataset, as in Table 10. FSL-based models outperformed traditional methods by substantial margins. CNN coupled with LSTM+CRF tagger performed the best in the traditional group, with an average of  $F1_{word}@3$  ( $F1_{BERT}@3$ ) of 46% (52%). VAT improved the performance of the linear tagger as its perturbations mitigated the model's biases towards specific words. FastSR obtained the highest performance, with an  $F1_{word}@3$  ( $F1_{BERT}@3$ ) of 61% (67%), about 15% higher than LSTM+CRF and 4% than ProtoNER. Notably, FastSR surpassed the other models in PRC by over 11%. Lastly, all the models achieved higher  $F1_{BERT}$  than

$F1_{word}$ , because  $F1_{BERT}$  accepts words that are semantically similar with, not necessarily the same as, annotated fragments.

Similar to sentence classification, with the sequence taggers trained on WD, we selected FastSR and the top three best-performing benchmarking models to test their reusability on the COVID dataset, as shown in Table 11. The reused FastSR models yielded performance comparable to the WD results. It surpassed the benchmarks by a substantial margin (8%-28%) in  $F1@3$  and PRC, highlighting FastSR models' reusability across literature domains.

**Table 11. Sequence Tagging for PICO Fragment Extraction on COVID Dataset**

| Models <sup>[1]</sup> / metrics (%)                                    | P <sub>word</sub><br>@3 | R <sub>word</sub><br>@3 | F1 <sub>word</sub> @3 & diff. |          | P <sub>BERT</sub><br>@3 | R <sub>BERT</sub><br>@3 | F1 <sub>BERT</sub> @3 & diff. |          | PRC & diff. |         |
|------------------------------------------------------------------------|-------------------------|-------------------------|-------------------------------|----------|-------------------------|-------------------------|-------------------------------|----------|-------------|---------|
| Average of country/population/sample size (reused model trained on WD) |                         |                         |                               |          |                         |                         |                               |          |             |         |
| LSTM_CRF                                                               | 63.83                   | 27.44                   | 38.38                         | 28.13*** | 66.98                   | 35.05                   | 46.02                         | 27.76*** | 55.42       | 4.95*** |
| Fine-tuned BioBERT                                                     | 60.25                   | 28.74                   | 38.91                         | 27.6***  | 64.18                   | 39.67                   | 49.03                         | 24.75*** | 51.45       | 8.92*** |
| ProtoNER                                                               | 66.93                   | 46.55                   | 54.91                         | 11.6***  | 71.91                   | 60.09                   | 65.48                         | 8.3***   | 51.93       | 8.44*** |
| FastSR                                                                 | 67.22                   | 65.81                   | 66.51                         | -        | 72.44                   | 75.17                   | 73.78                         | -        | 60.37       | -       |
| Risk factor (newly trained)                                            |                         |                         |                               |          |                         |                         |                               |          |             |         |
| LSTM_CRF                                                               | 55.41                   | 46.45                   | 50.53                         | 15.08*** | 58.52                   | 61.49                   | 59.97                         | 14.87*** | 54.42       | 4.18*** |
| Fine-tuned BioBERT                                                     | 56.10                   | 39.04                   | 46.04                         | 19.57*** | 59.68                   | 54.54                   | 56.99                         | 17.85*** | 51.19       | 7.41*** |
| ProtoNER                                                               | 58.11                   | 54.09                   | 56.03                         | 9.58***  | 63.60                   | 63.62                   | 63.61                         | 11.23*** | 56.27       | 2.33*   |
| FastSR                                                                 | 65.25                   | 65.98                   | 65.61                         | -        | 73.00                   | 76.78                   | 74.84                         | -        | 58.60       | -       |

**Note:** Significance levels are indicated as follows: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . <sup>[1]</sup> Only the WD model with the highest F1<sub>word</sub>@3 from each tagger category (see Table 10) was selected to test their reusability. VAT tagger was excluded because the coupled VAT classifier was not chosen for sentence classification due to performance degradation from adversarial training (see Table 8).

**Table 12. Sequence Tagging for PICO Fragment Extraction on EBM-NLP Dataset**

| Models / metrics (%)           | P <sub>word</sub><br>@3 | R <sub>word</sub><br>@3 | F1 <sub>word</sub> @3 & diff. |          | P <sub>BERT</sub><br>@3 | R <sub>BERT</sub><br>@3 | F1 <sub>BERT</sub> @3 & diff. |          | PRC & diff. [2] |          |
|--------------------------------|-------------------------|-------------------------|-------------------------------|----------|-------------------------|-------------------------|-------------------------------|----------|-----------------|----------|
| 1. Traditional sequence tagger |                         |                         |                               |          |                         |                         |                               |          |                 |          |
| Linear Layer                   | 51.13                   | 29.32                   | 34.01                         | 32.07*** | 61.19                   | 47.23                   | 51.32                         | 24.17*** | 40.63           | 31.14*** |
| CRF                            | 50.69                   | 30.17                   | 34.55                         | 31.53*** | 62.88                   | 49.29                   | 53.20                         | 22.29*** | 38.44           | 33.33*** |
| LSTM                           | 70.40                   | 30.72                   | 37.66                         | 28.42*** | 76.13                   | 39.77                   | 47.83                         | 27.66*** | 61.79           | 9.97***  |
| LSTM + CRF                     | 68.7                    | 46.56                   | 51.15                         | 14.93*** | 75.73                   | 58.25                   | 63.41                         | 12.08*** | 66.27           | 5.49***  |
| 2. VAT (SSL) [1]               | 64.19                   | 38.67                   | 45.16                         | 20.92*** | 71.00                   | 54.00                   | 59.50                         | 15.99*** | 55.93           | 15.84*** |
| 3. LLMs                        |                         |                         |                               |          |                         |                         |                               |          |                 |          |
| Fine-tuned BioBERT             | 76.18                   | 40.97                   | 47.4                          | 18.68*** | 80.36                   | 51.44                   | 57.74                         | 17.75*** | 65.28           | 6.49***  |
| GPT-4 Few-Shot                 | 34.73                   | 34.08                   | 32.86                         | 33.22*** | 57.32                   | 58.65                   | 55.28                         | 20.21*** | -               | -        |
| 4. FSL taggers                 |                         |                         |                               |          |                         |                         |                               |          |                 |          |
| ProtoNER                       | 68.10                   | 53.48                   | 56.80                         | 9.28***  | 77.28                   | 65.34                   | 69.33                         | 6.16***  | 64.31           | 7.45***  |
| Nearest neighbors              | 70.25                   | 57.79                   | 62.56                         | 3.52***  | 78.3                    | 70.37                   | 73.88                         | 1.62***  | 69.29           | 2.48***  |
| 5. FastSR (ours)               | 66.00                   | 68.21                   | 66.08                         | -        | 77.75                   | 73.97                   | 75.49                         | -        | 71.77           | -        |

**Note:** Significance levels are indicated as follows: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . <sup>[1]</sup> Virtual adversarial training on the model "linear layer" on top of BioBERT embeddings. <sup>[2]</sup> PRC was not calculated for GPT-4 because the predictive probabilities generated at the word level were found to be inconsistent and unreliable, leading to potentially misleading results.

The EBM-NLP benchmarking shows comparable results, as illustrated in Table 12. FastSR outperformed the benchmarking models across all metrics. The F1<sub>word</sub>@3 was above 66% (vs. 60% in WD). LSTM+CRF still led the traditional tagger groups with an F1<sub>word</sub>@3 around 51% (vs. 46 in WD). The gains can be attributed to the coarse granularity of the PICO elements and consistent linguistic patterns found in abstracts. Nevertheless, FastSR exceeded LSTM+CRF by 15% in terms of F1<sub>word</sub> measures, and the FSL benchmarks by a large margin in recall R<sub>word</sub>@3 (10~15%), with only a minor loss in precision (<4%). Interestingly, GPT-4's performance significantly declined in fragment tagging compared to sentence classification (33.22% vs. 12.12% in F1), while fine-tuned BioBERT's gap to FastSR remained consistent (18.68% vs. 15.10% in F1). This is likely because fragment tagging requires both preceding and

following context, where GPT-4's unidirectional self-attention falls short, indicating its struggle with complex NLP tasks demanding holistic text comprehension.

### Ablation Analysis

We now turn to a thorough ablation study on our novel extensions of FSL. As shown in Figure 1e, our novel components (vs. routine FSL) included (1) global context prototypes (Module B) to enhance metric-based learning, (2) attention-based fragment prototypes (Module C) for semantic correspondence of PICO fragments, and (3) a fragment-attended query representation (Module D) to share fragment prototypes across classification and tagging.

**Table 13. Ablation Analysis for Sentence Classification on WD Dataset**

| Models / metrics (%)                                                                            | P@3   | R@3   | F1@3 & diff. |          | PRC & diff. |          |
|-------------------------------------------------------------------------------------------------|-------|-------|--------------|----------|-------------|----------|
| FastSR full model                                                                               | 72.34 | 74.09 | 73.00        | -        | 72.61       | -        |
| M1: Full minus global context                                                                   | 69.63 | 69.89 | 69.36        | -3.64*** | 70.13       | -2.49*** |
| M2: Full minus attention                                                                        | 68.14 | 70.10 | 68.70        | -4.30*** | 68.48       | -4.13*** |
| M3: Full minus joint learning, or<br>M4: Full minus (attention + joint learning) <sup>[1]</sup> | 66.13 | 71.67 | 68.26        | -4.74*** | 69.34       | -3.28*** |
| M5: Full minus (global context + attention)                                                     | 69.58 | 68.41 | 68.50        | -4.50*** | 69.32       | -3.29*** |
| M6: Full minus (global context + joint learning) <sup>[2]</sup>                                 | 70.26 | 61.93 | 65.40        | -7.60*** | 66.39       | -6.22*** |
| M7: Hierarchical MTL architecture                                                               | 67.25 | 71.04 | 68.80        | -4.19*** | 68.95       | -3.66*** |
| M8: Section summarization as alternative global context                                         | 69.51 | 72.66 | 70.73        | -2.27*** | 70.52       | -2.09*** |
| M9: Reducing training samples by 1/3                                                            | 69.97 | 69.60 | 68.95        | -4.05*** | 70.98       | -1.63**  |

**Note:** Significance levels are indicated as follows: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . <sup>[1]</sup> Without joint learning, the attention only acts on sequence tagging. Thus, for sentence classification, these two configurations are equivalent (see Figure 1e). <sup>[2]</sup> For sentence classification, M5 is equivalent to the benchmarking model ProtoNet (see Figure 1e).

We first assessed the impact of each component using the WD dataset. We removed one component from the full model at a time to create three new models (M1-M3). In M1, we excluded the global context prototypes from the sentence prediction by setting  $\alpha_{con} = 0$  in Equation (12). In M2, we ignored semantic correspondence by generating fragment-related representations using word embedding averaging, as used in ProtoNER (Huang et al., 2020). In M3, we disabled joint learning by removing fragment-attended query representations, i.e.,  $\alpha_{frag} = 0$  in Equation (12) and eliminating the regularization term in Equation (19). Moreover, we simultaneously removed two components to analyze their potential interactions (M4-M6). In addition, we tested alternative designs for the components. As described in Task Complexity 4 section, we replaced the modular MTL architecture with a hierarchical one in M7, where sequence tagging was enabled only when a sentence  $q$  was positively classified first, i.e.,  $pw_t^j = p(c_j|w_t) * p(c_j|q)$  in Equation (15). In M8, we replaced global context with an unsupervised graph-based summarization model (Zheng & Lapata, 2019), as discussed in Task Complexity 2 section. Finally, we evaluated FastSR with a further reduced dataset. We applied stratified sampling to select 2/3 of the labeled samples to create 10 subsets and train models on each subset.

The ablation performance for the WD dataset is presented in Table 13. Removing context prototypes led to a 3.64% decrease in F1@3, excluding attention produced a 4.30% loss, and disabling joint learning resulted in a 4.74% drop. Additional reductions occurred when two components were simultaneously removed (M5-M6). The attention mechanisms that emphasize the core concepts of PICO fragments generate significant benefits because they make query representations and support set prototypes more meaningful for metric-based learning. Joint learning, which shares this multifaceted knowledge across the classification and tagging tasks, further

improves each task. The alternative MTL architecture and global context also led to a 2-4% reduction in the metrics. Finally, a one-third reduction in the training samples resulted in a 3-4% decrease in F1 and only a 1.63% decrease in PRC. This marginal decline underscores the robustness of FastSR in dealing with a more pronounced scarcity of labeled datasets.

Table 14 shows the ablation results of sequence tagging. First, removing global context prototypes led to a reduction of 2-3% in all F1 and PRC measures, demonstrating the importance of sentence global context for identifying the PICO text fragments. Second, excluding attention mechanisms led to a drop of 5.18% drop in  $F1_{word}@3$  and 3-4% in the other measures, indicating the value of semantic correspondence for PICO fragments in ML design. Third, disabling joint learning resulted in a loss of 3.76% in  $F1_{word}@3$  and 5.30% in PRC, showcasing the value of sentence classification in fragment tagging. Similarly, removing two components simultaneously (M4-M5) resulted in a further reduction in performance. The alternative MTL architecture and global context exhibited inferior performance, compared to our proposed design. Finally, a 1/3 reduction in training samples led to a slight 3-5% decline across all metrics.

We performed similar ablation analyses on the EBM-NLP dataset. Without global context, we tested two models, M2 and M3. We omitted M7 because the result was similar to the WD case. As shown in Table 15, removing the attention mechanisms for sentence classification resulted in a 1-2% drop in all metrics, and disabling joint learning led to a 2% decrease in F1@3 and 3% in PRC. For sequence tagging, the model's  $F1_{word}@3$  was reduced by 3% after removing the attention and 5% after disabling joint learning. PRC is reduced by over 2% after excluding each of the components.

**Table 14. Ablation Study of Sequence Tagging for PICO Fragment Extraction on WD Dataset**

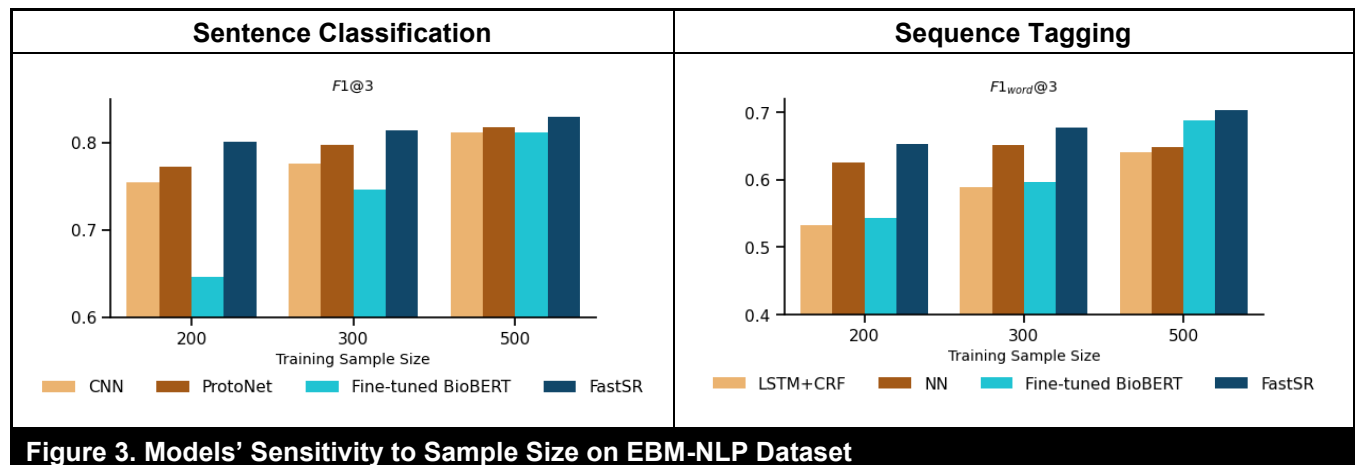
| Models / metrics                                                | Top three         |                   |                            |          |                   |                   |                            | PRC & diff. |       |
|-----------------------------------------------------------------|-------------------|-------------------|----------------------------|----------|-------------------|-------------------|----------------------------|-------------|-------|
|                                                                 | P <sub>word</sub> | R <sub>word</sub> | F1 <sub>word</sub> & diff. |          | P <sub>BERT</sub> | R <sub>BERT</sub> | F1 <sub>BERT</sub> & diff. |             |       |
| FastSR full model                                               | 71.04             | 55.77             | 60.79                      | -        | 74.48             | 63.28             | 67.19                      | -           | 62.08 |
| M1: Full minus global context                                   | 64.86             | 54.19             | 57.37                      | -3.41*** | 69.14             | 61.08             | 63.71                      | -3.48***    | 59.84 |
| M2: Full minus attention                                        | 67.54             | 49.84             | 55.60                      | -5.18*** | 73.05             | 58.78             | 64.02                      | -3.17***    | 57.70 |
| M3: Full minus joint learning                                   |                   |                   |                            |          |                   |                   |                            |             |       |
| M6: Full minus (global context + joint learning) <sup>[1]</sup> | 68.17             | 52.01             | 57.03                      | -3.76*** | 71.99             | 61.44             | 64.92                      | -2.27***    | 56.78 |
| M4: Full minus (attention + joint learning) <sup>[2]</sup>      | 56.83             | 58.08             | 57.17                      | -3.62*** | 66.91             | 62.36             | 64.27                      | -2.92***    | 50.12 |
| M5: Full minus (global context + attention)                     | 69.77             | 47.21             | 54.71                      | -6.08*** | 73.30             | 55.67             | 62.12                      | -5.07***    | 58.81 |
| M7: Hierarchical MTL architecture                               | 66.26             | 55.74             | 58.90                      | -1.89*   | 70.69             | 62.59             | 65.11                      | -2.09**     | 57.03 |
| M8: Alternative global context                                  | 66.64             | 51.78             | 56.83                      | -3.96*** | 70.22             | 59.86             | 63.61                      | -3.59***    | 59.48 |
| M9: Reducing training samples by 1/3                            | 68.29             | 50.69             | 55.83                      | -4.96*** | 73.05             | 58.83             | 63.42                      | -3.77***    | 59.75 |

**Note:** Significance levels are indicated as follows: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . <sup>[1]</sup> Without joint learning, global context only acts on sentence classification. Thus, for sequence tagging, M3 and M6 are equivalent (see Figure 1e). <sup>[2]</sup> For sequence tagging, M4 is equivalent to ProtoNER (see Figure 1e).

**Table 15. Ablation Study on EBM-NLP Dataset**

| (A) Sentence classification     |  |                         |                         |                               |          |                         |                         |                               |          |                   |
|---------------------------------|--|-------------------------|-------------------------|-------------------------------|----------|-------------------------|-------------------------|-------------------------------|----------|-------------------|
| Models / metrics <sup>[1]</sup> |  | P@3                     |                         | R@3                           |          | F1@3 & diff.            |                         | PRC & diff.                   |          |                   |
| FastSR full model               |  | 82.14                   |                         | 77.70                         |          | 79.71                   | -                       | 81.54                         |          | -                 |
| M2: Full minus attention        |  | 80.04                   |                         | 77.34                         |          | 78.54                   | -1.16***                | 79.71                         |          | -1.82***          |
| M3: Full minus joint learning   |  | 78.31                   |                         | 77.21                         |          | 77.65                   | -2.06***                | 78.50                         |          | -3.04***          |
| (B) Sequence tagging            |  |                         |                         |                               |          |                         |                         |                               |          |                   |
| Models / metrics                |  | P <sub>word</sub><br>@3 | R <sub>word</sub><br>@3 | F1 <sub>word</sub> @3 & diff. |          | P <sub>BERT</sub><br>@3 | R <sub>BERT</sub><br>@3 | F1 <sub>BERT</sub> @3 & diff. |          | PRC & diff.       |
| FastSR full model               |  | 66.00                   | 68.21                   | 66.08                         | -        | 77.75                   | 73.97                   | 75.49                         | 0        | 71.77<br>-        |
| M3: Full minus attention        |  | 68.16                   | 60.54                   | 62.95                         | -3.13*** | 78.64                   | 68.7                    | 72.92                         | -2.57*** | 69.63<br>-2.14*** |
| M2: Full minus joint learning   |  | 71.18                   | 55.17                   | 61.03                         | -5.05*** | 79.2                    | 66.15                   | 71.69                         | -3.8***  | 69.38<br>-2.38*** |

**Note:** Significance levels are indicated as follows: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

**Figure 3. Models' Sensitivity to Sample Size on EBM-NLP Dataset**

An intriguing question is FastSR's sensitivity to various sample sizes. In the WD dataset, we assessed its sensitivity by reducing the sample size by 1/3 (see model M9 in Tables 13-14). The EBM-NLP dataset afforded us an opportunity to assess its sensitivity to larger sample sizes. We selected the best-performing model in each category, based on our previous evaluation, and trained them on 30 randomly sampled subsets of 200, 300, and 500 training samples. Figure 3 shows the average  $F1_{@3}$  and  $F1_{word@3}$  scores of these models. FastSR consistently outperformed the benchmarks, although the performance gap decreased as the sample size increased, demonstrating the sample efficiency of FastSR and delineating the boundary conditions under which FastSR is effective. We also conducted an error analysis, demonstrating how FastSR's design improves extraction performance, as noted in Appendix H.

## SR Automation

Computational design science research emphasizes integrating ML models into human decision processes, i.e., creating a human-machine process (Berente et al., 2021). Thus, the goal of our qualitative evaluation is to shed light on when, how, and for whom FastSR can expedite SR. We conducted this evaluation in

four aspects: report generation, qualitative feedback, and practical benefits and implications on SR guidelines.

**Report generation:** We demonstrated FastSR's efficacy by comparing its SR reports for the WD dataset with those from leading benchmark methods and SR experts. Using top-one sentences from FastSR and ProtoNet and manually extracted PICO elements, we generated four comparative plots (Figure 4). Figure 4a shows the distribution of studies by WD manifestations (P element), and Figure 4b presents the distribution of interventions. These figures reveal FastSR's closer alignment with SR expert-generated plots, unlike ProtoNet, which missed numerous articles in each category, affecting the completeness, validity, and recency of medical evidence. Figure 4c focuses on WD study sample sizes in China. SR experts found eight studies (sample sizes 1 to 691), shown in blue bubbles. FastSR retrieved seven, while ProtoNet only three, missing the pivotal 2015 study with a sample size of 691. Figure 4d examines trientine studies, an alternative treatment for WD patients intolerant of penicillamine. ProtoNet missed the highest-quality evidence from a clinical trial, whereas FastSR identified diverse study types. This comparison illustrates FastSR's proficiency in retrieving key studies and providing comprehensive, current, and high-quality evidence in specialized SR domains.

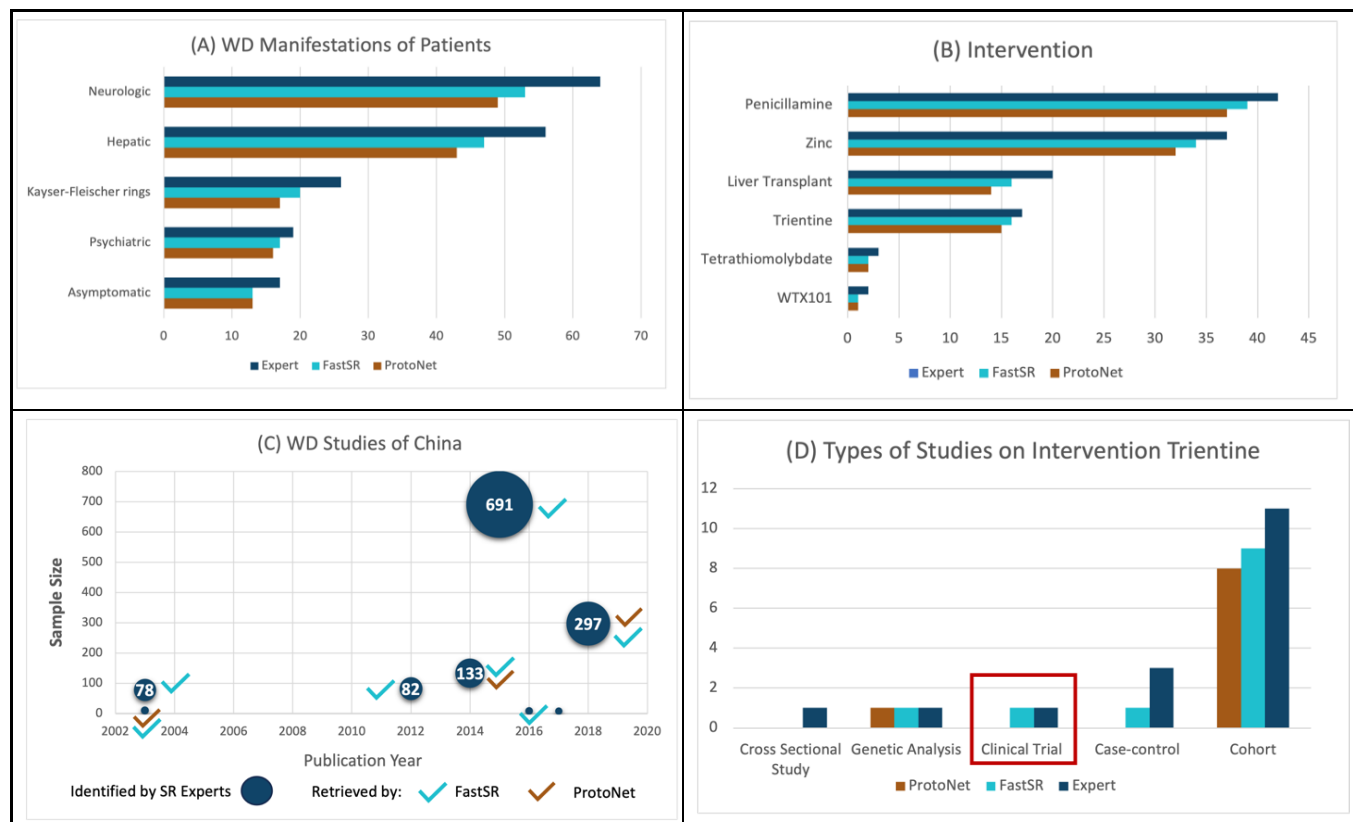


Figure 4. Retrieved Research Articles per PICO Elements on WD Dataset

| Table 16. Estimates of Time Reduction by FastSR |                                                                                                                                                                                                                                                   |                                        |                                         |                                         |        |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|-----------------------------------------|-----------------------------------------|--------|
| Breakdown tasks                                 | Explanation                                                                                                                                                                                                                                       | Estimated time (hours)                 |                                         |                                         |        |
|                                                 |                                                                                                                                                                                                                                                   | FastSR                                 | ProtoNet                                | CNN                                     | Manual |
| 1. Data annotation                              | Assuming each article contains three instances of a PICO element, and two SR experts read 30 articles to gather sufficient labeled samples for training.                                                                                          | 60<br>(~ 30 articles)                  | 60<br>(~ 30 articles)                   | 60<br>(~ 30 articles) <sup>[1]</sup>    | N/A    |
| 2. Model training                               | Experts tune ML/DL models and scan the entire literature to retrieve PICO data.                                                                                                                                                                   | ~ 40 (5 days)                          |                                         |                                         |        |
| 3. Validation of retrieved PICO elements        | Given the WD test set of 110 articles, two SR experts perform data extraction for one article, leading to 220 hours. For ML solutions, one SR expert validates the top three sentences of each PICO class retrieved by each model. <sup>[1]</sup> | 10<br>(~1200 sentences ≈ 10 articles)  | 12<br>(~1500 sentences ≈ 12 articles)   | 13<br>(~1600 sentences ≈ 13 articles)   | 220    |
| 4. Resolving false predictions                  | Resolving a false prediction would require SR experts the same amount of time as screening a full article (Shemilt et al., 2016). We assume that a false top-1 prediction necessitates full article screening by one expert.                      | 43<br>(43 articles, P@1: 81% R@1: 80%) | 58<br>(58 articles, P@1: 77%, R@1: 70%) | 63<br>(63 articles, P@1: 79%, R@1: 64%) | NA     |
| Total                                           | 110 test articles                                                                                                                                                                                                                                 | 153                                    | 170                                     | 176                                     | 220    |
|                                                 | Imputed time for the entire project, 550 articles                                                                                                                                                                                                 | 365                                    | 450                                     | 480                                     | 1100   |
|                                                 | Reduction on the total SR project time by FastSR compared to benchmarks                                                                                                                                                                           | -                                      | ~20%                                    | ~25%                                    | ~65%   |

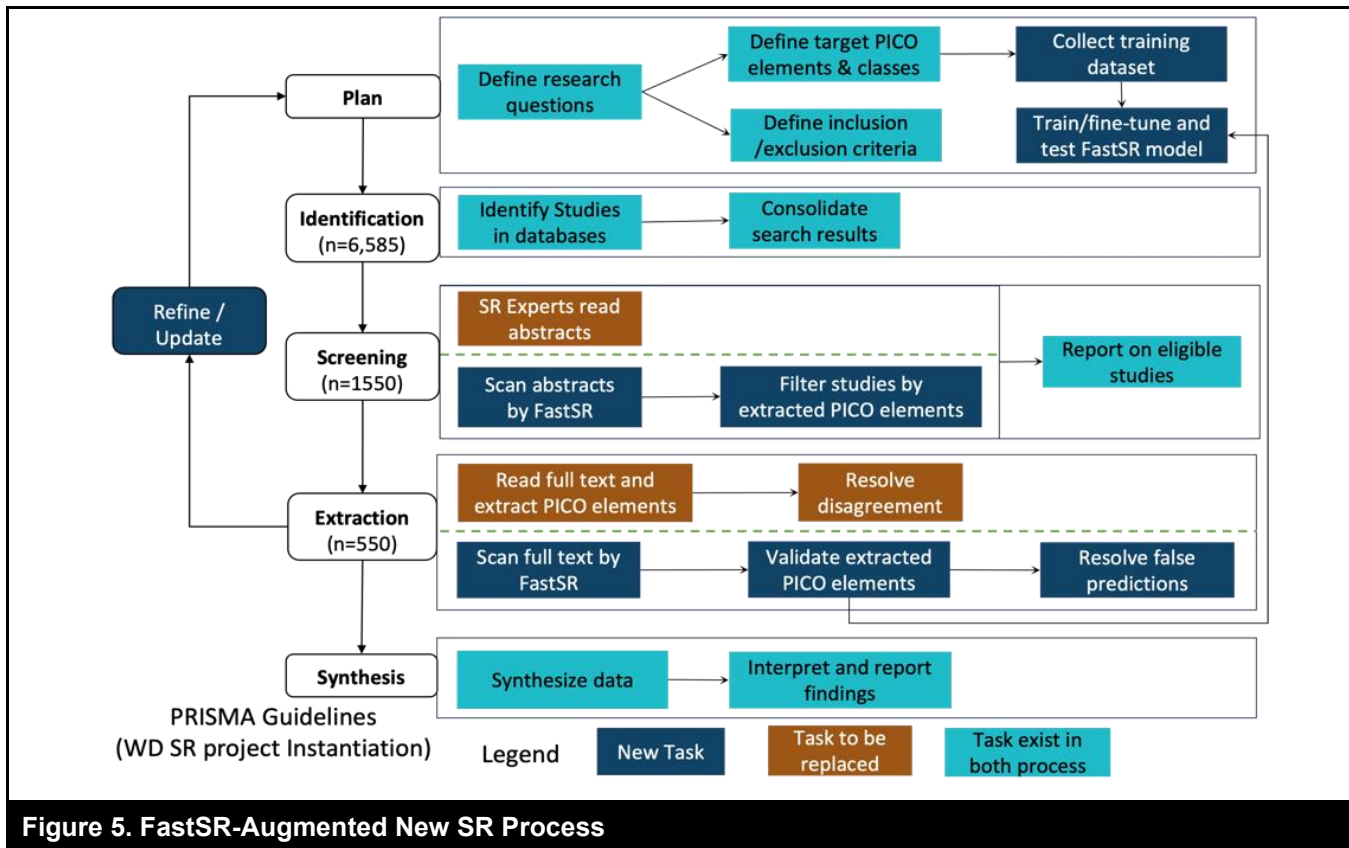
**Note:** Ideally, a few hundred articles should be annotated for improved performance, as suggested by (Kitchens et al., 2018). <sup>[1]</sup> Due to the collocation of multiple PICO elements in one sentence, the number of unique sentences may vary by ML solutions.

**Qualitative feedback:** We then conducted interviews with SR experts who have interacted with FastSR to assess FastSR's *importance*, *accessibility*, and *suitability* (Lukyanenko et al., 2019; Rosemann & Vessey, 2008). We summarize part of the qualitative feedback in Appendix Table F1. Specifically, FastSR was considered *important* and *useful*, especially for PICO element extraction: "Remarkably, FastSR identified a few sentences that could be overlooked in a condensed synopsis." They also mentioned its significant impact on evidence synthesis: "With significant time savings in PICO extraction, I can focus more on summarizing and visualizing findings from multiple articles." In terms of *accessibility*, they applauded the simplicity of presenting sentences of PICO elements but suggested that a user interface would allow easier navigation across PICO elements and articles. Regarding *suitability*, the participants felt that FastSR is particularly suitable for time-sensitive SR tasks (e.g., COVID-19). They expressed a desire to test this tool in their upcoming SR projects. They did, however, recommend a thorough revision of the SR protocol to guide the inclusion of automation tools into SR processes.

**Economic benefits due to time reduction in SR data extraction:** We estimated the economic benefits of FastSR-augmented SR using the WD SR project as a case study. Table 16 outlines the specific extraction steps and time required for four methods: FastSR, the best FSL method (ProtoNet), the

top traditional DL method (CNN), and manual SR. Our estimates are based on a project with 550 articles, averaging 120 sentences per article, an annotation time of one hour per article (Nussbaumer-Streit et al., 2021), and an SR expert cost of \$100 per hour. Manual SR typically requires at least two annotators for data extraction, a process referred to as double screening (Higgins et al., 2019). In contrast, for ML-augmented SR, a single screening, where a single expert validates the ML extraction results, is recommended (Shemilt et al., 2016).

In the four-step process in Table 16, data annotation and model training constituted a one-time investment of 100 hours for all three ML-augmented SR methods. The time spent on validation and resolving false predictions varied with model performance. For instance, during validation, while all models retrieved the top three sentences per PICO class, FastSR reduced the number of unique sentences retrieved (1,200 vs. 1,600 sentences) by capturing multiple PICO elements in a single sentence. Time spent on resolving false predictions depends on model performance. We assume that an SR expert would have to screen the full text to see whether the retrieved sentence is incorrect (false positive) or lacks the target PICO element (false negative). With FastSR achieving 81% P@1 and 80% R@1 across 110 test articles, around 43 articles (21 for false positives and 22 for false negatives) would necessitate full-text screening.



**Figure 5. FastSR-Augmented New SR Process**

Overall, this analysis shows that FastSR significantly reduces time and cost due to its efficient automation and superior performance compared to benchmarking models. With single screening, FastSR achieves a 65% time reduction relative to double-screening manual SR, leading to a cost savings of \$73,500 for a project with 550 articles. Even with double screening, FastSR saves about 45% of the time compared to manual SR. Compared to benchmark models, FastSR saves 20-25% time, irrespective of the screening method. This aligns with research indicating that ML solutions can cut SR project workloads by 30%-70% (O'Mara-Eves et al., 2015; Shemilt et al., 2016). Considering the annual publication of over 4,000 SR reports (Grimshaw et al., 2013), effective tools like FastSR could yield substantial economic gains.

**Implications on SR guidelines:** Accelerated data extraction has the potential to transform the conventional five-stage SR process (Page et al., 2021) into an iterative, flexible, and more efficient approach. As outlined in Table 1, the screening stage (screening abstracts to determine article eligibility) and the extraction stage (full-text annotation) are labor-intensive and face challenges due to insufficient support of robust IT artifacts. FastSR's remarkable performance with both abstracts and full-text articles suggests its potential to semi-automate these two stages, as illustrated in Figure 5.

Specifically, during the planning stage, SR experts annotate a limited number of training examples. Utilizing this concise training dataset, FastSR is capable of initially screening abstracts containing PICO elements to determine article eligibility. Subsequently, FastSR can efficiently extract PICO elements from the screened full-text articles. The SR experts can then validate the extracted sentences and text fragments, addressing any false predictions.

While SR experts are essential for validating results and addressing false predictions, this semi-automated approach can significantly reduce the time required for SR because SR experts can focus on the extracted top sentences rather than painstakingly reviewing the full-text articles. Moreover, with the expedited screening and extraction stages, they can further examine research questions and refine search criteria, which are critical for the entire SR project. Finally, this semi-automatic approach also has the potential to mitigate errors and biases that may arise from automation methods. Through validation, SR experts gain insights into the prediction rationale and common errors and biases. This knowledge can be collectively shared with model developers, enabling them to fine-tune FastSR and improve the interpretability of the models.

## Discussion and Conclusion

In this section, we discuss the implications of our design artifacts, the takeaways from designing ML solutions for process automation/augmentation (Padmanabhan et al., 2022), and the generalizability of FastSR.

### Computational design for high-expertise and low-label environments—A deep understanding of problem context:

Many real-world process automation/augmentation environments are rife with high-expertise and limited training data (Padmanabhan et al., 2022). Following Samtani et al. (2023), we tackled this design challenge by first obtaining a thorough understanding of the application environment and then searching for a knowledge base to guide our design. Specifically, based on our review of the literature and expert interviews, we identified four types of task complexity and proposed design requirements to customize FSL to SR automation. Our design framework, artifact, prototype, and evaluation can guide future IS computational design science researchers to address design in a similar environment.

**Making machines learn like humans:** Human sensemaking involves synergistically integrating simple components into semantic meaning. Compositionality theory emphasizes the need to find the appropriate components and interrelations to accomplish this synergistic effect. FastSR utilizes a rich theory-guided domain adaptation (Abbasi et al., 2024) to identify design requirements that delineate the necessary components and their interrelations. We designed multiple representations and prototypes within an FSL framework to implement the design requirements, resulting in a synergistic effect where various design components contributed to the overall operational utility. Overall, our design has implications for managing AI and human-in-the-loop work by delineating how to involve human expertise in human-machine process automation/augmentation environments (Berente et al., 2021).

### Customized evaluation strategy and generalizability:

Evaluation is the cornerstone of design science research. We argue that artifacts geared towards complex sociotechnical environments necessitate a customized and multifaceted evaluation strategy. Accordingly, we developed a comprehensive strategy to assess FastSR's overall operational utility, including experiments that shed light on the accuracy and quality of data extraction (in comparison with state-of-the-art benchmark models), downstream case studies that highlight dashboard decision-making value and potential time savings for experts, and interview-based qualitative feedback on usefulness. Moreover, we conducted detailed ablation analyses on FastSR components to assess the efficacy of satisfying design requirements as an advantageous outcome. Furthermore, by using three datasets, we evaluated the generalizability of

FastSR across multiple disease domains, a wide spectrum of PICO data elements, different text styles (abstract vs. full text), and variations in sample size. Overall, we show that FastSR can substantially reduce SR experts' workload and shift their cognitive load from mechanical data extraction to high-value-added evidence synthesis. We are hopeful our evaluation strategy, which emphasizes design efficacy/validity, generalizability, and boundary conditions, may inform future AI/ML-related design research embedded in sociotechnical environments.

We also acknowledge several limitations of this study, thereby identifying future research opportunities. First, we chose to concentrate on demonstrating generalizability within the medical SR context, leaving the exploration of its application to other domains for future work. Second, FastSR assumes predefined PICO classes established by medical professionals, but future work could focus on developing methods to automatically identify new, disease-specific PICO classes. Third, PICO classes often have a hierarchical structure. For example, the Intervention class can include subcategories like surgical and educational interventions. Future work could explore prototype designs that might better reflect this hierarchy to accommodate small or rare subclasses. Fourth, while ALMs underperform in data extraction tasks compared to BERT-based models, they might be useful for downstream tasks like evidence synthesis. Future work could explore integrating both types of LLMs into different stages of the SR process. Finally, whereas our problem-driven design framework remains stable, individual ML components can be replaced as new technologies emerge. Future research could focus on integrating advanced ML components into our design.

In conclusion, we propose a customized FSL solution with several novel extensions to integrate human annotation logic into a computational framework that requires minimal labeling effort. Our comprehensive, multifaceted evaluation showcases FastSR's advantages in data extraction and SR automation compared to baseline models. Our results have important implications for designing computational artifacts in high-expertise, low-label environments.

## Acknowledgments

The authors extend their gratitude to the Genesis Research Group for providing the training datasets and offering invaluable domain knowledge consultations. We also sincerely thank Yu Hong, Chen Liao, Colin Navickas, Leyan Tang, and Siwei Wang for their meticulous efforts in data labeling and validation. Rong Liu and Jingjing Li contributed equally to this research and are thus co-first authors. Ahmed Abbasi was funded by the National Science Foundation (NSF) (Grants 2240347 and IIS-2039915) and a Kemper Faculty Award.

## References

- Abbasi, A., Parsons, J., Pant, G., Sheng, O. R. L., & Sarker, S. (2024). Pathways for design research on artificial intelligence. *Information Systems Research*, 35(2), 441-459. <https://doi.org/10.1287/isre.2024.editorial.v35.n2>
- Abbasi, A., Sarker, S., & Chiang, R. H. (2016). Big data research in information systems: Toward an inclusive research agenda. *Journal of the Association for Information Systems*, 17(2), i-xxxii. <https://doi.org/10.17705/1jais.00423>
- Banerjee, A., Merugu, S., Dhillon, I. S., Ghosh, J., & Lafferty, J. (2005). Clustering with Bregman divergences. *Journal of Machine Learning Research*, 6(10), 1705-1749.
- Bansal, T., Jha, R., & McCallum, A. (2020a). Learning to few-shot learn across diverse natural language classification tasks. In *Proceedings of the 28th International Conference on Computational Linguistics* (pp. 5108-5123). <https://doi.org/10.18653/v1/2020.coling-main.448>
- Bansal, T., Jha, R., Munkhdalai, T., & McCallum, A. (2020b). Self-supervised meta-learning for few-shot natural language classification tasks. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing* (pp. 522-534). <https://doi.org/10.18653/v1/2020.emnlp-main.38>
- Bao, Y., Wu, M., Chang, S., & Barzilay, R. (2019). Few-shot text classification with distributional signatures. In *Proceedings of the International Conference on Learning Representations*. <https://openreview.net/forum?id=H1emfT4twB>
- Beltagy, I., Lo, K., & Cohan, A. (2019). Scibert: A pretrained language model for scientific text. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing* (pp. 3615-3620). <https://doi.org/10.18653/v1/D19-1371>
- Berente, N., Gu, B., Recker, J., & Santhanam, R. (2021). Managing artificial intelligence. *MIS Quarterly*, 45(3), 1433-1450. <https://doi.org/10.25300/MISQ/2021/16274>
- Borah, R., Brown, A. W., Capers, P. L., & Kaiser, K. A. (2017). Analysis of the time and workers needed to conduct systematic reviews of medical interventions using data from the PROSPERO registry. *BMJ Open*, 7(2), Article 012545. <https://doi.org/10.1136/bmjopen-2016-012545>
- Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., & Askell, A. (2020). Language models are few-shot learners. In *Proceedings of the 34th International Conference on Neural Information Processing Systems* (pp. 1877-1901).
- Chen, Q., Du, J., Hu, Y., Keloth, V. K., Peng, X., Raja, K., Zhang, R., Lu, Z., & Xu, H. (2024a). Large language models in biomedical natural language processing: Benchmarks, baselines, and recommendations. arXiv. <http://arxiv.org/abs/2305.16326>
- Chen, S., Zhang, Y., & Yang, Q. (2024b). Multi-task learning in natural language processing: An overview. *ACM Computing Surveys*, 56(12), 1-32. <https://doi.org/10.1145/3663363>
- Cheng, K.-H., Chou, S.-Y., & Yang, Y.-H. (2019). Multi-label few-shot learning for sound event recognition. In *Proceedings of the 21st IEEE International Workshop on Multimedia Signal Processing*. <https://doi.org/10.1109/MMSP.2019.8901732>
- Clarke, L., Clarke, M., & Clarke, T. (2007). How useful are Cochrane reviews in identifying research needs? *Journal of Health Services Research and Policy*, 12(2), 101-103. <https://doi.org/10.1258/135581907780279648>
- Cohan, A., Dernoncourt, F., Kim, D. S., Bui, T., Kim, S., Chang, W., & Goharian, N. (2018). A discourse-aware attention model for abstractive summarization of long documents. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. <https://doi.org/10.18653/v1/N18-2097>
- Davis, J., & Goadrich, M. (2006). The relationship between precision-recall and ROC curves. In *Proceedings of the 23rd International Conference on Machine Learning* (pp. 233-240). <https://doi.org/10.1145/1143844.11438>
- Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional Transformers for language understanding. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. <https://doi.org/10.18653/v1/N19-1423>
- DeYoung, J., Lehman, E., Nye, B., Marshall, I., & Wallace, B. C. (2020). Evidence inference 2.0: More data, better models. In *Proceedings of the 19th SIGBioMed Workshop on Biomedical Language Processing* (pp. 123-132). <https://doi.org/10.18653/v1/2020.bionlp-1.13>
- Djulgovic, B., & Guyatt, G. H. (2017). Progress in evidence-based medicine: A quarter century on. *The Lancet*, 390(10092), 415-423. [https://doi.org/10.1016/S0140-6736\(16\)31592-6](https://doi.org/10.1016/S0140-6736(16)31592-6)
- Dong, Y., Jiang, X., Liu, H., Jin, Z., Gu, B., Yang, M., & Li, G. (2024). Generalization or memorization: Data contamination and trustworthy evaluation for large language models. In *Findings of the Association for Computational Linguistics* (pp. 12039-12050). <https://doi.org/10.18653/v1/2024.findings-acl.716>
- Dopierre, T., Gravier, C., & Logerais, W. (2021). A neural few-shot text classification reality check. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics* (pp. 935-943). <https://doi.org/10.18653/v1/2021.eacl-main.79>
- Finn, C., Abbeel, P., & Levine, S. (2017). Model-agnostic meta-learning for fast adaptation of deep networks. In *Proceedings of the 34th International Conference on Machine Learning* (pp. 1126-1135).
- Gao, T., Fisch, A., & Chen, D. (2021). Making pre-trained language models better few-shot learners. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing*.
- Grimshaw, J., Craig, J., Tovey, D., & Wilson, M. (2013). The Cochrane collaboration 20 years in. *CMAJ*, 185(13), 1117-1118. <https://doi.org/10.1503/cmaj.131251>
- Guyatt, G., Rennie, D., Meade, M., & Cook, D. (2002). *Users' guides to the medical literature: A manual for evidence-based clinical practice*. AMA Press Chicago.
- He, Z., Wang, Z., Wei, W., Feng, S., Mao, X., & Jiang, S. (2020). A Survey on recent advances in sequence labeling from deep learning models. arXiv. <https://arxiv.org/abs/2011.06727>
- Higgins, J., Thomas, J., Chandler, J., Cumpston, M., Li, T., Page, M., & Welch, V. (2019). *Cochrane handbook for systematic reviews of interventions*. John Wiley & Sons.
- Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735-1780. <https://doi.org/10.1162/neco.1997.9.8.1735>
- Hu, Y., Chen, Q., Du, J., Peng, X., Keloth, V. K., Zuo, X., Zhou, Y., Li, Z., Jiang, X., & Lu, Z. (2024). Improving large language

- models for clinical named entity recognition via prompt engineering. *JAMIA*, 31(9), 1812-1820. <https://doi.org/10.1093/jamia/ocad259>
- Huang, J., Li, C., Subudhi, K., Jose, D., Balakrishnan, S., Chen, W., Peng, B., Gao, J., & Han, J. (2020). *Few-shot Named entity recognition: A comprehensive study*. arXiv. <https://arxiv.org/abs/2012.14978>
- Jones, A. P., Conroy, E., Williamson, P. R., Clarke, M., & Gamble, C. (2013). The use of systematic reviews in the planning, design and conduct of randomised trials: A retrospective cohort of NIHR HTA funded trials. *BMC Medical Research Methodology*, 13(1), 1-6. <https://doi.org/10.1186/1471-2288-13-50>
- Kim, Y. (2014). Convolutional neural networks for sentence classification. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing* (pp. 1746-1751). <https://doi.org/10.3115/v1/D14-1181>
- Kiritchenko, S., de Bruijn, B., Carini, S., Martin, J., & Sim, I. (2010). ExaCT: Automatic extraction of clinical trial characteristics from journal publications. *BMC Medical Informatics and Decision Making*, 10(1), Article 56. <https://doi.org/10.1186/1472-6947-10-56>
- Kitchens, B., Dobolyi, D., Li, J., & Abbasi, A. (2018). Advanced customer analytics: Strategic value through integration of relationship-oriented big data. *Journal of Management Information Systems*, 35(2), 540-574. <https://doi.org/10.1080/07421222.2018.1451957>
- Klein, G., Moon, B., & Hoffman, R. R. (2006). Making sense of sensemaking 1: Alternative perspectives. *IEEE Intelligent Systems*, 21(4), 70-73. <https://doi.org/10.1109/MIS.2006.75>
- Lafferty, J., McCallum, A., & Pereira, F. C. (2001). Conditional random fields: Probabilistic models for segmenting and labeling sequence data. In *Proceedings of the 18th International Conference on Machine Learning* (pp. 282-289).
- Lake, B. M., Salakhutdinov, R., & Tenenbaum, J. B. (2015). Human-level concept learning through probabilistic program induction. *Science*, 350(6266), 1332-1338. <https://doi.org/10.1126/science.aab3050>
- Lee, J., Yoon, W., Kim, S., Kim, D., Kim, S., So, C. H., & Kang, J. (2020). BioBERT: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics*, 36(4), 1234-1240. <https://doi.org/10.1093/bioinformatics/btz682>
- Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., K ttler, H., Lewis, M., Yih, W., & Rock tschel, T. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Proceedings of the 34th International Conference on Neural Information Processing Systems* (pp. 9459-9474).
- Lewis, P., Stenetorp, P., & Riedel, S. (2021). question and answer test-train overlap in open-domain question answering datasets. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics*.
- Li, J., Larsen, K., & Abbasi, A. (2020). TheoryOn: A design framework and system for unlocking behavioral knowledge through ontology learning. *MIS Quarterly*, 44(4), 1733-1772. <https://doi.org/10.25300/MISQ/2020/15323>
- Li, Y., & Caragea, C. (2019). Multi-task stance detection with sentiment and stance lexicons. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing* (pp. 6299-6305). <https://doi.org/10.18653/v1/D19-1657>
- Liu, P., Yuan, W., Fu, J., Jiang, Z., Hayashi, H., & Neubig, G. (2023). Pre-train, prompt, and predict: A systematic survey of prompting methods in natural language processing. *ACM Computing Surveys*, 55(9), 1-35. <https://doi.org/10.1145/3560815>
- Loshchilov, I., & Hutter, F. (2019). Decoupled weight decay regularization. In *Proceedings of the International Conference on Learning Representations*.
- Lukyanenko, R., Parsons, J., Wiersma, Y. F., & Maddah, M. (2019). Expecting the unexpected: Effects of data collection design choices on the quality of crowdsourced user-generated content. *MIS Quarterly*, 43(2), 623-648. <https://doi.org/10.25300/MISQ/2019/14439>
- Luo, R., Sun, L., Xia, Y., Qin, T., Zhang, S., Poon, H., & Liu, T.-Y. (2022). BioGPT: Generative pre-trained Transformer for biomedical text generation and mining. *Briefings in Bioinformatics*, 23(6), bbac409. <https://doi.org/10.1093/bib/bbac409>
- Luong, M. T., Pham, H., & Manning, C. D. (2015). Effective approaches to attention-based neural machine translation. In *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing* (pp. 1412-1421). <https://doi.org/10.18653/v1/D15-1166>
- Maass, W., Parsons, J., Purao, S., Storey, V., & Woo, C. (2018). Data-driven meets theory-driven research in the era of big data: Opportunities and challenges for information systems research. *Journal of the Association for Information Systems*, 19(12), 1253-1273. <https://doi.org/10.17705/1jais.00526>
- Marshall, I., Kuiper, J., & Wallace, B. (2016). RobotReviewer: Evaluation of a system for automatically assessing bias in clinical trials. *JAMIA*, 23(1), 193-201. <https://doi.org/10.1093/jamia/ocv044>
- Marshall, I., & Wallace, B. (2019). Toward systematic review automation: A practical guide to using machine learning tools in research synthesis. *Systematic Reviews*, 8(1), 163. <https://doi.org/10.1186/s13643-019-1074-9>
- Miyato, T., Dai, A. M., & Goodfellow, I. (2017). Adversarial training methods for semi-supervised text classification. In *Proceedings of the 5th International Conference on Learning Representations*. [https://openreview.net/forum?id=r1X3g2\\_xl](https://openreview.net/forum?id=r1X3g2_xl)
- Nussbaumer-Streit, B., Ellen, M., Klerings, I., Sfetcu, R., Riva, N., Mahmi c-Kaknjo, M., Poulentzas, G., Martinez, P., Baladia, E., & Ziganshina, L. E. (2021). Resource use during systematic review production varies widely: A scoping review. *Journal of Clinical Epidemiology*, 139, 287-296. <https://doi.org/10.1016/j.jclinepi.2021.05.019>
- Nye, B., Li, J. J., Patel, R., Yang, Y., Marshall, I., Nenkova, A., & Wallace, B. C. (2018). A corpus with multi-level annotations of patients, interventions and outcomes to support language processing for medical literature. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics* (pp. 197-207). <https://doi.org/10.18653/v1/P18-1019>
- Oliver, A., Odena, A., Raffel, C., Cubuk, E. D., & Goodfellow, I. J. (2018). Realistic evaluation of deep semi-supervised learning algorithms. In *Proceedings of the 32nd International Conference on Neural Information Processing Systems* (pp. 3239-3250).
- O'Mara-Eves, A., Thomas, J., McNaught, J., Miwa, M., & Ananiadou, S. (2015). Using text mining for study identification in systematic reviews: A systematic review of current approaches. *Systematic Reviews*, 4(1), Article 5. <https://doi.org/10.1186/2046-4053-4-5>
- Padmanabhan, B., Fang, X., Sahoo, N., & Burton-Jones, A. (2022). Editor's comments: Machine learning in information systems research. *MIS Quarterly*, 46(1), iii-xix.

- Page, M., Moher, D., Bossuyt, P., Boutron, I., Hoffmann, T., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., & Brennan, S. E. (2021). PRISMA 2020 explanation and elaboration: Updated guidance and exemplars for reporting systematic reviews. *BMJ*, 372, Article 160. <https://doi.org/10.1136/bmj.n160>
- Pelletier, F. J. (1994). The principle of semantic compositionality. *Topoi*, 13(1), 11-24. <https://doi.org/10.1007/BF00763644>
- Peng, C., Yang, X., Chen, A., Smith, K. E., PourNejatian, N., Costa, A. B., Martin, C., Flores, M. G., Zhang, Y., & Magoc, T. (2023). A study of generative large language model for medical research and healthcare. *NPJ Digital Medicine*, 6(1), Article 210. <https://doi.org/10.1038/s41746-023-00958-w>
- Pronovost, P. J. (2013). Enhancing physicians' use of clinical guidelines. *JAMA*, 310(23), 2501-2502. <https://doi.org/10.1001/jama.2013.281334>
- Qiu, R., Wei, X., Zhao, M., Zhong, C., Zhao, C., Hu, J., Li, M., Huang, Y., Han, S., He, T., & Chen, J. (2020). Outcome reporting from protocols of clinical trials of coronavirus disease 2019 (COVID-19): A review. medRxiv. <https://doi.org/10.1101/2020.03.04.20031401>
- Radford, A., Narasimhan, K., Salimans, T., & Sutskever, I. (2018). *Improving language understanding by generative pre-training* (Working paper). Available at [https://cdn.openai.com/research-covers/language-unsupervised/language\\_understanding\\_paper.pdf](https://cdn.openai.com/research-covers/language-unsupervised/language_understanding_paper.pdf)
- Rai, A. (2017). Editor's comments: Diversity of design science research. *MIS Quarterly*, 41(1), iii-xviii.
- Richter, A. N., & Khoshgoftaar, T. M. (2020). Sample size determination for biomedical big data with limited labels. *Network Modeling Analysis in Health Informatics and Bioinformatics*, 9(1), 1-13. <https://doi.org/10.1007/s13721-020-0218-0>
- Rios, A., & Kavuluru, R. (2018a). EMR coding with semi-parametric multi-head matching networks. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. <https://doi.org/10.18653/v1/N18-1189>
- Rios, A., & Kavuluru, R. (2018b). Few-shot and zero-shot multi-label learning for structured label spaces. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, 3132-3142. <https://doi.org/10.18653/v1/D18-1352>
- Rosemann, M., & Vessey, I. (2008). Toward improving the relevance of information systems research to practice: The role of applicability checks. *MIS Quarterly*, 32(1), 1-22. <https://doi.org/10.2307/25148826>
- Russell, D. M., Stefik, M. J., Pirolli, P., & Card, S. K. (1993). The cost structure of sensemaking. In *Proceedings of the INTERACT and CHI Conference on Human Factors in Computing Systems* (pp. 269-276).
- Saito, T., & Rehmsmeier, M. (2015). the precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS One*, 10(3), Article e0118432.
- Samtani, S., Zhu, H., Padmanabhan, B., Chai, Y., Chen, H., & Nunamaker, J. (2023). Deep learning for information systems research. *Journal of Management Information Systems*, 40(1), 271-301. <https://doi.org/10.1080/07421222.2023.2172772>
- Schmidt, L., Olorisade, B. K., McGuinness, L. A., Thomas, J., & Higgins, J. P. (2025). Data extraction methods for systematic review (semi) automation: A living systematic review. *F1000Research*, 10, Article 401. <https://doi.org/10.12688/f1000research.51117.2>
- Schmidt, L., Weeds, J., & Higgins, J. (2020). Data mining in clinical trial text: Transformers for classification and question answering tasks. In *Proceedings of the 13th International Joint Conference on Biomedical Engineering Systems and Technologies* (pp. 83-94). <https://doi.org/10.5220/0008945700830094>
- Seo, M., Kembhavi, A., Farhadi, A., & Hajishirzi, H. (2017). Bidirectional attention flow for machine comprehension. In *Proceedings of the 5th International Conference on Learning Representations*. <https://openreview.net/forum?id=HJ0UKP9ge>
- Shemilt, I., Khan, N., Park, S., & Thomas, J. (2016). Use of cost-effectiveness analysis to compare the efficiency of study identification methods in systematic reviews. *Systematic Reviews*, 5(1), Article 140. <https://doi.org/10.1186/s13643-016-0315-4>
- Smith, V., Clarke, M., Williamson, P., & Gargon, E. (2015). Survey of new 2007 and 2011 Cochrane reviews found 37% of prespecified outcomes not reported. *Journal of Clinical Epidemiology*, 68(3), 237-245. <https://doi.org/10.1016/j.jclinepi.2014.09.022>
- Snell, J., Swersky, K., & Zemel, R. (2017). Prototypical networks for few-shot learning. In *Proceedings of the 31st Conference on Neural Information Processing Systems* (pp. 4080-4090).
- Song, W., Song, Z., Liu, L., & Fu, R. (2020). Hierarchical multi-task learning for organization evaluation of argumentative student essays. In *Proceedings of the 29th International Joint Conference on Artificial Intelligence* (pp. 3875-3881). <https://doi.org/10.24963/ijcai.2020/536>
- Spasic, I., & Nenadic, G. (2020). Clinical text data in machine learning: Systematic review. *JMIR Medical Informatics*, 8(3), Article 17984. <https://doi.org/10.2196/17984>
- van Engelen, J. E., & Hoos, H. H. (2020). A survey on semi-supervised learning. *Machine Learning*, 109(2), 373-440. <https://doi.org/10.1007/s10994-019-05855-6>
- Vinyals, O., Blundell, C., Lillicrap, T., Kavukcuoglu, K., & Wierstra, D. (2016). Matching networks for one shot learning. In *Proceedings of the 30th International Conference on Neural Information Processing Systems* (pp. 3637-3645).
- Vinyals, O., Fortunato, M., & Jaitly, N. (2015). Pointer networks. In *Proceedings of the 29th Conference on Neural Information Processing Systems* (pp. 2692-2700).
- Wang, L., Lo, K., Chandrasekhar, Y., Reas, R., Yang, J., Burdick, D., Eide, D., Funk, K., Katsis, Y., Kinney, R., Li, Y., Liu, Z., Merrill, W., Mooney, P., Murdick, D., Rishi, D., Sheehan, J., Shen, Z., Stilson, B., ... Kohlmeier, S. (2020a). *CORD-19: The COVID-19 Open Research dataset*. arXiv. <https://arxiv.org/abs/2004.10706>
- Wang, Y., Yao, Q., Kwok, J. T., & Ni, L. M. (2020b). Generalizing from a few examples: A survey on few-shot learning. *ACM Computing Surveys*, 53(3), 1-34. <https://doi.org/10.1145/3386252>
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L. and Polosukhin, I., 2017. Attention is all you need. In *Proceedings of the 31st International Conference on Neural Information Processing Systems* (pp. 6000-6060).
- Wiseman, S., & Stratos, K. (2019). Label-agnostic sequence labeling by copying nearest neighbors. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics* (pp. 5363-5369). <https://doi.org/10.18653/v1/P19-1533>
- Xiao, W., & Carenini, G. (2019). Extractive summarization of long documents by combining global and local context. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing* (pp. 3011-3021). <https://doi.org/10.18653/v1/D19-1298>

- Xu, H., Liu, B., Shu, L., & Yu, P. (2019). Open-world learning and application to product classification. In *Proceedings of the World Wide Web Conference* (pp. 3413-3419). <https://doi.org/10.1145/3308558.3313644>
- Yang, X., Song, Z., King, I., & Xu, Z. (2022). A survey on deep semi-supervised learning. *IEEE Transactions on Knowledge and Data Engineering*, 35(9), 8934-8954. <https://doi.org/10.1109/TKDE.2022.3220219>
- Yu, M., Guo, X., Yi, J., Chang, S., Potdar, S., Cheng, Y., Tesauro, G., Wang, H., & Zhou, B. (2018). Diverse few-shot text classification with multiple metrics. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. <https://doi.org/10.18653/v1/N18-1109>
- Zhang, M.-L., Li, Y.-K., Liu, X.-Y., & Geng, X. (2018). Binary relevance for multi-label learning: An overview. *Frontiers of Computer Science*, 12(2), 191-202. <https://doi.org/10.1007/s11704-017-7031-7>
- Zhang, T., Kishore, V., Wu, F., Weinberger, K. Q., & Artzi, Y. (2020). BERTScore: Evaluating text generation with BERT. In *Proceedings of the 8th International Conference on Learning Representations*. <https://openreview.net/forum?id=SkeHuCVFDr>
- Zhang, X., Zhao, J., & LeCun, Y. (2015). Character-level Convolutional Networks for Text Classification. In *Proceedings of the 29th International Conference on Neural Information Processing Systems* (pp. 649-657).
- Zheng, H., & Lapata, M. (2019). Sentence centrality revisited for unsupervised summarization. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics* (pp. 6236-6247). <https://doi.org/10.18653/v1/P19-1628>

## About the Authors

**Rong Liu** is an associate professor of information systems at the College of Business at Florida International University. She has also worked as a research staff member at the IBM T. J. Watson Research Center and as an associate professor at Stevens Institute of Technology. She received her Ph.D. in Business Administration from Penn State University. Her research interests include deep learning, natural language processing, fintech, blockchain, and business process management. She has authored more than 60 journal and conference publications and holds 20 patents. ORCID: <https://orcid.org/0000-0001-7176-1999>

**Jingjing Li** is the Andersen Alumni Associate Professor of Commerce at the McIntire School of Commerce, University of Virginia. She earned her Ph.D. in business administration from the University of Colorado at Boulder. Her research centers on artificial intelligence and big data analytics, with applications in healthcare, marketing, management, and public policy. Dr. Li's work has been published in leading journals, including *MIS Quarterly*, *Information Systems*

*Research*, *Journal of Management Information Systems*, *Journal of Marketing*, *Strategic Management Journal*, *Review of Economics and Statistics*, and *ACM Transactions on Information Systems*. She has received prestigious awards for her research, teaching, and service, such as the INFORMS Design Science Award, CIST Best Paper Award, WITS Best Paper and Prototype Awards, MISQ Outstanding Associate Editor Award, and Poets & Quants Best Undergraduate Professor Award. Her projects have been supported by the U.S. National Science Foundation, Virginia 4-VA Research Grant, Amazon, Google, Microsoft, the Jefferson Trust, and the UVA Data Analytics Center. Before joining McIntire, Dr. Li worked as a scientist at Microsoft, where she developed large-scale machine learning solutions for various flagship products. ORCID: <https://orcid.org/0000-0003-1967-0090>

**Marko Zivkovic** is a theoretical physicist by training who has spent over 14 years advancing innovation and technology in real-world evidence (RWE) generation, optimization, and communication. With extensive expertise in real-world data (RWD) analytics and health economics and outcomes research (HEOR), he has worked with diverse U.S. and global data sources across multiple therapeutic areas to uncover insights that drive impactful change. As the executive vice president (EVP) of Product and Innovation and head of AI Products at Genesis Research Group, an international life sciences consultancy, Dr. Zivkovic leads the development of transformative, AI-powered solutions designed to tackle complex challenges in evidence generation and decision-making in an effective, transparent, and efficient manner. His work emphasizes commitment to blending traditional methodologies with cutting-edge technologies, accelerating time-to-insight, and delivering innovative approaches that shape the future of the biopharmaceutical, biotech, and healthcare industries.

**Ahmed Abbasi** is the Joe and Jane Giovanini Professor of IT, Analytics, and Operations (ITAO) at the Mendoza College of Business of the University of Notre Dame. He received his Ph.D. in information systems from the Artificial Intelligence Lab at the University of Arizona, M.B.A. and B.S. degrees in information technology from Virginia Tech, and an M.S. degree from Columbia University. Ahmed has 20 years of experience pertaining to human-centered analytics. His research has been funded by over a dozen grants from the U.S. National Science Foundation and industry partners such as Amazon Web Services, eBay, Microsoft, and Oracle. He has also received the IEEE Technical Achievement Award, INFORMS Design Science Award, and IBM Faculty Award for his work at the intersection of machine learning and design. Ahmed has published over 100 articles in top journals and conferences and has won AIS Top Publication and MIS Quarterly Best Paper awards. His work has been featured in various media outlets, including *The Wall Street Journal*, *Harvard Business Review*, *Associated Press*, *WIRED*, and CBS. Ahmed serves on the editorial board for various IS, ACM, and IEEE journals. ORCID: <https://orcid.org/0000-0001-7698-7794>

# Appendix A

## PICO Elements of Wilson Disease SR Project

| Table A1. PICO Elements of Wilson Disease SR Project |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                 |                                                                                                                                                                                                           |                                                                                                                                                |                                                                                             |
|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Category                                             | PICO data elements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Example PICO elements extracted from the literature                                                                             |                                                                                                                                                                                                           |                                                                                                                                                |                                                                                             |
| Study & patient characteristics (P, I, C)            | First author, year of publication, title, data source type, data source name, study design, study period, study length, country, region, study sponsor, population definition, study sample size, intervention, intervention sample size, intervention dose/schedule, comparator, comparator sample size, comparator dose/schedule, age at enrollment, age at Wilson disease onset/diagnosis, duration of wilson disease, sex, ethnicity, race, wilson disease manifestations, genotype, baseline clinical measures, other measures | <b>Population definition</b>                                                                                                    | <b>Intervention dose/schedule</b>                                                                                                                                                                         | <b>Wilson disease manifestations</b>                                                                                                           | <b>Baseline clinical measures</b>                                                           |
|                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | WD patients                                                                                                                     | Group 1: DPA 2-5 mg/kg/day, zinc sulfate maintained at a single daily dose of 50-75 mg after initial period of dose of 75150 mg/day; Group 2 and 3: DPA max 7 mg/kg/day; zinc sulfate 150-225 mg/day; ... | Hepatomegaly: Asymptomatic: 81.5%; Hepatic: 62.9%; Combined: 55.6%; Neurological: 33.3% Splenomegaly: G1: 62.9%; G3: 66.6%; Aymptomatic: 33.3% | Positive Family history: 56/72 (77.8%) Elevated urinary copper excretion: 19/25 (84.2%) ... |
| Burden of illness (P)                                | Incidence, prevalence, mortality, other measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Incidence</b>                                                                                                                | <b>Prevalence</b>                                                                                                                                                                                         | <b>Mortality</b>                                                                                                                               |                                                                                             |
|                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | The average annual incidence rate was 1.44 per million person-years                                                             | The average annual prevalence was 17.93 per million                                                                                                                                                       | Death: 26 (12.3%) The median time from diagnosis to death for these 26 patients was 4.2 years (IQR 2.1 to 5.0 years).                          |                                                                                             |
| Efficacy (O)                                         | liver endpoints, biomarkers, neurological measures, psychiatric measures, Kayser Fleischer rings, 24-hour urinary copper excretion, spot urinary copper, Clinical Global Impression scales, other measures                                                                                                                                                                                                                                                                                                                          | <b>Liver transplantation</b>                                                                                                    | <b>Serum non-ceruloplasmin copper concentration</b>                                                                                                                                                       | <b>Hemoglobin level (Hg)</b>                                                                                                                   | <b>Alanine aminotransferase (ALT)</b>                                                       |
|                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Graft survival (%): Children: 1 year: Fulminant: 87.5; Chronic: 100 p=0.25; 5 year: Fulminant: 82.5; Chronic: 100 p=0.25        | Post-transplant values of serum copper were relatively low (mean=74.8 6 15.7 µg/dL);                                                                                                                      | Mean Hg: Start of Zinc therapy: Normal dose: 150; Higher dose: 129; Pvalue: 0.01 ...                                                           | Mean=50.89±32.29 U/L, p=0.448                                                               |
| Safety (O)                                           | Safety measures including adverse events, serious adverse events, nephrotoxicity measures, treatment compliance, discontinuation, other measures                                                                                                                                                                                                                                                                                                                                                                                    | <b>Adverse events</b>                                                                                                           | <b>Serious adverse events</b>                                                                                                                                                                             | <b>Serum creatinine (SCr)</b>                                                                                                                  | <b>Blood urea nitrogen (BUN)</b>                                                            |
|                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Treatment with WTX101 was generally well tolerated with most reported adverse events being mild (grade 1) to moderate (grade 2) | Death: All patients: NR; hepatic presentation: 12 (9); Neurologic presentation: 5 (8); Asymptomatic siblings: NR; P=ns                                                                                    | 0-1 yrs follow up: Mean 0.8 ± 0.2 mg/dL (N=9); 1-2 yrs follow up: Mean 0.9 ± 0.2 mg/dL (N=10); ...                                             | N (mg/dl) Year 1: 33; Mean: 10.4 Year 2: 16; Mean: 11.0 ...                                 |

|                                                          |                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                |                                                                                                                                                      |                                                                                                                                                           |                                                                                                                                                               |
|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Quality of life and patient reported outcomes (O)</b> | PHQ-9, UWDRS, SF-36, GAS-WD, MMAS8, TSQM, MMSE, HAMD, WHO-QoL-BREF, NSS, DRS, WAIS, RAVLT, FAS, WMS, Benton Visual Retention Test, Raven's Progressive Matrices, Wisconsin Card Sorting Test, Digit Span Memory Test, Boston Naming Test, Verbal Fluency Test, Trail Making Test, Intelligence Structure, Perceptual Speed Test, Perceptual Maze Test, other assessments | <b>Personal Health Questionnaire-9 Depression Scale (PHQ-9)</b>                                                                | <b>Unified WD Rating Scale Part II (UWDRS)</b>                                                                                                       | <b>Short Form-36 Health Survey questionnaire (SF-36)</b>                                                                                                  | <b>Global Assessment Scale for Wilson Disease (GAS for WD)</b>                                                                                                |
|                                                          |                                                                                                                                                                                                                                                                                                                                                                          | All WD patients showed mild depressive symptoms                                                                                | Patients showed continued improvements in mean UWDRS disability score (2.2 vs 4.1)                                                                   | Mean (SD):<br>PF (physical function): 81.1 (26.8)<br>RF (role functioning physical): 68.8 (41.3) ...                                                      | Used for evaluation of neurological involvement but results are not presented                                                                                 |
| <b>Health economics (P)</b>                              | Health economic measures including: healthcare resource utilization, costs, budget impact, cost-effectiveness, and quality-adjusted life years                                                                                                                                                                                                                           | <b>HCRU estimate</b>                                                                                                           | <b>Cost measure</b>                                                                                                                                  | <b>Budget impact and estimate</b>                                                                                                                         | <b>Cost-effectiveness measure</b>                                                                                                                             |
|                                                          |                                                                                                                                                                                                                                                                                                                                                                          | Median duration of hospitalization was 11 days                                                                                 | Length of stay coincided with an average cost of care increase of 113% [adjusted for inflation, $P < 0.001$ ]                                        | Cost pressure on NHS England: Year 1 £4,008.1k; Year 2 £4,008.1k; ...                                                                                     | There is no published evidence of cost effectiveness                                                                                                          |
| <b>Diagnostic guidelines (I)</b>                         | Criteria for diagnosis, biomarkers, differential diagnosis, misdiagnoses, timing of diagnosis                                                                                                                                                                                                                                                                            | <b>Criteria</b>                                                                                                                | <b>Biomarkers</b>                                                                                                                                    | <b>Differential diagnoses</b>                                                                                                                             | <b>Common misdiagnoses</b>                                                                                                                                    |
|                                                          |                                                                                                                                                                                                                                                                                                                                                                          | MRI Assessment Scale; tiers 1-3 graded MRI changes from copper toxicosis while tier 4 graded changes from manganese deposition | CuEXC and REC reference ranges established for 6 different age groups; important pre-requisite for using in diagnosis and monitoring of WD treatment | It appears that the physician may be able to use 24-hr urine copper excretion, nonceruloplasmin plasma copper, and copper uptake as adequate tools ...    | Mental retardation N=2; hysteria N=1; personality disorder N=1; frontal lobe syndrome N=1; multiple sclerosis N=1; cryptogenic cirrhosis N=2; viral hepatitis |
| <b>Treatment guidelines (I)</b>                          | Treatment recommended, recommending organization, guideline or recommendation, goal of therapy, line of treatment                                                                                                                                                                                                                                                        | <b>Intervention</b>                                                                                                            | <b>Recommending organization</b>                                                                                                                     | <b>Guideline/ recommendation</b>                                                                                                                          |                                                                                                                                                               |
|                                                          |                                                                                                                                                                                                                                                                                                                                                                          | Tetrathiomolybdate                                                                                                             | European Medicines Agency CHMP assessment report                                                                                                     | During pregnancy, drugs should be administered twice daily (DPA 600 mg/d, trientine 600 mg/d, zinc 100 mg/d) and treatments should never be discontinued. |                                                                                                                                                               |
| <b>Genetics (P)</b>                                      | <i>Genetic characterization, genotype/phenotype correlation</i>                                                                                                                                                                                                                                                                                                          | <b>Genetic characterization</b>                                                                                                |                                                                                                                                                      | <b>Genotype-phenotype correlation</b>                                                                                                                     |                                                                                                                                                               |
|                                                          |                                                                                                                                                                                                                                                                                                                                                                          | Among these 9 out 23 patients had a novel mutation in exon 21 (Gln1399Arg)                                                     |                                                                                                                                                      | Mutation: Associated manifestation [c.51+4A>T]/[c.1934T>G]: Neurological, ...                                                                             |                                                                                                                                                               |

## Appendix B

### Summary of Annotation Process for COVID Dataset

Three people were recruited to work on tagging the annotation, reviewing the consensus, and summarizing the working contents. Four categories were included for the annotation regarding COVID-19 publication: risk factor (train and test), population, sample size, and country, which came from different sentences and paragraphs.

Annotator 1, Senior Biostatistician / Senior Investigator from the Department of Aging and Health in the School of Medicine of Johns Hopkins University, worked on tagging the annotation for 84 sentences of risk factors (train) and 142 sentences of risk factors (test) as well as reviewing and summarizing.

Annotator 2, Ph.D. student and graduate research assistant from the Department of Health Behavior and Policy in the School of Medicine of Virginia Commonwealth University worked on tagging the annotation for 84 sentences of risk factors (train), 122 sentences of population, 110 sentences of sample size, and 131 sentences of country.

Annotator 3, a Data Analyst from CFG Merchant Solutions, worked on tagging the annotation for 142 sentences of risk factors (test), 122 sentences of population, 110 sentences of sample size, and 131 sentences of country.

The working process is listed below:

- For each category, two people tagged the annotation for comparison, and comments were added for those with issues.
- For the tagged results, Annotator 1 reviewed the matching as well as the comments.
- After the review and correction, Annotator 1 created an EXCEL including the annotation with issues/comments.
- Annotator 1 summarized all the information, including category, total number of records, number of records with comments/issues, matching percentage, etc., in the table below.

| Table B1. Annotation Summary |          |             |             |             |                       |               |
|------------------------------|----------|-------------|-------------|-------------|-----------------------|---------------|
| Category                     | #Records | Annotation  |             | Review      | # Issues/<br>comments | Matching<br>% |
| Risk Factor (Train)          | 84       | Annotator 1 | Annotator 2 | Annotator 3 | 1                     | 86.00%        |
| Risk Factor (Test)           | 142      | Annotator 1 | Annotator 3 | Annotator 2 | 3                     | 84.00%        |
| Population                   | 122      | Annotator 2 | Annotator 3 | Annotator 1 | 10                    | 86.07%        |
| Sample Size                  | 110      | Annotator 2 | Annotator 3 | Annotator 1 | 16                    | 95.45%        |
| Country                      | 131      | Annotator 2 | Annotator 3 | Annotator 1 | 1                     | 84.73%        |

## Appendix C

### Pretrained Model for Sentence Global Context

We trained the model (in Figure 2B) with PubMed sections as inputs and their corresponding section titles as targets. There is a total of 13 frequently used section titles. For computation efficiency, we only took up to the first 20 sentences for each section and a maximum of 40 words for each sentence. In total, we kept 50,000 words. For each BiLSTM layer, we used 200 memory units. We trained the model with a sample of 56,000 sections and tested it on about 14,000 sections. The model achieved outstanding performance, with an average F1 score of 91% across section titles, as shown in Table C1.

| Table C1. Broad Context Model Performance |             |             |             |         |
|-------------------------------------------|-------------|-------------|-------------|---------|
| Section titles                            | Precision   | Recall      | F1-score    | Support |
| Introduction                              | 0.93        | 0.90        | 0.91        | 2000    |
| Results and discussion                    | 0.90        | 0.90        | 0.90        | 2000    |
| Conclusion                                | 0.88        | 0.91        | 0.89        | 2000    |
| Methods                                   | 0.90        | 0.88        | 0.89        | 2000    |
| Case report                               | 0.97        | 0.98        | 0.97        | 2000    |
| Disclosure                                | 0.98        | 0.97        | 0.98        | 2000    |
| Supplementary material                    | 0.91        | 0.73        | 0.81        | 215     |
| Limitations                               | 0.88        | 0.90        | 0.89        | 402     |
| Objectives                                | 0.94        | 0.85        | 0.89        | 443     |
| Abbreviations                             | 0.96        | 0.94        | 0.95        | 148     |
| Experiments                               | 0.85        | 0.79        | 0.82        | 553     |
| Figures and tables                        | 0.74        | 0.69        | 0.71        | 137     |
| Measures                                  | 0.82        | 0.81        | 0.81        | 165     |
| <b>Average</b>                            | <b>0.92</b> | <b>0.91</b> | <b>0.91</b> |         |

# Appendix D

## Performance Metric Calculation

*Article retrieval effectiveness by top sentences.* The objective of sentence classification is to retrieve relevant articles (containing at least one relevant sentence) for each PICO data element (Kiritchenko, et al., 2010). For each article, once an element-bearing sentence is found for a PICO element (e.g., sample size), there is no need for SR experts to continue examining other sentences for the same element. Thus, an SR automation tool only needs to recommend a few sentences ( $N \leq 3$ ) to retrieve the PICO elements of a class. We adopted top- $N$  precision ( $P@N$ ), recall ( $R@N$ ), and F1 ( $F1@N$ ) to evaluate the accuracy of the recommended sentences. Specifically, let  $ST_N^j$  be the top- $N$  sentences by prediction probabilities  $ps^j$  ( $ps^j \geq 0.5$ , the default probability threshold), and  $ST_{true}^j$  be the true sentences for class  $j$  in an article. We calculate  $P@N = \frac{|ST_N^j \cap ST_{true}^j|}{\min(N, |ST_N^j|)}$ ,  $R@N = \frac{|ST_N^j \cap ST_{true}^j|}{\min(N, |ST_{true}^j|)}$ , and  $F1@N = \frac{2 * P@N * R@N}{(P@N + R@N)}$  for each article.<sup>11</sup> In addition, we also calculate the area under the precision-recall curve (PRC), a threshold invariant metric. We prefer PRC over AUC because each article only has a few element-bearing sentences, and AUC can be inflated for highly imbalanced datasets (Davis & Goadrich, 2006).

*Sequence Tagging.* Once top- $N$  sentences are identified, we proceed to extract PICO text fragments by the probability  $pw$  generated for each word in the sentences (see Equation 15). The widely used metric for text fragment extraction is F1 scores (Schmidt et al., 2020), which measures the word overlap between tagged fragments ( $\hat{f}$ ) in top- $N$  sentences and corresponding ground-truth fragments ( $f$ ).<sup>12</sup> The precision is defined as  $P_{word}@N = \frac{|f \cap \hat{f}|}{|\hat{f}|}$ , i.e., the number of overlapping words between  $f$  and  $\hat{f}$  divided by the total number of words in  $\hat{f}$ . The recall is calculated as  $R_{word}@N = \frac{|f \cap \hat{f}|}{|f|}$ , and  $F1_{word}@N = \frac{2 * P_{word}@N * R_{word}@N}{(P_{word}@N + R_{word}@N)}$ . For example, if  $f =$  “pediatric patients with WD” and  $\hat{f} =$  “Patients with WD who were under 14 years old”,  $P_{word}@1 = \frac{3}{9}$ ,  $R_{word}@1 = \frac{3}{4}$ ,  $F1_{word}@1 = \frac{6}{13}$ . Although  $f$  and  $\hat{f}$  have similar meanings, their F1 score is less than 0.5. Similarly, we also report the overall PRC metric for this task.

As illustrated by the previous example, the lexical-based F1 score is not an appropriate evaluation metric because PICO fragment tagging prioritizes semantic relevance over lexical matching. To address this issue, BERTScore has been proposed to evaluate the semantic equivalence between two sequences (Zhang et al., 2020). Let  $f = \{w_1, w_2, \dots, w_M\}$  and  $\hat{f} = \{\hat{w}_1, \hat{w}_2, \dots, \hat{w}_{\hat{M}}\}$  be two sequences of normalized BERT word embeddings. The precision is defined as  $P_{BERT}@N = \frac{1}{|\hat{f}|} \sum_{\hat{w}_j \in \hat{f}} (\max_{w_i \in f} w_i^T \hat{w}_j)$ , recall as  $R_{BERT}@N = \frac{1}{|f|} \sum_{w_i \in f} (\max_{\hat{w}_j \in \hat{f}} w_i^T \hat{w}_j)$ , and  $F1_{BERT}@N = \frac{2 * P_{BERT}@N * R_{BERT}@N}{(P_{BERT}@N + R_{BERT}@N)}$ , where  $w_i^T \hat{w}_j$  measures the cosine similarity between  $w_i$  and  $\hat{w}_j$ . BERTScore seeks to optimally match each embedding with the most similar one in the other sequence. In the previous example, “pediatric” can be best matched with “patient,” with a cosine similarity of 0.86, and the best match of “14” is “pediatric,” with a similarity of 0.66. As a result, we obtain  $P_{BERT}@1 = 0.81$ ,  $R_{BERT}@1 = 0.94$ ,  $F1_{BERT}@1 = 0.87$ , which are more aligned with SR experts’ annotation. We adopted both word- and semantic-based metrics for tagging evaluation.

<sup>11</sup> The recommendation literature usually defines  $P@N = \frac{|ST_N^j \cap ST_{true}^j|}{N}$  and  $R@N = \frac{|ST_N^j \cap ST_{true}^j|}{|ST_{true}^j|}$ . As the number of true sentences varies by article, the metrics can have different ranges. For example, when  $|ST_{true}^j| = 1$ ,  $R@N$  is capped to 1; when  $|ST_{true}^j| = 5$ ,  $R@N$  is capped to  $N/5$ . This makes it difficult to measure a model’s average retrieval capability across articles. Thus, we use truncated  $P@N$  and  $R@N$ , which are ranged in  $[0, 1]$  for all articles.

<sup>12</sup>  $f$  is the set of labeled fragments from up to  $N$  true sentences that are best matched with  $ST_N^j$ .

# Appendix E

## Description of Model Implementation

All the deep learning models were implemented using PyTorch. The detailed parameter setting is shown in Table E1. The maximum length of a sentence was 100 words. Following Devlin et al. (2018), we used the average hidden states of the last four layers of BioBERT as our word embedding with a dimension of 768. For CNN units, we chose four filter sizes with 20 filters each size:  $[1, 3, 5, 7] \times 100$ . The hidden states of BiLSTM layers and the linear projection layers were set to 100. For FastSR, the  $k$ -shot was set to 5. The impact of  $k$  on sentence classification is in Table E2. The batch size was 32. A 4-fold cross-validation was applied to these baseline models for hyperparameter tuning. We used batch normalization, dropout, and early stopping to avoid overfitting. With NVIDIA RTX A4500 20G GPU, FastSR can be trained within about 30 minutes. For most of the baseline models, we adopted the original implementations provided by the authors, as listed in Table E3. The pseudocode of FastSR can be found in an anonymously shared online folder: [https://osf.io/tsdza/?view\\_only=1c0507e9e88d4bce8f7b63c530880e78](https://osf.io/tsdza/?view_only=1c0507e9e88d4bce8f7b63c530880e78)

**Table E1. Parameter Settings in Experiments across Three Datasets**

| Parameter                             | Value                                                                                                         | Applied models                                                            |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Sentence length                       | 100                                                                                                           | All                                                                       |
| Pretrained embedding                  | BioBERT                                                                                                       | All deep learning models unless stated otherwise                          |
| CNN layer filter sizes and number     | [1, 3, 5, 7], 20                                                                                              | CNN, LSTM + CNN, all FSL-based models                                     |
| BiLSTM layer hidden states            | 100 for each direction                                                                                        | LSTM, LSTM + CNN, BiLSTM tagger, BiLSTM + CRF tagger                      |
| Linear layer projection               | 100                                                                                                           | BiLSTM tagger, BiLSTM + CRF tagger, FastSR (tagging), ProtoNER, NN tagger |
| K-shot                                | 5                                                                                                             | FastSR, ProtoNet, Matching Net, ProtoNER, NN tagger                       |
| Batch size                            | 32                                                                                                            | All deep learning models                                                  |
| Multi-label classification conversion | One-vs-the-rest                                                                                               | All models                                                                |
| Regularization                        | Drop out, Batch Normalization                                                                                 | All deep learning models                                                  |
| Distance measure                      | L2: ProtoNet, ProtoNER;<br>Cosine distance: Matching Net, Nearest Neighbor (NN)                               |                                                                           |
| Unlabeled dataset for VAT (SSL)       | WD: Sections with all labeled sentences removed<br>EBM-NLP: All sampled sentences except the training subset. |                                                                           |
| Global context module                 | Enabled for FastSR trained on WD and COVID, disabled for EBM-NLP dataset                                      |                                                                           |

**Table E2. Sensitivity analysis of key parameters K on WD Sentence Classification**

| K-shot                       | 2     | 3    | 4     | 5     | 6     | 7     |
|------------------------------|-------|------|-------|-------|-------|-------|
| Sentence classification F1@1 | 77.44 | 77.5 | 77.65 | 79.42 | 77.73 | 78.71 |

**Table E3. Model Implementation References**

|                                               |                                                                                                                                                                                                                                                       |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ProtoNet                                      | <a href="https://github.com/jakesnell/prototypical-networks/tree/master">https://github.com/jakesnell/prototypical-networks/tree/master</a>                                                                                                           |
| ProtoNER                                      | <a href="https://github.com/few-shot-NER-benchmark/BaselineCode/tree/main/src">https://github.com/few-shot-NER-benchmark/BaselineCode/tree/main/src</a>                                                                                               |
| Matching network                              | <a href="https://github.com/gitabcworld/MatchingNetworks">https://github.com/gitabcworld/MatchingNetworks</a>                                                                                                                                         |
| Nearest neighbor                              | <a href="https://github.com/swiseman/neighbor-tagging">https://github.com/swiseman/neighbor-tagging</a>                                                                                                                                               |
| VAT (SSL)                                     | <a href="https://github.com/WangJiuniu/adversarial_training">https://github.com/WangJiuniu/adversarial_training</a>                                                                                                                                   |
| MAML                                          | <a href="https://github.com/dragen1860/MAML-Pytorch">https://github.com/dragen1860/MAML-Pytorch</a>                                                                                                                                                   |
| FSL_DS                                        | <a href="https://github.com/YujiaBao/Distributional-Signatures">https://github.com/YujiaBao/Distributional-Signatures</a>                                                                                                                             |
| L2AC                                          | <a href="https://github.com/howardhsu/Meta-Open-World-Learning">https://github.com/howardhsu/Meta-Open-World-Learning</a>                                                                                                                             |
| CRF                                           | <a href="https://github.com/sgrvinod/a-PyTorch-Tutorial-to-Sequence-Labeling">https://github.com/sgrvinod/a-PyTorch-Tutorial-to-Sequence-Labeling</a>                                                                                                 |
| Fine-tuned BioBERT                            | <a href="https://huggingface.co/docs/transformers/training">https://huggingface.co/docs/transformers/training</a>                                                                                                                                     |
| Document summarization by sentence centrality | <a href="https://github.com/mswellhao/PacSum">https://github.com/mswellhao/PacSum</a><br>Instead of training sentence embedding, we applied SentenceBERT to generate sentence embedding ( <a href="https://www.sbert.net">https://www.sbert.net</a> ) |
| Hierarchical BiLSTM                           | <a href="https://github.com/armancohan/long-summarization">https://github.com/armancohan/long-summarization</a>                                                                                                                                       |

## Appendix F

### Qualitative Feedback for FastSR Sentence and Fragment Length Summary Statistics ■

**Table F1. Summary of the Qualitative Feedback for FastSR**

| Applicability | Quotes Related to FastSR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Importance    | <ul style="list-style-type: none"> <li>• “I no longer need to read through the entire article to summarize the relevant PICO elements. I can immediately locate the information I’m seeking based on FastSR’s sentences.”</li> <li>• “Remarkably, FastSR identified a few sentences that could be overlooked in a condensed synopsis.”</li> <li>• “This tool is a great help for the time-consuming element extraction process. The tool can highlight multiple PICO data elements hidden in a sentence so I can check them all at once.”</li> <li>• “With significant time savings in PICO extraction, I can focus more on summarizing and visualizing findings from multiple articles.”</li> </ul> |
| Accessibility | <ul style="list-style-type: none"> <li>• “For each identified sentence, the tool shows the paragraph encompassing it and indicates the most likely section where it appears. Such information helps me validate the sentence.”</li> <li>• “This tool can present the results in multiple views: multiple sentences per element, multiple elements per sentence, or aggregated elements per article. It would be valuable if a user interface could render these views.”</li> </ul>                                                                                                                                                                                                                   |
| Suitability   | <ul style="list-style-type: none"> <li>• “The transfer learning capability can really jumpstart new SR projects. I want to use the tool to scan articles in a new literature domain to have a quick view of study design, sample size, intervention, and other aspects of the literature.”</li> <li>• “The tool can reflect the iterative annotation that I usually perform for SR. With an initial scan of the literature, I can validate the sentences and fine-tune the annotation until I’m happy with the results.”</li> </ul>                                                                                                                                                                  |

**Table F2. Sentence and Fragment Length (Words) Summary Statistics**

|                    | Dataset | Count | Mean  | SD    | Min | 25% | 50% | 75% | Max |
|--------------------|---------|-------|-------|-------|-----|-----|-----|-----|-----|
| Training sentences | WD      | 660   | 22.24 | 14.43 | 4   | 12  | 19  | 28  | 110 |
|                    | COVID   | 474   | 25.72 | 13.95 | 3   | 16  | 23  | 33  | 140 |
|                    | EBM-NLP | 200   | 25.70 | 13.83 | 6   | 16  | 22  | 32  | 91  |
| Fragments          | WD      | 418   | 6.28  | 4.68  | 1   | 3   | 5   | 8   | 42  |
|                    | COVID   | 83    | 9.73  | 18.30 | 1   | 2   | 6   | 9   | 129 |
|                    | EBM-NLP | 341   | 3.83  | 4.01  | 1   | 1   | 2   | 5   | 30  |

# Appendix G

## Benchmarking FastSR Against Leading LLMs: A Comparative Analysis

### LLMs Overview

LLMs include masked language models (MLMs) like BERT (Devlin et al., 2019), which use Transformer encoders (Vaswani et al., 2017) to predict masked tokens with bidirectional self-attention, and autoregressive language models (ALMs) like GPT (Radford et al., 2018), which use Transformer decoders for text generation with unidirectional self-attention. The former excel in extractive tasks (Chen et al., 2024), while the latter are ideal for generative tasks (Brown et al., 2020). Both models can be pretrained or fine-tuned with domain-specific data to adapt to particular tasks. For instance, BioBERT (Lee et al., 2020) and SciBERT (Beltagy et al., 2019) are pretrained on biomedical texts and can be fine-tuned for BioNLP tasks. BioGPT (Luo et al., 2022) and GatorTronGPT (Peng et al., 2023) are pretrained on medical or clinical texts and can be fine-tuned for medical content generation. Fine-tuning large models like GPT-4 (Brown et al., 2020) typically requires extensive, high-quality labeled data. However, SR tasks often have limited labels per class, making fine-tuning less effective (Schmidt et al., 2021). ALMs can also perform tasks without further training through few-shot prompting, but they are highly sensitive to prompt wording, example selection, and domain specificity, making them more challenging for precise tasks (Liu et al., 2023). Additionally, ALMs are more susceptible to data contamination than MLMs—where test data overlaps with pre-training data, inflating performance (Dong et al., 2024). This is because ALMs, with their sequential generation process, are more likely to reuse memorized content, especially in open-domain datasets. To illustrate these limitations, we benchmarked FastSR against fine-tuned BioBERT, fine-tuned GPT models, and zero-shot/few-shot prompted GPT across both open-domain and close-domain datasets. Table G1 is an overview of the LLMs included in our benchmarking models. In our previous version, we reported the performance of fine-tuned BioBERT.

| Table G1. Categories of LLMs for Medical Domain |                            |            |               |                                                                     |                                                                                                                              |
|-------------------------------------------------|----------------------------|------------|---------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Type                                            | Model                      | Base model | # Params      | Pretrained dataset                                                  | Applications in medical domain                                                                                               |
| MLM                                             | BioBERT (open source)      | BERT       | 110 million   | PubMed abstracts, PMC full-text articles                            | Text classification, named entity recognition, relation extraction,                                                          |
| ALM                                             | BioGPT (open source)       | GPT-2      | 347 million   | 15 million PubMed abstracts                                         | End-to-end relation extraction, question answering, document classification                                                  |
|                                                 | GatorTronGPT (open source) | GPT-3      | 20 billion    | 277 billion words from UF Health clinical text and the Pile dataset | End-to-end relation extraction, question answering                                                                           |
|                                                 | GPT-3.5 (API)              |            | 175 billion   | Diverse text data (books, articles, websites, social media, codes)  | Text classification, named entity recognition, relation extraction, question answering, text summarization, creative writing |
|                                                 | GPT-4 (API)                |            | ~1.8 trillion |                                                                     |                                                                                                                              |

### Evaluation Strategy

We explored two prevalent methods—few-shot prompting and fine-tuning—used in the existing literature to adapt ALMs for the SR context (Chen et al., 2024; Luo et al., 2022; Peng et al., 2023). Given that GPT-3.5 and GPT-4 are charged based on API calls, with costs directly related to the number of testing examples, we began with the EBM-NLP dataset, which has the fewest testing sentences, as shown in Table G2.

For the EBM-NLP dataset, we tested all possible combinations of models and methods for sentence classification, except for GPT-4, as OpenAI’s fine-tuning service for GPT-4 was not available at the time of the experiments in July 2024. We then conducted sequence tagging. In the SR context, the success of sequence tagging is highly dependent on the performance of sentence classification. If sentences are extracted incorrectly, all subsequent text fragments will be incorrect. Likewise, if any sentences are missed, the corresponding fragments will also be absent, leading to incomplete results. Both BioGPT and GatorTronGPT performed significantly worse, with at least a 46% performance gap in F1 score compared to FastSR. Fine-tuned GPT-3.5 also showed a 9% performance gap compared to FastSR. Therefore, we focused on the top-performing model in sentence classification—GPT-4 with few-shot prompting—for the sequence tagging evaluations.

Data contamination presents a significant challenge when evaluating LLMs (Brown et al., 2020). This occurs when models are exposed to testing data during pretraining, leading to artificially inflated performance. The issue is particularly pronounced in open-domain datasets like EBM-NLP (Lewis et al., 2021) and closed-source models (Dong et al., 2024). To obtain more reliable evaluations, we extended our benchmarking to closed-domain datasets, specifically the WD and COVID datasets, as recommended by existing literature (Gao et al., 2021). These datasets consist of full-text articles from credential-restricted academic platforms, making them less susceptible to web crawling and thus less likely to be included in pretraining datasets.

**Table G2. Evaluation Strategy for LLMs**

| Data                          | Public access | # Class | Train sent.       | Test art. /sent. | Tasks                   | Model                         | Method                  |
|-------------------------------|---------------|---------|-------------------|------------------|-------------------------|-------------------------------|-------------------------|
| <b>EBM-NLP</b><br>(Abstracts) | High          | 3       | 255               | 200/<br>2044     | Sentence classification | GPT-4                         | Few-shot                |
|                               |               |         |                   |                  |                         | GPT-3.5, BioGPT, GatorTronGPT | Few-shot, fine-tuned    |
|                               |               |         |                   |                  | Sequence tagging        | GPT-4                         | Few-shot <sup>[1]</sup> |
| <b>WD</b><br>(Full-Texts)     | Low           | 6       | 418               | 110/<br>12,320   | Sentence classification | GPT-4                         | Few-shot                |
| <b>COVID</b><br>(Full-Texts)  | Low           | 4       | 83 <sup>[2]</sup> | 62/<br>8464      | Sentence classification | GPT-4                         | Few-shot                |

**Note:** <sup>[1]</sup> GPT-4 fine-tuning service was not available at the time of the experiments in July 2024. <sup>[2]</sup> The COVID dataset contains four classes, three of which overlap with the WD dataset, and one new class, "Risk Factor," with 83 labeled sentences.

Given the extensive number of testing sentences in the WD and COVID datasets, we focused on GPT-4 with few-shot prompting, the top performer. A 32% performance gap between the ALMs and FastSR in sentence classification indicated that successful sequence tagging was unlikely. Additionally, API costs had already approached \$4,660. Considering these factors, we decided not to proceed with sequence tagging for these datasets.

Table G2 offers an overview of the evaluation strategy for LLMs across three datasets and two tasks. We also employed the following experiment setups across all datasets and tasks. First, we experimented with various prompt variations on a subset of sentences, iterating at least three times per task to identify the most effective prompting strategy. Second, to ensure reproducibility, we used specific snapshots of GPT-3.5 (gpt-3.5-turbo-0125) and GPT-4 (gpt-4-turbo-2024-04-09) and also set the LLM temperature to 0.7 to maintain consistent randomness in content generation. Third, for fine-tuning, EBM-NLP and WD utilized all available training data, while COVID used the training examples from WD for the three overlapping classes, combined with COVID-specific data for the "Risk Factor" class. This approach aligns with the experimental setup used for FastSR and other benchmark models on the COVID dataset. Finally, due to the high cost associated with GPT models, we did not generate multiple splits but conducted an article/sentence-level pairwise t-test to calculate statistical significance. Nonetheless, the substantial performance margin between these models and FastSR clearly demonstrates the effectiveness of our approach. Detailed prompting and fine-tuning strategies are described in the following section.

## Prompting and Fine-Tuning Strategy

### Sentence Classification

As mentioned, we applied few-shot prompting and fine-tuning to all models—BioGPT, GatorTronGPT, GPT-3.5, and GPT-4—for sentence classification tasks on EBM-NLP datasets and selected the top-performing model for the WD and COVID datasets. Following Chen et al. (2024), we adopted a prompt engineering strategy to improve the ALM's ability to understand the context and generate the expected output. This strategy organizes prompts around components, including Task Description, Examples, Input, and Output specifications (see Table G3). Additionally, to facilitate the calculation of the precision-recall curve (PRC), we generated predictive probabilities for each label (Saito & Rehmsmeier, 2015). In the context of SR, where a sentence may correspond to multiple PICO classes, a multilabel classification scenario is required. Thus, the predictive probabilities for each label are independently assessed and do not need to sum to 1.

Selecting examples for few-shot prompting is essential for reducing biases and preventing overfitting (Gao et al., 2021). Both the order and composition of these examples significantly impact model performance (Liu et al., 2023). We opted to randomly sample five examples for the EBM-NLP dataset per label for three key reasons: (1) more examples can sometimes degrade performance due to the model's sensitivity to input length and context window size (Brown et al., 2020; Gao et al., 2021); (2) using one to five examples per label aligns with standard practices in few-shot prompting studies, particularly in the medical domain (Chen et al., 2024); and (3) limiting the number of examples helps control API costs, which increase with context window size. For labels with fewer than five examples, we used sampling with replacement to ensure balanced representation and reduce sampling bias. A fixed random seed (42) was chosen to ensure reproducibility. For the WD and COVID datasets, we selected one example per class to manage the high API costs associated with large context window.

For fine-tuning, we adapted the pretrained BioGPT (Luo et al., 2022) and GatorTronGPT (Peng et al., 2023) models for sentence classification using labeled examples from each dataset. The dataset was split into 80% for training and 20% for validation. We used the Hugging Face Transformers library (v4.18.0) with PyTorch (v1.10.0) as our framework. Fine-tuning was performed on 4 NVIDIA A100 GPUs for 3 epochs. The AdamW optimizer (Loshchilov & Hutter, 2019) was employed with a learning rate of 5e-5, epsilon of 1e-8, and weight decay of 0.01. The batch size was set to 8 per GPU. The time for fine-tuning on one task took approximately 15 to 30 minutes.

We utilized OpenAI's fine-tuning service for GPT-3.5. As before, the labeled data was split into 80% training and 20% testing. The data was formatted according to OpenAI's fine-tuning specifications, with each example containing a prompt and a completion. The fine-tuning process was initiated through OpenAI's API, with default hyperparameters recommended for classification tasks: a learning rate multiplier of 0.1, 3 epochs, and a batch size of 8. The fine-tuning for one task took approximately 10 minutes. After fine-tuning, we applied the same prompting strategies but without examples to obtain the sentence classification results.

**Table G3. Prompt for Sentence Classification**

| Prompt component                | Content                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Task description                | Classify the sentence to determine if it mentions information about the following labels: 'interventions,' 'outcomes,' and 'participants.' Each sentence can have multiple labels. For each label, generate a classification probability that indicates the likelihood of the information related to the label being '1' (mentioned) in the text. |
| Label definition                | Interventions include surgical, physical, educational, pharmacological, control, psychological, and other types of intervention.<br>Participants include condition, age, sample size, and sex.<br>Outcomes include physical outcomes, pain, adverse effects, mental outcomes, and mortality.                                                      |
| Examples for few-shot prompting | Example input: The study included 100 patients with type 2 diabetes who received metformin treatment for 6 months, resulting in improved glycemic control.<br>Example output: {'interventions': 0.9, 'outcomes': 0.8, 'participants': 0.95}                                                                                                       |
| Input                           | Input text: {text}                                                                                                                                                                                                                                                                                                                                |
| Output specification            | Output specification: The output should be in JSON format, where the keys are the labels, and the values are the classification probabilities. Use a JSON format like {'interventions': 0.6, 'outcomes': 0.7, 'participants': 0.6}.                                                                                                               |

## Sequence Tagging

Given that the success of sequence tagging is closely tied to the performance of sentence classification, we selected the top-performing model in sentence classification—GPT-4 few-shot prompting—for our sequence tagging evaluations. Following (Hu et al., 2024), we developed a comprehensive prompting strategy that includes a task description, fragment markup guide, fragment definitions, examples, and input/output specifications. During the selection of few-shot examples, we noticed a significant imbalance in fragment distribution among subtypes (e.g., surgical, physical, and educational interventions). To address this, after performing random sampling per label, we adjusted the examples to ensure that each subtype was represented by at least one example in the few-shot set. To ensure reproducibility, we used a fixed random seed (42).

**Table G4 Prompt for Sequence Tagging**

| Prompt component               | Content                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Task description               | Generate an HTML version of the input text, marking up specific fragments related to "interventions," "participants," and "outcomes." Use HTML <span> tags to highlight these fragments. Each <span> should have a class attribute indicating the type of intervention.                                                                                                                                                                        |
| Fragment markup guide          | Use <span class= 'intervention'> to denote an intervention.                                                                                                                                                                                                                                                                                                                                                                                    |
| Fragment definition            | Intervention includes surgical, physical, educational, pharmacological, control, psychological, and other types of intervention.<br>Participants include condition, age, sample size, and sex.<br>Outcomes include physical outcomes, pain, adverse effects, mental outcomes, and mortality.                                                                                                                                                   |
| Examples in few-shot prompting | Example input: Thirty patients were treated with a simple shoulder sling and 30 patients with ESIN within 3 days after trauma.<br>Example output: <span class= 'participants'>Thirty patients</span> were treated with a <span class= 'interventions'>simple shoulder sling</span> and <span class= 'participants'>30 patients</span> with <span class= 'interventions'>ESIN</span> within 3 days after <span class= 'outcomes'>trauma</span>. |
| Input                          | Input text: {text}                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Output specification           | Provide the marked-up HTML version of the input text.                                                                                                                                                                                                                                                                                                                                                                                          |

## Results

We begin by discussing the results of sentence classification and sequence tagging on the EBM-NLP dataset, an open-domain dataset. Next, we analyze the sentence classification performance on closed-domain datasets. Finally, we conclude with insights drawn from these evaluations.

### EBM-NLP

Table G5 shows that open-source models like BioGPT and GatorTronGPT perform significantly worse in sentence classification compared to larger proprietary models like GPT-3.5 and GPT-4. Fine-tuning leads to a better performance than few-shot prompting. GPT-4 few-shot prompting demonstrates superior performance with the highest F1 (67.59%) and PRC (61.49%), likely due to their advanced architectures and larger size. However, FastSR outperforms all models, with a 12.12% (20.05%) performance gain in F1 (PRC) over GPT-4 few-shot prompting, highlighting FastSR's robustness even amidst the development of large language models.

| Table G5. Sentence Classification Benchmarking on EBM-NLP Dataset |       |       |              |          |            |          |
|-------------------------------------------------------------------|-------|-------|--------------|----------|------------|----------|
| Models / metrics (%)                                              | P@3   | R@3   | F1@3 & Diff. |          | PRC & Diff |          |
| BioGPT                                                            |       |       |              |          |            |          |
| Few-shot                                                          | 8.84  | 11.32 | 9.93         | 69.78*** | 37.66      | 43.88*** |
| Fine-tuned                                                        | 90.26 | 16.05 | 25.38        | 54.33*** | 70.56      | 10.98*** |
| GatorTronGPT                                                      |       |       |              |          |            |          |
| Few-shot                                                          | 31.62 | 33.33 | 32.05        | 47.66*** | 39.12      | 42.42*** |
| Fine-tuned                                                        | 52.90 | 28.85 | 33.09        | 46.62*** | 67.82      | 13.72*** |
| GPT-3.5 Turbo                                                     |       |       |              |          |            |          |
| Few-shot                                                          | 50.09 | 55.36 | 52.59        | 27.12*** | 44.96      | 36.58*** |
| Fine-tuned                                                        | 58.28 | 59.37 | 58.72        | 20.99*** | 49.66      | 31.88*** |
| GPT-4 Turbo                                                       |       |       |              |          |            |          |
| Few-shot                                                          | 63.70 | 72.19 | 67.59        | 12.12*** | 61.49      | 20.05*** |
| Fine-tuned BioBERT                                                | 85.01 | 58.00 | 64.61        | 15.10*** | 77.65      | 3.89***  |
| FastSR                                                            | 82.14 | 77.70 | 79.71        | -        | 81.54      | -        |

| Table G6. Sequence Tagging for PICO Fragment Extraction on EBM-NLP Dataset |                      |                      |                               |          |                      |                      |                              |
|----------------------------------------------------------------------------|----------------------|----------------------|-------------------------------|----------|----------------------|----------------------|------------------------------|
| Models /Metrics (%)                                                        | P <sub>word</sub> @3 | R <sub>word</sub> @3 | F1 <sub>word</sub> @3 & diff. |          | P <sub>BERT</sub> @3 | R <sub>BERT</sub> @3 | F1 <sub>BERT</sub> @3 & diff |
| <b>GPT-4 Few-Shot</b>                                                      | 34.73                | 34.08                | 32.86                         | 33.22*** | 57.32                | 58.65                | 55.28 20.21***               |
| <b>Fine-tuned BioBERT</b>                                                  | 76.18                | 40.97                | 47.40                         | 18.68*** | 80.36                | 51.44                | 57.74 17.75***               |
| <b>FastSR</b>                                                              | 66.00                | 68.21                | 66.08                         | -        | 77.75                | 73.97                | 75.49 -                      |

**Note:** Significance levels are indicated as follows: \*p < 0.10, \*\*p < 0.05, \*\*\*p < 0.01.

When applying GPT models to fragment extraction, the performance gap between GPT models and FastSR becomes more pronounced, with a 33.22% difference in word-level F1 and a 20.21% gap in BERTScore. This gap is likely because fragment extraction requires a deep understanding of context in both directions. In contrast, GPT models rely on a unidirectional self-attention mechanism that only considers preceding words, thereby limiting their effectiveness in such tasks.

The comparison between GPT models and BioBERT further highlights this limitation. Despite being a much smaller model in terms of parameters (millions versus GPT-4's trillions), BioBERT outperforms GPT-4's best few-shot variant by 14.54% in word-level F1. This underscores the importance of bidirectional context processing in complex NLP tasks like fragment extraction, where the ability to understand the context from both directions is more critical than sheer model size.

Interestingly, compared to BioBERT and FastSR, GPT models demonstrate more significant gains when evaluated with BERTScore versus word-level F1 score. For instance, GPT-4's few-shot performance improves by 22.42% when evaluated with BERTScore relative to its word-level F1 score, whereas BioBERT shows only a 10.34% improvement. This narrows the gap between BioBERT and GPT-4 to just 2.46% in BERTScore, compared to a 14.54% gap in word-level F1. Since BERTScore evaluates fragment matching based on semantic similarity rather than strict word-by-word alignment, this suggests that GPT models are less precise in lexical matching but are more adept at semantic understanding, further supporting the fact that GPT models may be more effective for generative tasks rather than for extractive tasks that require precise identification of lexical boundaries.

## WD and COVID

We then evaluated GPT models' sentence classification performance on the WD (Table G7) and COVID datasets (Table G8), both of which have low public accessibility and minimal exposure to data contamination. Similar to the open-domain EBM-NLP dataset, GPT 4 significantly outperformed GPT 3.5 by a large margin. However, GPT models significantly underperform FastSR, with a 33.77% (43.67%) gap in F1 (PRC) on the WD dataset and an average 31.55% (32.15%) gap in F1 (PRC) on the COVID datasets. These findings suggest that GPT's performance on the EBM-NLP dataset may suffer from severe data contamination, where the annotated abstracts in the EBM-NLP dataset were likely included in the pretraining data.

**Table G7. Sentence Classification Benchmarking on WD Dataset**

| Models / metrics (%) | P@3   | R@3   | F1@3 & diff. |          | PRC & diff. |          |
|----------------------|-------|-------|--------------|----------|-------------|----------|
| GPT 3.5 Fine-tuned   | 58.49 | 25.45 | 25.00        | 48.00*** | 19.72       | 52.89*** |
| GPT 4 Few-shot       | 31.22 | 55.26 | 39.23        | 33.77*** | 28.94       | 43.67*** |
| Fine-tuned BioBERT   | 63.63 | 66.35 | 64.17        | 8.83***  | 63.77       | 8.84***  |
| FastSR               | 72.34 | 74.09 | 73.00        | -        | 72.61       | -        |

Note: Significance levels are indicated as follows: \*p < 0.10, \*\*p < 0.05, \*\*\*p < 0.01.

**Table G8. Sentence Classification Benchmarking on COVID Dataset**

| Models / metrics (%)        | P@3   | R@3   | F1@3 & diff. |          | PRC & diff. |          |
|-----------------------------|-------|-------|--------------|----------|-------------|----------|
| Reused                      |       |       |              |          |             |          |
| GPT 3.5 Fine-tuned          | 37.26 | 47.76 | 41.86        | 36.58*** | 30.77       | 44.99*** |
| GPT 4 Few-shot              | 37.24 | 50.89 | 43.01        | 35.43*** | 33.33       | 42.43*** |
| Fine-tuned BioBERT          | 66.23 | 58.64 | 62.21        | 16.23*** | 65.01       | 10.75*** |
| FastSR                      | 75.59 | 81.53 | 78.44        | -        | 75.76       | -        |
|                             |       |       |              |          |             |          |
| Risk factor (newly trained) |       |       |              |          |             |          |
| GPT 3.5 Fine-tuned          | 47.76 | 41.36 | 44.33        | 32.63*** | 44.96       | 30.33*** |
| GPT4 Few-shot               | 45.90 | 55.86 | 50.40        | 26.56*** | 53.43       | 21.86*** |
| Fine-tuned BioBERT          | 57.44 | 62.65 | 59.93        | 17.03*** | 58.00       | 17.29*** |
| FastSR                      | 71.73 | 83.02 | 76.96        | -        | 75.29       | -        |

Note: Significance levels are indicated as follows: \*p < 0.10, \*\*p < 0.05, \*\*\*p < 0.01.

In contrast, the BioBERT model appears more robust to data contamination, with a 24.94% (34.83%) performance gain in F1 (PRC) on the WD dataset and an average 14.43% (18.13%) decrease in F1 (PRC) on the COVID dataset, possibly due to its smaller pretraining dataset and its ability to genuinely understand text semantics through bidirectional self-attention. Due to the lower performance in sentence classification and the high API costs, we did not proceed with sequence tagging for GPT models.

## Discussion

Based on the benchmarking analyses, we identified the following takeaways:

**FastSR's superiority in complex SR tasks:** GPT-4 surpasses its predecessors like GPT-3.5 and domain-specific models due to its advanced architecture and extensive training data. FastSR consistently outperforms GPT models across various datasets, particularly in complex SR tasks requiring precise text comprehension. In our EBM-NLP dataset experiments, FastSR demonstrated a significantly greater advantage in sequence tagging (33.22% higher F1) compared to sentence classification (12.12% higher F1). This disparity likely arises from GPT models' use of unidirectional self-attention, which generates text sequentially by considering only the preceding context. In contrast, FastSR leverages BioBERT for base word embeddings, effectively capturing context from both directions and giving it a distinct edge in tasks where full-context understanding is crucial.

**GPT's challenges with closed-domain datasets:** GPT models perform poorly on closed-domain datasets. In our experiments with the open-domain EBM-NLP dataset, GPT models' sentence classification performance was 12% behind FastSR. However, on closed-domain datasets, the performance gap widened significantly, with GPT models showing over 30% degradation. This disparity likely arises from data contamination, where open-domain datasets and their labels may have been included in GPT models' pretraining, artificially boosting performance on related tasks. This suggests that GPT models without significant customizations, such as retrieval augmented generation (Lewis et al., 2020), may struggle in real-world SR tasks, where most datasets are closed-domain, comprising full-text articles from credential-restricted academic platforms.

**Limited control in GPT models:** GPT models offer less direct control over output performance compared to traditional machine learning approaches, often requiring extensive experimentation to achieve desired results. This is due to several factors: the opaque nature of GPT's complex neural network architecture, especially in proprietary models, which makes it difficult to interpret how specific inputs lead to certain outputs; the autoregressive design, where each token is generated sequentially based on preceding tokens, complicating control over longer outputs; and high sensitivity to prompt formulation, where small wording changes can cause significant output variations. These factors make predicting and controlling GPT models' behavior challenging, particularly in tasks like SR data extraction, where accuracy and consistency are crucial for evidence-based decision-making. The lack of control in GPT outputs could lead to inconsistencies in data extraction, compromising the integrity of SR.

**High cost and low efficiency in GPT models:** The cost of using proprietary GPT models, despite their strong performance, can be high, especially compared to FastSR or BERT-based models. GPT models' costs increase with the number of testing sentences due to API calls, whereas FastSR's costs are primarily incurred during the initial training phase, with minimal expenses for generating additional content. In our experiments with models like GPT-4, costs ranged from \$1.5 to \$4 per article for generating candidate sentences and fragments for SR experts to review. However, the lack of control over GPT outputs often requires further refinement of these initial results, increasing costs even more. Additionally, GPT's lower performance in specific tasks means SR experts must review more articles to verify results than they would with FastSR, further raising manual costs. Moreover, GPT models like GPT-4 can have variable response times, sometimes taking several seconds or more to generate a typical response for an SR task. This latency can become more pronounced during high-traffic periods, leading to potential delays in time-sensitive applications. For instance, in our experiments, testing on a large dataset like WD took 25 hours with GPT models, compared to just 2 hours with FastSR. Overall, FastSR emerges as a more cost-effective and time-efficient solution compared to GPT models.

# Appendix H

## Error Analysis

To further demonstrate the value of the proposed design components, we ran an error analysis comparing FastSR to the best models in each method group (i.e., CNN / LSTM+CRF for the DL and ProtoNet / ProtoNER for FSL) in the WD dataset. Table H1 contains a few tagged examples. Ex. 1 shows the importance of global context: both benchmarks failed to retrieve relevant sentences. Ex.2 demonstrates the value of the attention mechanism: while the sentences were correct, DL missed one number (lower recall), and routine FSL identified additional words (lower precision). Ex.3 highlights the value of joint learning by showing how different types of errors can interact to reduce extraction performance. For example, the DL model failed to account for fragments when classifying sentences, despite a correct identification of the section.

**Table H1. Error Analysis for Sentence Classification and Sequence Tagging on WD Dataset**

| Comparison                                | Sentences and fragments                                                                                                                                      | Section              | Reasons                                                 |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------------------------------------------------|
| <b>Ex 1. Perspective</b>                  |                                                                                                                                                              |                      |                                                         |
| Ground-truth                              | The enrolled patients were <b>prospectively</b> examined at the Department of Neurosciences ...                                                              | Patients and methods |                                                         |
| CNN<br>LSTM + CRF                         | It would be interesting to test this hypothesis in a <b>prospective</b> study on a larger sample of patients.                                                | Discussion           | Incorrect section                                       |
| ProtoNet<br>ProtoNER                      | It would be interesting to test this hypothesis in a <b>prospective</b> study on a larger sample of patients.                                                | Discussion           | Incorrect section                                       |
| FastSR                                    | The enrolled patients were <b>prospectively</b> examined at the Department of Neurosciences ...                                                              | Patients and methods |                                                         |
| <b>Ex. 2. Sample size</b>                 |                                                                                                                                                              |                      |                                                         |
| Ground-truth                              | <b>Fifty-one</b> treatment-naïve NWD patients were ...                                                                                                       | Abstract             |                                                         |
| CNN<br>LSTM + CRF                         | <b>Fifty-one</b> treatment-naïve NWD patients were subjected to detailed clinical evaluation.                                                                | Abstract             | Correct section and sentences but incorrect tagging     |
| ProtoNet<br>ProtoNER                      | <b>Fifty-one</b> treatment-naïve NWD patients were subjected to detailed clinical evaluation.                                                                | Abstract             | Correct section and sentence but incorrect tagging      |
| FastSR                                    | <b>Fifty-one</b> treatment-naïve NWD patients were ...                                                                                                       | Abstract             |                                                         |
| <b>Ex.3. Intervention/dosage/delivery</b> |                                                                                                                                                              |                      |                                                         |
| Ground-truth                              | The presymptomatic patients were treated with <b>50 mg of elemental zinc sulfate twice a day, separated from food by at least 1 hour</b> , from the time ... | Methods              |                                                         |
| CNN<br>LSTM + CRF                         | Three short tandem repeat (STR) markers, D13S133, D13S301, and D13S314, were ...                                                                             | Methods              | Correct section but incorrect sentence (nothing tagged) |
| ProtoNet<br>ProtoNER                      | <b>Zinc is recommended for use as maintenance therapy</b> or for the rare patient who is <b>intolerant of trientine or penicillamine.</b>                    | Comment              | Incorrect section                                       |
| FastSR                                    | The presymptomatic patients were treated with <b>50 mg of elemental zinc sulfate twice a day</b> , separated from food by at least 1 hour, from the time ... | Methods              | Partially retrieved, only miss "Delivery"               |

**Table H2. Error Analysis for Sentence Classification and Sequence Tagging on EBM-NLP Dataset**

| Comparison          | Sentences and Fragments                                                                                                                                                                | Subcategory                | Reasons                                                                                                                                   |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Ex 1. Participants  |                                                                                                                                                                                        |                            |                                                                                                                                           |
| Ground-truth        | The 101 eligible cases (bestatin: 48, control: 53) were analyzed; the bestatin group achieved longer remission than the control group and a statistically significant longer survival. | Participants - Sample Size |                                                                                                                                           |
| CNN LSTM + CRF      | It would be interesting to test this hypothesis in a prospective study on a larger sample of patients.                                                                                 |                            | Missed the entire fragment because this pattern is not present in the total 17 training samples of this subcategory                       |
| ProtoNet ProtoNER   | It would be interesting to test this hypothesis in a prospective study on a larger sample of patients.                                                                                 |                            |                                                                                                                                           |
| FastSR              | The 101 eligible cases (bestatin: 48, control: 53) were analyzed; the bestatin group achieved longer remission than the control group and a statistically significant longer survival. |                            | Captured the total sample size, but missed the samples for treatment and control groups                                                   |
| Ex. 2. Intervention |                                                                                                                                                                                        |                            |                                                                                                                                           |
| Ground-truth        | CONCLUSIONS The Junior Detective Training Program appeared to be effective in enhancing the social skills and emotional understanding of children with AS.                             | Intervention - Educational |                                                                                                                                           |
| CNN LSTM + CRF      | CONCLUSIONS The Junior Detective Training Program appeared to be effective in enhancing the social skills and emotional understanding of children with AS.                             |                            | Missed the entire fragment because this type of treatment is underrepresented in the total 16 training samples of this subcategory        |
| ProtoNet ProtoNER   | CONCLUSIONS The Junior Detective Training Program appeared to be effective in enhancing the social skills and emotional understanding of children with AS.                             |                            | Only captured a portion of the fragment                                                                                                   |
| FastSR              | CONCLUSIONS The Junior Detective Training Program appeared to be effective in enhancing the social skills and emotional understanding of children with AS.                             |                            | Only captured a portion of the fragment                                                                                                   |
| Ex.3. Outcome       |                                                                                                                                                                                        |                            |                                                                                                                                           |
| Ground-truth        | Tryptophan depletion led to a significant increase in behaviors such as whirling, flapping, pacing, banging and hitting self, rocking, and toe walking (P < .05).                      | Outcome - Mental           |                                                                                                                                           |
| CNN LSTM + CRF      | Tryptophan depletion led to a significant increase in behaviors such as whirling, flapping, pacing, banging and hitting self, rocking, and toe walking (P < .05).                      |                            | Missed part of the fragment perhaps because this outcome pattern is underrepresented in the total 15 training samples of this subcategory |
| ProtoNet ProtoNER   | Tryptophan depletion led to a significant increase in behaviors such as whirling, flapping, pacing, banging and hitting self, rocking, and toe walking (P < .05).                      |                            |                                                                                                                                           |
| FastSR              | Tryptophan depletion led to a significant increase in behaviors such as whirling, flapping, pacing, banging and hitting self, rocking, and toe walking (P < .05).                      |                            |                                                                                                                                           |

